

Tunduma Water Supply Project, Intake, Raw Water transmission main; Water Treatment Plant; Treated water transmission main; Ikana and Uhuru Park Tunduma Storage Tanks; Nkangamo Booster Pump Station, Access roads, SCADA and Interconnecting main distribution pipes from Uhuru Park Tunduma Storage tanks

Part 2 – Employer's Requirements and Specifications

Section VII – Employer's Requirements and Specifications

A Employer's Requirements

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List of Abbreviations

BR	Balancing Reservoir
DCI	Ductile Cast Iron
DD	Detailed Design
DN	Diameter Nominal
DNP	Defects Notification Period
DWPS	Drinking Water Pump station
DWTP	Drinking Water Treatment Plant
DS	Dissolved Solids
E&M	Electro-mechanical
ESIA	Environmental and Social Impact Assessment
ESHS-AP	Environmental, Social, Health and Safety Action Plan
ESMP	Environmental and Social Management Plan
EN	European Norm
FC	Fecal Coliforms
FD	Final Design
FIDIC	Fédération Internationale des Ingénieurs-conseils
FS	Feasibility Study
GCF	Green Climate Fund
GIS	Geographic Information System
HDPE	High Density Polyethylene
H&S	Health and Safety
KfW	Kreditanstalt für Wiederaufbau
LS	Lump Sum
NPV	Net Present Value
OHTL	Overhead Transmission Line
O&M	Operation and Maintenance
PACl	Poly Aluminum Chloride
P&DB	Plant and Design Build
PLC	Programmable Logical Control
PR	Primary Reservoir
PVC	Polyvinyl Chloride
PS	Pump station
RAP	Resettlement Action Plan
RC	Reinforced Concrete

RWPS	Raw Water Pump station
SoW	Scope of Works
TANESCO	Tanzania Electric Supply Company
TOC	Total Organic Carbon
ToR	Terms of Reference
VFD	Variable Frequency Drive
WHO	World Health Organization

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1. Introduction

1.1 General

The Employer's Requirements presented herein define the location of the Site and the Works, specify the purpose of the Works and particular requirements for the completed Works, including functional requirements, quality, performance and scope, requirements for the supply of certain items (such as consumables), special obligations and other details generally in accordance with the Conditions of Contract for Plant and Design Build, Second Edition 2017, published by FIDIC.

The technical specifications are deemed to form and shall be construed as an integral part of the Employer's Requirements.

1.2 Project Objective

The purpose of the project

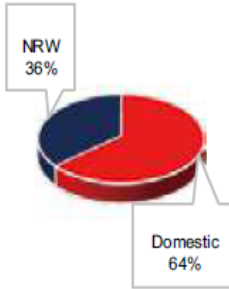
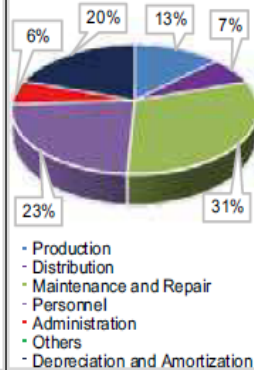
Tunduma Water Supply Project, Phase 1: Intake, Grit tank, Raw Water transmission main; Water Treatment Plant; Treated water transmission main; Storage Tanks; Access roads, SCADA and Interconnecting main distribution pipes from Storage tanks with the intention to develop a reliable water source, design and construct a water supply for Tunduma town, from river Momba as a source.

The project's objective of Phase 1 of this project is the turnkey construction of

- i. the Water Treatment Plant, including intake, raw water transmission main and one treatment line, and connecting pump main;
- ii. PS and treated water pumping main from drinking water reservoir to storage tanks;
- iii. Access roads

1.3 Current status

According to EWURA performance report 2022/2023, the class and capacity of Tunduma WSSA is as follow

TUNDUMA WSSA PROFILE				2022/23			
EWURA CLASS Provisional		LICENCE NO: PWSSSL/05/2023			CATEGORY: C		
General Data About Water Utility		Tariff Structure					
			Domestic	Institution	Industry	Commercial	Other
Total population (Nos)	21,930	Water					
Length of water network (km)	98.11	(TZS/m ³)	420	500	na	655	na
Water Demand (m ³)	7,570,794	Sewerage					
Water Production (m ³)	74,438	(TZS/m ³ of drinking water)					
Water Production Capacity (m ³ /d)	1,741.00		na	na	na	na	na
Direct Water Service Coverage (%)	40						
Total Water Connections (Nos)	906						
Total Active Connections (Nos)	906	The charge at water kiosk is TZS 50 per 20 litres container					
Domestic Connections (Nos)	840						
Operational Kiosks (Nos)	18	Water source distribution	Annual Water Use		Composition of O&M costs		
Sewerage Connections (Nos)	-				 - Production - Distribution - Maintenance and Repair - Personnel - Administration - Others - Depreciation and Amortization		
Metering Ratio (%)	100						
Non Revenue Water (%)	36.4						
Number of Staff (Nos)	20						
Staff/1000 connections (ratio)	22						
Service hours (hour)	6						
Sewerage Coverage (%)	na						
Working ratio	2.85						
Operating ratio	3.56						
Total Income (million TZS)	1,768.87						
Total O&M Expenses (million TZS)	385.63						
E.coli compliance (%)	100						
Turbidity compliance (%)	100						

1.4 Beneficiary

Beneficiary of the project is the population within the area covered by the project.

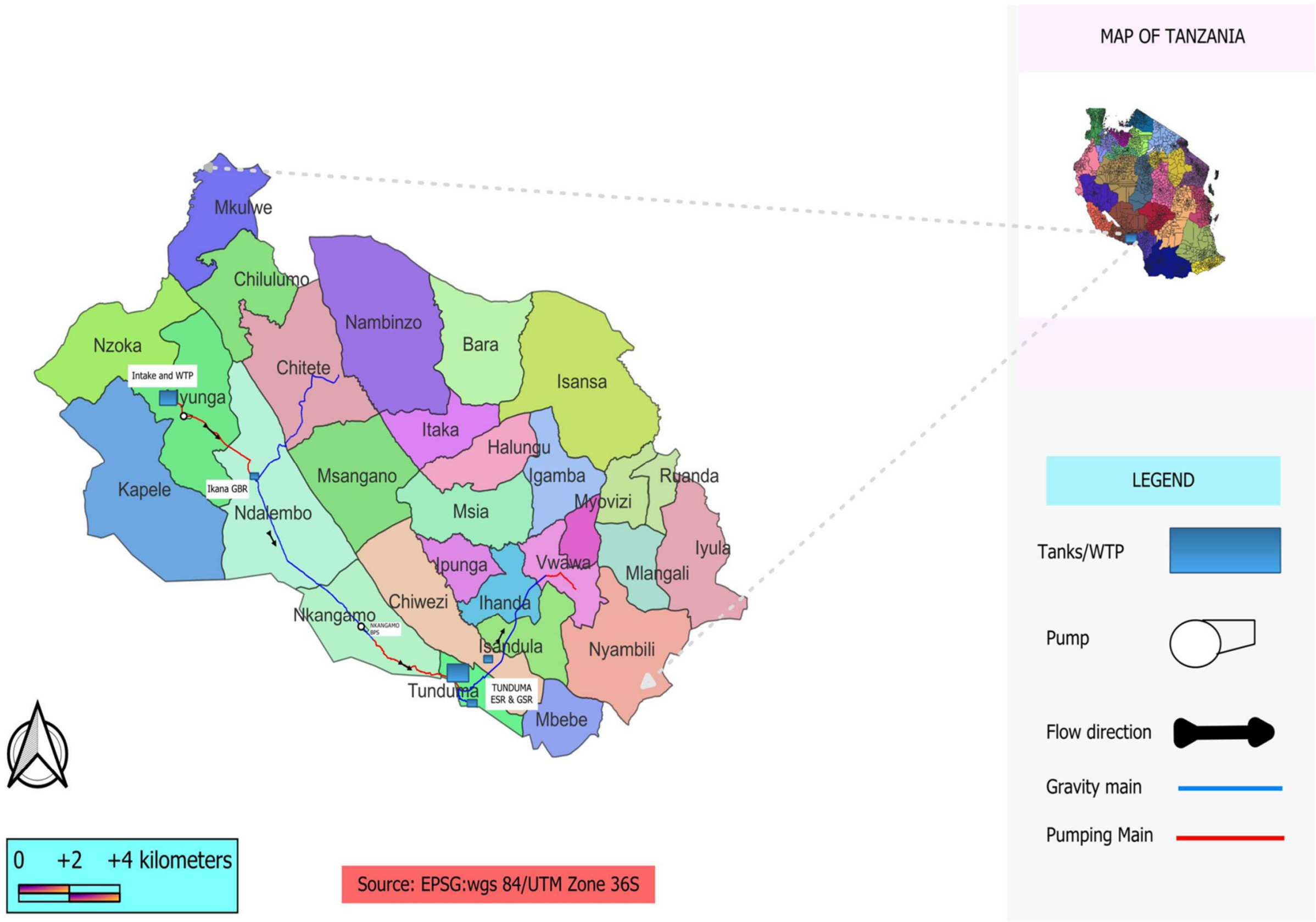
1.5 Notice of other Contracts

Currently there is ongoing Feasibility study under KfW and contractor should take into account inputs from the FS

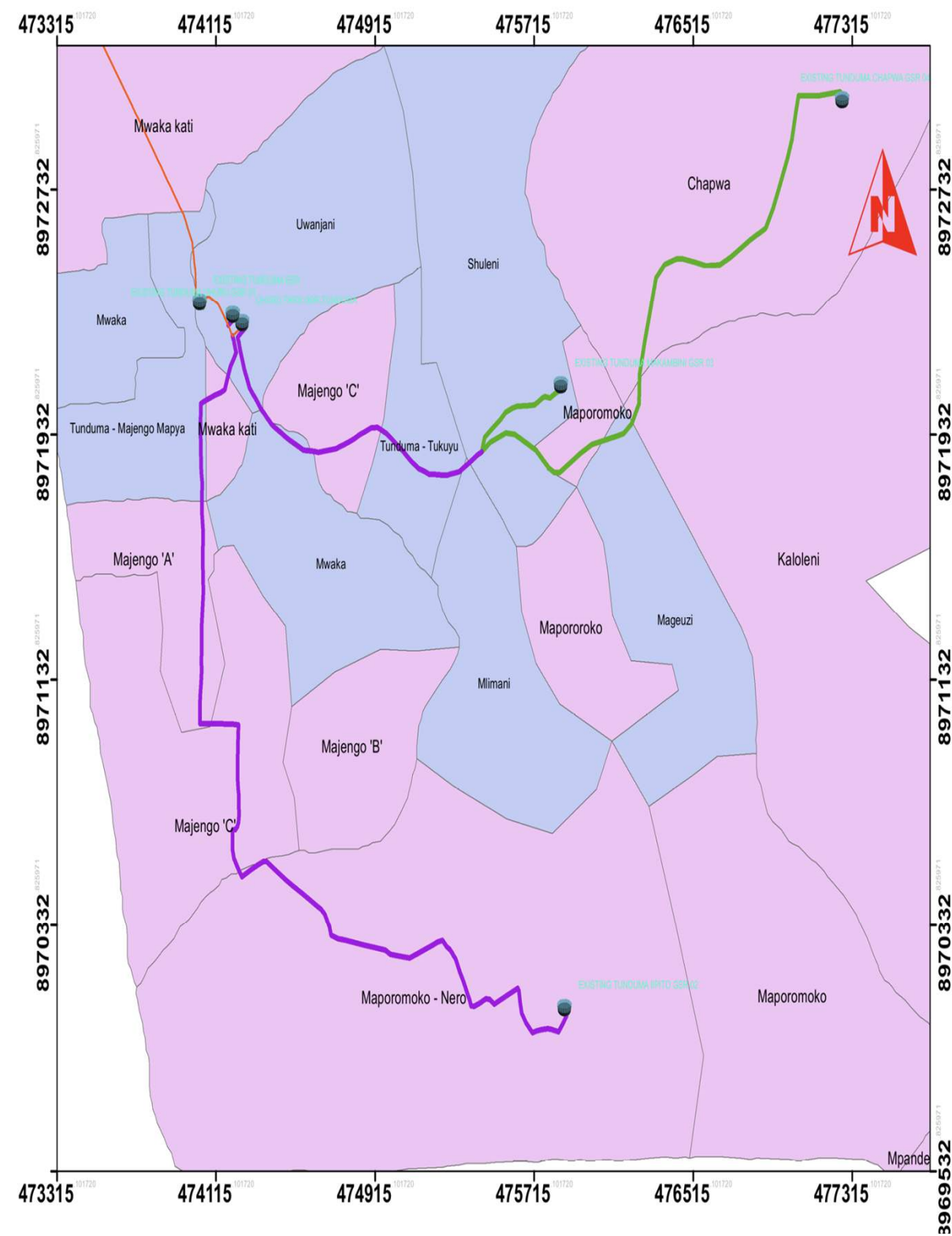
1.6 Site Locations

Tunduma is a town in Songwe Region, Tanzania, on the border between Tanzania and Zambia. It has border posts for both the Tanzam Highway and the TAZARA railway (for which it has a station) linking the two countries. It is located 103 km southwest of Mbeya. It is also the junction for the tarmac road which runs via Sumbawanga through the remote far western districts of Tanzania to Kasulu and Kibondo in the north-west. According to the 2022 census, it has a population of 219,309.[1] Tunduma can be the most famous border city within SADC countries and it is informally known as the gateway to East Africa with the neighbour country of Zambia. Once you are in Tunduma, you can enter the town of Nakonde, in Zambia either on foot or other means of transportation.

PROJECT NAME: TUNDUMA WATER SUPPLY PROJECT
LOCATION OF TUNDUMA WARDS FROM TANZANIA MAP



PROJECT NAME: TUNDUMA WATER SUPPLY PROJECT
AREA SURVED BY THE PROJECT



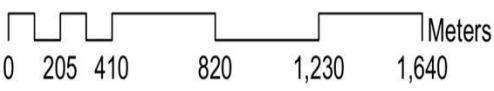
NOTES

1. This drawing should be read together with other drawings

Coordinate System: WGS 1984 UTM Zone 36S
Projection: Transverse Mercator
Datum: WGS 1984
False Easting: 500,000.0000
False Northing: 10,000,000.0000
Central Meridian: 33.0000
Scale Factor: 0.9996
Latitude Of Origin: 0.0000
Units: Meter

Legend

- PROPOSED STEEL-DN600-PN10/16/25
- PROPOSED STEEL-DN150-PN10/16
- PROPOSED STEEL-DN100-PN10/16



SUPPLIED TUNDUMA VILLAGES		
SUPPLIED BY TUNDUMA GBR AND EXISTING TANKS		
SN	NAME OF WARD	NAME OF VILLAGE
26	SOGEA	HALFLODONI
27	SOGEA	IKULU
28	MAKAMBINI	MAKAMBINI
29	MAKAMBINI	SOWETO
30	MAKAMBINI	KANANI
31	MAKAMBINI	TOKYO
32	TUNDUMA	TUKUYU
33	TUNDUMA	SIKANYIKA
34	TUNDUMA	IHANDA
35	TUNDUMA	TUNDUMA KATI
36	TUNDUMA	UNYAMWANGA
37	KATETE	KATETE
38	KATETE	MLENDA
39	KATETE	NAMANZANYA
40	KATETE	NAKAUMBA
41	KATETE	FURA
42	MPEMBA	MPEMBA
43	MPEMBA	MPEMBA
44	MPEMBA	NANSELE
45	MPEMBA	KATENIELE
46	MPEMBA	KOKOTO
47	MPEMBA	NANDANGA
48	MPEMBA	MSINDE
49	CHWEZI	CHWEZI
50	CHWEZI	NSANEYA

SUPPLIED TUNDUMA VILLAGES		
SUPPLIED BY TUNDUMA GBR AND EXISTING TANKS		
SN	NAME OF WARD	NAME OF VILLAGE
1	MAIENGO	MAIENGO A
2	MAIENGO	MAIENGO B
3	MAIENGO	MAIENGO C
4	MAIENGO	MAIENGO MAPYA
5	MPANDE	MPANDE KATI
6	MPANDE	MACHINIUNI
7	MPANDE	IRITO
8	MPANDE	MJIM WEMA
9	MPANDE	TAZARA
10	CHAPWA	CHAPWA
11	CHAPWA	ISANZO
12	CHAPWA	IGAWA
13	CHAPWA	MSAPANIA
14	CHAPWA	KESLIA
15	MAPOROMOKO	SHULENI
16	MAPOROMOKO	MAGEUZI
17	MAPOROMOKO	MUMANI
18	MAPOROMOKO	MAPOROMOKO
19	KALOLENI	NELO
20	KALOLENI	KALOLENI
21	KALOLENI	DANIDA
22	KALOLENI	CUSTOM
23	KALOLENI	MIGOMBANI
24	SOGEA	SOGEA
25	SOGEA	JAKAYA

SUPPLIED TUNDUMA VILLAGES		
SUPPLIED BY TUNDUMA GBR AND EXISTING TANKS		
SN	NAME OF WARD	NAME OF VILLAGE
51	CHWEZI	NAMOLE
52	CHWEZI	NIUMBA
53	CHWEZI	MSAMBA II
54	MUUNGANO	MUUNGANO
55	MUUNGANO	SATUNI
56	MUUNGANO	NAZARETI
57	MUUNGANO	MKOMBOZI
58	MUUNGANO	KISANGANI
59	MWAKAKATI	MWAKAKATI
60	MWAKAKATI	MELELANI
61	MWAKAKATI	SONGAMBELE
62	MWAKAKATI	GOROFANI
63	CHIPAKA	ZYGWILA
64	CHIPAKA	MWL NYERERE
65	CHIPAKA	KATETE
66	CHIPAKA	MSONGWA
67	CHIPAKA	CHIPAKA
68	UNCATEGORISED	KILUMA HEWA
69	UNCATEGORISED	MSASANI
70	UNCATEGORISED	MBAGALA
71	UNCATEGORISED	UWANJANI

1.7 Topographical site conditions

The topographical survey has been carried out by the Employer. The information is not warranted by the Employer, and the Contractor shall conduct detailed topographical survey and make use of and interpret the same on his own responsibility.

1.8 Geological and Geotechnical Site Conditions

The Red Sandstone Group (RSG) in the Rukwa Rift Basin of southwestern Tanzania represents one of the only well-exposed, fossiliferous Cretaceous–Paleogene continental sedimentary sequences in sub-equatorial Africa. The significance of the RSG for reconstructing the paleoenvironmental and paleoclimatic history of African ecosystems during these critical time periods has been obfuscated by long-standing confusion and debate over the age of the deposits. Detailed stratigraphic, sedimentologic, and paleontologic investigations of the RSG conducted between 2002 and 2008 have produced a wealth of new fossil discoveries and data on lithofacies, alluvial architecture, sedimentary provenance, clay mineralogy and geochronology that resolve the long-standing debate over the age of these deposits. This study confirms the existence of an extensive middle Cretaceous sequence, herein named the Galula Formation, and subdivided into the Mtuka and Namba members. Moreover, we document the existence of a previously unrecognized late Paleogene continental sequence termed the Nsungwe Formation, which is divided into the Utengule and Songwe members. The Galula Formation represents a 600-3000 m thick sequence of amalgamated, braided fluvial deposits that were deposited across a large braidplain system via multiple parallel channels that had their source in the highlands of Malawi and Zambia. The middle Cretaceous Dinosaur Beds of Malawi are hypothesized to be at least partially correlative with the Galula Formation, and represent proximal deposits of this large, northwest flowing, trunk stream system. A moderately diverse terrestrial vertebrate fauna, including multiple species of dinosaurs, crocodyliforms, turtles, fishes and mammals have been recovered, along with a sparse aquatic molluscan fauna. Lithofacies and clay mineralogy indicate that Cretaceous paleoclimate ameliorated during deposition of the Galula Formation, transitioning from tropical semi-arid to tropical humid conditions.



1.10 storage tanks

1.11 Pipeline

1-6

More details of the geotechnical site conditions of the construction sites can be conducted during the working construction. Site investigations have to be carried out on behalf of the Employer. The information is not warranted by the Employer, and the Contractor shall make use of and interpret the same on his own responsibility.

1.12 Climate

Characterized by a Cwa Köppen climate classification, the Tunduma district in Tanzania is enveloped by a humid subtropical climate with predominantly dry winters. Due to its geographical placement, this area experiences significant climatological diversities throughout the year.

A year-round investigation of climate parameters reveals notable variances in climatic patterns. Average high temperatures range from 22.2°C (72°F) to 28.9°C (84°F), largely maintaining a consistent composure in the high 20s°C. The lower end of the temperature spectrum fluctuates between a mild 10.7°C (51.3°F) and a more temperate 17.3°C (63.1°F).

Humidity is generally substantial for most months, with values oscillating from a high of 88% to a low of 54%. A considerable decrease appears between the first four months and the following four, with increased aridity noticed from May to August. Precipitation follows a similar pattern. Copious amounts of rainfall are common during the initial months, with January being the wettest with 257mm (10.12") of rainfall. Contrastingly, the period from May to September is markedly dry.

When focused on wind speed, moderately gentle winds blow through the year with wind speeds peaking at 14.2km/h (8.8mph) in October. Conversely, the lesser wind flows are observed in February at a pace of 5.8km/h (3.6mph).

1.13 Power Supply and Telecommunications

Currently there is no power supply system close to the envisaged construction sites, which need to be upgraded where necessary. The Contractor has to provide stand-by power for all construction equipment and the temporary site facilities.

During the construction of WTP, storage tanks and other facilities the power supply lines to the sites need to be constructed. In order to finance these construction activities Provisional sums have been provided, which shall cover the costs. It is proposed that the Contractor shall contract the Tanzania Electric Supply Company (TANESCO) as nominated sub-contractor. Payments for these works shall be made under the Provisional Sums.

So TANESCO will be responsible for design and construction of 11kV overhead electrical transmission lines to the sites. These lines shall terminate in a substation at each site. Capacity of these lines shall be designed and constructed to serve the WTP with the capacity of 20,000m³/d and Intake 21,000m³/d

Tanzania Telecommunications Company Limited (TTCL) is the oldest and largest fixed line telecommunications company in Tanzania. The company comes forth from the former Tanzania Posts and Telecommunications Corporation in 1993. TTCL was wholly owned by the Government of Tanzania until the partial privatization of the company in 2001. In addition, there are several mobile telephone providers offering a competitive and good service including international roaming. Access to the internet is available and offered by several providers,

however the systems lack reliability and speed of data transfer. It is anticipated optical fibers from TTCL will be used for internet connection of SCADA and other systems.

2. Scope of Services and Supplies

2.1 Work Construction and Contract Limits

The design and constructions of the WTP with the effective capacity of 20,000 m³/d after deduction of less the 5% of water losses within the WTP.

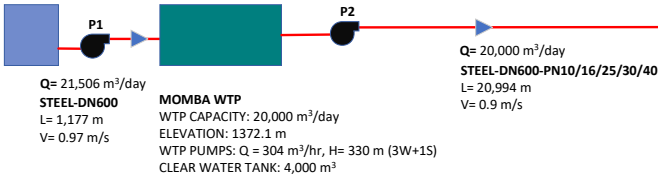
The Contractor will be responsible for the elaboration of the Detailed Design and the construction of the 20,000m³/d. Key features shall be displayed in Contractor's concept design, which will make sure that extension requirements of 97,000m³/d will be attended.

Figure 2: Schematic diagram of project waterworks

PROJECT NAME: CONSTRUCTION, TESTING AND COMMISSIONING OF WATER SUPPLY SCHEMES TO TUNDUMA
FROM MOMBA(SAISI) RIVER

TUNDUMA WATER SUPPLY PROJECT DESIGN PROJECTION YEAR 2035		
S/N	PIPE SELECTION	LENGTH (m)
1	ABSTRACTION PIPE	
	STEEL-DN600-PN10	100
2	RAW WATER PIPE	
	STEEL-DN600-PN10	1,177
3	RUWASHO WTP TO IKANA GBR	
	STEEL-DN600-PN10	305
	STEEL-DN600-PN16	119
	STEEL-DN600-PN25	4,705
	STEEL-DN600-PN30	10,798
	STEEL-DN600-PN40	5,067
	SUB-TOTAL-02	20,994
4	FROM IKANA GBR TO NKANGAMO BPS	
	STEEL-DN600-PN10	2,877
	STEEL-DN600-PN16	13,247
	STEEL-DN600-PN25	22,380
	SUB-TOTAL-03	38,504
5	NKANGAMO BPS TO UHURU PARK TUNDUMA	
	STEEL-DN600-PN10	10,310
	STEEL-DN600-PN16	7,177
	STEEL-DN600-PN25	4,185
	SUB-TOTAL-04	21,672
6	FROM J-0001N TO EXISTING TUNDUMA_ESR	
	STEEL-DN100-PN10	38
7	FROM J-0002N TO EXISTING TUNDUMA_GSR 01	
	STEEL-DN150-PN10	40
8	FROM TUNDUMA GSR TO EXISTING TUNDUMA_IPITO GSR 02	
	STEEL-DN150-PN10	4,285
9	FROM TUNDUMA UHURU PARK TO JUNCTION J-0003N	
	STEEL-DN150-PN10	1,555
10	FROM J-0003N TO EXISTING TUNDUMA_MAKAMBINI GSR 03	
	STEEL-DN100-PN10	494
11	FROM J-0004N TO EXISTING TUNDUMA_CHAPWA GSR 04	
	STEEL-DN100-PN25	2,664
	SUB-TOTAL FROM ALL JOINTS	9,076
	GRAND-TOTAL	91,523

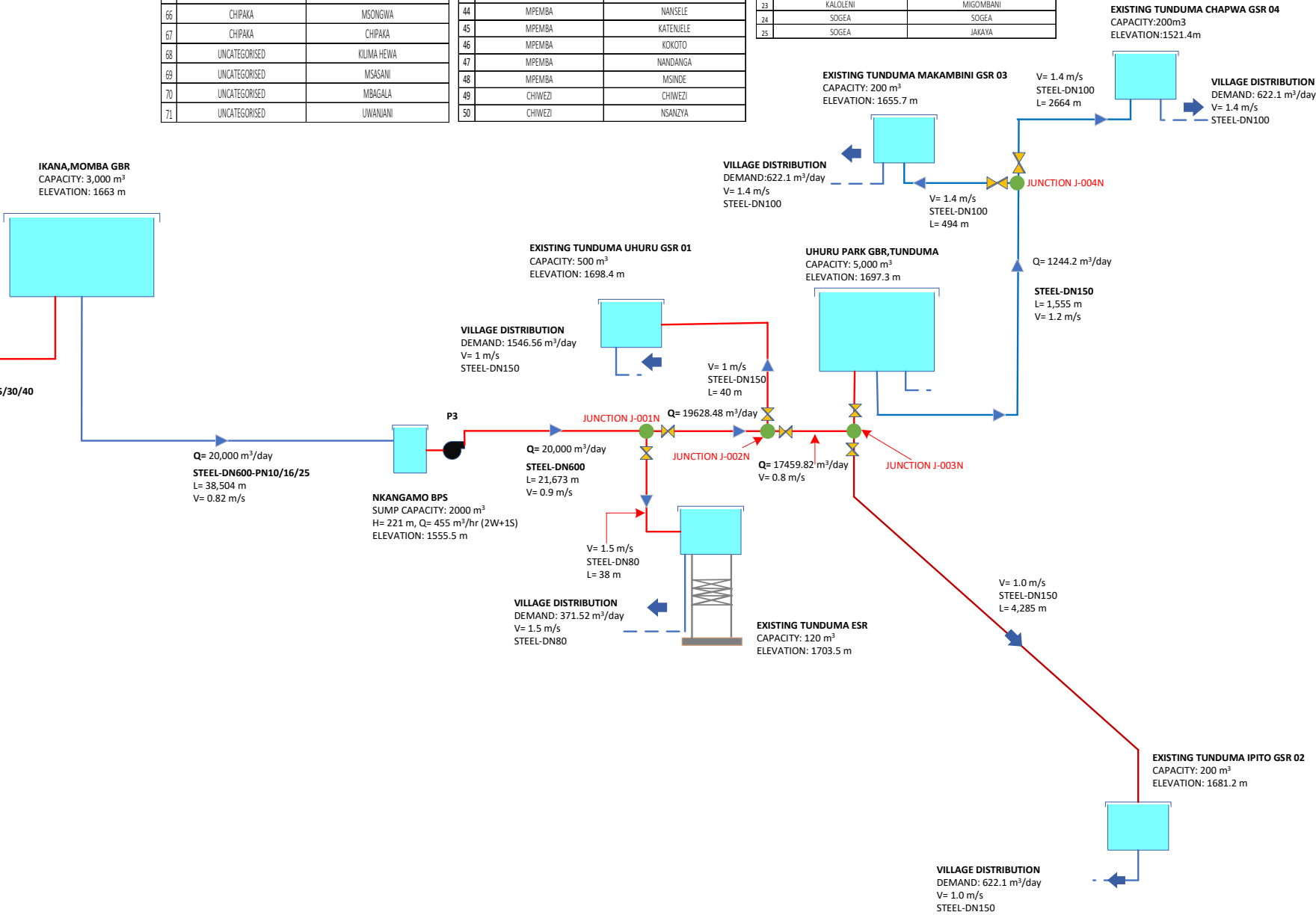
MOMBA INTAKE
INTAKE CAPACITY: 21,506 m³/day
ABSTRACTION & RAW WATER PIPE SIZE: DN600
ELEVATION: 1363 m
INTAKE LOW LIFT PUMP: Q= 978 m³/hr, H= 10 m (1W+1S)
INTAKE HIGH LIFT PUMP: Q= 978 m³/hr, H= 25 m (1W+1S)



SUPPLIED TUNDUMA VILLAGES		
SUPPLIED BY TUNDUMA GBR AND EXISTING TANKS		
S/N	NAME OF WARD	NAME OF VILLAGE
51	CHIWEZI	NAMOLE
52	CHIWEZI	NIUMBA
53	CHIWEZI	MSAMBA II
54	MUUNGANO	MUUNGANO
55	MUUNGANO	SAYUNI
56	MUUNGANO	NAZARETI
57	MUUNGANO	MKOMBOZI
58	MUUNGANO	KISANGANI
59	MWAKAKATI	MWAKAKATI
60	MWAKAKATI	MELELANI
61	MWAKAKATI	SONGAMBELE
62	MWAKAKATI	GOROFANI
63	CHIPAKA	ZYIGWILA
64	CHIPAKA	MWIL NYERERE
65	CHIPAKA	KATETE
66	CHIPAKA	MSONGWA
67	CHIPAKA	CHIPAKA
68	UNCATEGORISED	KILIMA HEWA
69	UNCATEGORISED	MSASANI
70	UNCATEGORISED	MBAGALA
71	UNCATEGORISED	UWANJANI

SUPPLIED TUNDUMA VILLAGES		
SUPPLIED BY TUNDUMA GBR AND EXISTING TANKS		
S/N	NAME OF WARD	NAME OF VILLAGE
26	SOGEA	HALFLODONI
27	SOGEA	IKULU
28	MAKAMBINI	MAKAMBINI
29	MAKAMBINI	SOWETO
30	MAKAMBINI	KANANI
31	MAKAMBINI	TOKYO
32	TUNDUMA	TURUYU
33	TUNDUMA	SIKANYIKA
34	TUNDUMA	IHANDA
35	TUNDUMA	TUNDUMA KATI
36	TUNDUMA	UNYAMWANGA
37	KATETE	KATETE
38	KATETE	MLENDA
39	KATETE	NAMANZANYIZA
40	KATETE	NAKAUMBA
41	KATETE	FURA
42	MPEMBA	MPEMBA
43	MPEMBA	MPEMBA
44	MPEMBA	NANSELE
45	MPEMBA	KATENUELE
46	MPEMBA	KOKOTO
47	MPEMBA	NANDANGA
48	MPEMBA	MSINDE
49	CHIWEZI	CHIWEZI
50	CHIWEZI	NSANZIA

SUPPLIED TUNDUMA VILLAGES		
SUPPLIED BY TUNDUMA GBR AND EXISTING TANKS		
S/N	NAME OF WARD	NAME OF VILLAGE
1	MAJENGO	MAJENGO A
2	MAJENGO	MAJENGO B
3	MAJENGO	MAJENGO C
4	MAJENGO	MAJENGO MAPYA
5	MPANDE	MPANDE KATI
6	MPANDE	MACHINIIONI
7	MPANDE	IPITO
8	MPANDE	MIIM WEMA
9	MPANDE	TAZARA
10	CHAPWA	CHAPWA
11	CHAPWA	ISANZO
12	CHAPWA	IGAWA
13	CHAPWA	MSAPANIA
14	CHAPWA	KESELA
15	MAPOROMOKO	SHULENI
16	MAPOROMOKO	MAGELOZI
17	MAPOROMOKO	MUMANI
18	MAPOROMOKO	MAPOROMOKO
19	KALOLENI	NELO
20	KALOLENI	KALOLENI
21	KALOLENI	DANIDA
22	KALOLENI	CUSTOM
23	KALOLENI	MIGOMBANI
24	SOGEA	SOGEA
25	SOGEA	JARAYA



2.2 Scope of Works

The following project components shall be designed and constructed:

- i. Intake, Grit tank and Raw water transmission pipe from Momba River
- ii. One treatment line of the WTP, including Drinking Water Storage Tank ("DWST") and Pump Station ("PS") at WTP pumping to storage tanks,

The Contractor's scope of works shall include all elements/aspects to ensure the proper implementation of the project, which includes:

- o Detailed topographical survey and soil investigation;
- o Preliminary and detailed design as described in Section 4;
- o Construction of the drinking Intake, raw water transmission pipe, water treatment plant, pump stations, transmission mains and reservoirs, together with all ancillary works;
- o Training of the Employer's personnel;
- o Commissioning and testing of the facilities;
- o Defects Notification Period of one year, following Taking Over;
- o Test after completion for six (6) months, following Taking Over.

Concerning the construction of the Power connections to the sites, the Contractor shall note:

The Contract provides provisional sums for the Power connections to the sites. It is proposed that the Contractor shall contract the Tanzania Electric Supply Company (TANESCO) as nominated sub-contractor and make them responsible for design and construction of 11kV overhead electrical transmission lines to the project sites.

These lines shall terminate in a substation at each site. Capacity of these lines shall be designed and constructed to serve the WTP with the capacity of 97,000m³/d. The Contractor is responsible for the coordination with TANESCO and proper connection of the transformer stations to the project sites. The following sites will have to be served:

- o Intake, WTP including PS, Storage Tanks and Booster Pumping Station

The scope of works is detailed hereunder as follows:

- o To carry out all necessary, complementary topographic surveys, raw water quality, soils investigations, laboratory analyses or related investigations, which the Contractor deems necessary, to supplement the data provided by the Employer. The Contractor shall satisfy himself concerning the complementary site investigations, which are his sole responsibility. No claim because of insufficient information will be accepted.
- o To prepare preliminary and detailed designs as specified in Section 4;
- o To supply all materials required for construction of the works. This includes manufacture, collection, transportation and delivery to site. The Contractor will be responsible for ensuring that all procedures are adequately covered and that the materials fully conform to the contract requirements. These responsibilities will include all necessary charges or duties related to

insurance, freight, taxes (including customs and excise duties, surcharges, etc.), and all testing and inspections for quality control.

- To provide all necessary staff (including designers, administrators, site supervision personnel) and workmen (including all necessary specialists, operators, tradesmen, artisans, etc., in addition to semi-skilled and unskilled workers) necessary for execution of the works, up to completion. The Contractor shall provide all suitable facilities and accommodation for the staff and workmen, and he shall make provision for all costs related to such provisions and for medical, re-location, taxes or other expenses.
- To provide all equipment, machinery, tools, etc., and related spares, maintenance and consumables necessary for implementation of the works.
- To provide all site offices, stores, workshops and facilities necessary for use by the Employer, Employer's Personnel and support staff, and for the Contractor himself, his support staff and his sub-contractors.
- To undertake all operations necessary to construct the Works. These operations shall include installation of main treatment, transport and storage units, blowers and compressors, chemical dosing equipment, main pump/motor sets, all cranes and hoists, drainage systems, surge protection systems, pressure relief or control valves, cooling water systems, as well as electrical plant and instrumentation and control equipment, etc., and construction of all concrete or other works, all pipework, together with all testing, commissioning and operation of the works as required.
- To commission equipment, start-up and trial-operate the facilities for a period of three months.
- The Contractor must train Employer's personnel in classroom (during construction) and on the plant (during commissioning period), in order to put them in the position to run the facilities later-on.
- To liaise with other Contractors on the site (if any) and to ensure harmonious co-operation with them, so that conflicts are avoided, and areas of common interest, constructional interface or potential overlaps are addressed without cost to the Employer or delays in completion.
- To prepare documentary records of the Works in the form of "as-built" drawings, schedules, etc. and to produce Manuals for Operation and Maintenance of the Works.
- A Quality Assurance Procedure shall be implemented in accordance with Clause 4.9 of the Conditions of Contract.

All the above works shall be performed in a professional way and with good engineering or constructional practice. Upon completion of the works the scheme shall be fully operational with minimum disruption or inconvenience to interested parties, including landowners, and there shall be no outstanding matters requiring attention.

3. Documents to be submitted by the Contractor

3.1 Work Program

The Work program shall show the logical sequence and correlation between the categories of activities (works) in form of a bar chart. The Work Program shall be produced using common scheduling software, compatible with MS-Project and in such a detail stipulated in CoC clause 8.3.

The works program shall show as a minimum:

- Pump installation at Intake and WTP inclusive of testing and commissioning
- Transmission pipes both raw and treated one and other water appurtenances
- Booster Pumping station and Storage tanks

the following work breakdown structure:

- mobilization;
- civil works construction
 - preparation of structural design, design approval,
 - equipment and material proposals and approval,
 - temporary works,
 - earth works,
 - concreting works, concrete curing period, period for water tightness testing.
 - architectural works,
 - steel works,
 - road works,
 - landscaping works,
 - pipe laying works, pipe road crossing, pipe pressure testing, pipe cleaning and disinfection;
- mechanical works (valves, pumps, and other major mechanical equipment)
 - submission of proposals for equipment and material and approval / ordering date,
 - manufacturing, delivery to Site,
 - installation, testing;
- electrical works (MV switchgear, transformers, LV switchgear and MCCs, diesel genset, and other major electrical equipment)
 - preparation of electrical execution design, design approval,
 - submission of proposals for equipment and material and approval / ordering date,
 - manufacturing, delivery to Site,

- installation, cabling, testing;
- ICA works (instruments, control and automation hardware, and other major ICA equipment)
 - preparation of ICA execution design, design approval,
 - submission of proposals for equipment and material and approval / ordering date,
 - manufacturing, delivery to Site,
 - installation, cabling, programming, testing;
- preparation and approval of as-built documentation, O&M Manuals, and Functional Design Specifications;
- preparation / submission / approval of commissioning / testing program of the entire system;
- defects notification period.

The time for completion of the works and sections thereof shall correspond to the periods prescribed in the Contract.

The work program shall show important milestones for design, construction, testing, commissioning, and contract commencement date and scheduled taking-over date.

The Work Program shall consider, when calculating periods for execution,

- the prevailing climatic conditions,
- the periods for approval of submittals by the Engineer / Employer,
- the periods required to obtain various permits and approvals from other stakeholders than the Engineer or Employer,
- getting exemption certificate for materials, equipment's etc.,
- ESHS requirements,
- Periods of work of sub-contractors,
- Periods of work of other contractors to be engaged by the Employer.

The work program shall be able to calculate the critical path and float time and to generate the rate of progress and the cumulative quantity or percentage of work expected to be achieved on each activity by the end of each month ("S-Curve").

The Contractor shall allocate resources in term of equipment's, plants and workforce and related cost to each activity in the Work Program and calculate the cash flow.

The Contractor shall be solely responsible for internal coordination and management of the execution design and construction process. Consent to the Contractor's program by the Engineer shall not relieve the Contractor of his duties and responsibilities under the Contract.

The program shall be updated when requested by the Engineer but as a minimum on a monthly basis (as an attachment to the monthly report, in paper and electronic format) to show actual progress and any revisions necessary to achieve completion.

The outline program should be as follows:

Table 1: Outline Program

Tasks (Duration)	Description
Total period for execution of works	36 months
Execution of Design work (6 months)	Detailed site investigation works, preparation of draft Execution Design for 20,000m ³ /d WTP, intake with the capacity of 21,000m ³ /transmission main pipelines, storage tanks, booster pumping station to Engineer, review by the Engineer, and preparation of final Execution Design. Staged submission and approval are recommended where necessary otherwise not.
Construction period, Commissioning and start-up (max. 18months)	Starts directly after the design period. This period includes the construction and completion of all physical works. Starts section -wise for each section of works completed (and thus may overlap with the construction period). Commissioning, start-up, testing and trial operation of all facilities shall commence only after all works have been completed.
Defects Notification Period (12 months)	Starts after construction period and includes commissioning and start-up. Remedy of minor defects and deficiencies, in accordance with the snag lists prepared by the Engineer.

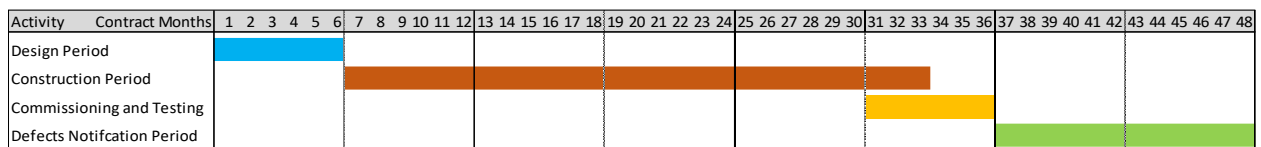


Figure 3: Tentative Work Programme

3.2 Insurance to be provided by the Contractor

The Contractor shall provide the following insurances:

3.2.1 The Works

The Contractor shall insure and keep insured in the joint names of the Contractor and the Employer from the Commencement Date until the date of the issue of the Taking-Over Certificate for the Works:

- the Works and Contractor's Documents, together with Materials and Plant for incorporation in the Works, for their full replacement value. The insurance

cover shall extend to include loss and damage of any part of the Works as a consequence of failure of elements defectively designed or constructed with defective material or

- b) an additional amount of fifteen percent (15%) of such replacement value (or such other amount as may be specified in the Contract Data) to cover any additional costs incidental to the rectification of loss or damage, including professional fees and the cost of demolition and removal of debris.

The insurance cover shall cover the Employer and the Contractor against all loss or damage from whatever cause arising until the issue of the Taking-Over Certificate for the Works. Thereafter, the insurance shall continue until the date of the issue of the Performance Certificate in respect of any incomplete work for loss or damage arising from any cause occurring before the date of the issue of the Taking-Over Certificate for the Works, and for any loss or damage occasioned by the Contractor in the course of any operation carried out by the Contractor for the purpose of complying with the Contractor's obligations under CoC Clause 11 (Defects after Taking Over) and CoC Clause 12 (Tests after Completion).

However, the insurance cover provided by the Contractor for the Works may exclude any of the following:

- I. the cost of making good any part of the Works which is defective (including defective material and workmanship) or otherwise does not comply with the Contract, provided that it does not exclude the cost of making good any loss or damage to any other part of the Works attributable to such defect or non-compliance;
- II. indirect or consequential loss or damage including any reductions in the Contract Price for delay;
- III. wear and tear, shortages and pilferages; and
- IV. unless otherwise stated in the Contract Data, the risks arising from Exceptional Events.

3.2.2 Goods

the Contractor shall insure, in the joint names of the Contractor and the Employer, the Goods and other things brought to Site by the Contractor to the extent specified and/or amount stated in the Contract Data (if not specified or stated, for their full replacement value including delivery to Site). The Contractor shall maintain this insurance from the time the Goods are delivered to the Site until they are no longer required for the Works.

3.2.3 Liability for breach of professional duty

To the extent that the Contractor is responsible for the design of part of the Permanent Works under CoC Sub-Clause 4.1 [Contractor's General Obligations], and/or any other design under the Contract, and consistent with the indemnities specified in CoC Clause 17 [Care of the Works and Indemnities]:

- (a) the Contractor shall effect and maintain professional indemnity (a) insurance against liability arising out of any act, error or omission by the Contractor in carrying out the Contractor's design obligations in an amount not less than that stated in the Contract Data (if not stated, the amount agreed with the Employer); and

- (b) if stated in the Contract Data, such professional indemnity insurance shall also indemnify the Contractor against liability arising out of any act, error or omission by the Contractor in carrying out the Contractor's design obligations under the Contract that results in the Works (or Section or Part or major item of Plant, if any), when completed, not being fit for the purpose(s) for which they are intended under CoC Sub-Clause 4.1 [Contractor's General Obligations].

The Contractor shall maintain this insurance for the period specified in the Contract Data.

3.2.4 Injury to persons and damage to property

The Contractor shall insure, in the joint names of the Contractor and the Employer, against liabilities for death or injury to any person, or loss of or damage to any property (other than the Works) arising out of the performance of the Contract and occurring before the issue of the Performance Certificate, other than loss or damage caused by an Exceptional Event.

The insurance policy shall include a cross-liability clause such that the insurance shall apply to the Contractor and the Employer as separate insureds.

Such insurance shall be affected before the Contractor begins any work on the Site and shall remain in force until the issue of the Performance Certificate and shall be for not less than the amount stated in the Contract Data (if not stated, the amount agreed with the Employer).

3.2.5 Injury to employees

The Contractor shall affect and maintain insurance against liability for claims, damages, losses and expenses (including legal fees and expenses) arising out of the execution of the Works in respect of injury, sickness, disease or death of any person employed by the Contractor or any of the Contractor's other personnel.

The Employer and the Engineer shall also be indemnified under the policy of insurance, except that this insurance may exclude losses and claims to the extent that they arise from any act or neglect of the Employer or of the Employer's Personnel.

The insurance shall be maintained in full force and effect during the whole time that the Contractor's Personnel are assisting in the execution of the Works. For any person employed by a Subcontractor, the insurance may be affected by the Subcontractor, but the Contractor shall be responsible for the Subcontractor's compliance with this Sub-Clause.

3.2.6 Other insurances required by Laws and by local practice

The Contractor shall provide all other insurances required by the Laws of the countries where (any part of) the Works are being carried out, at the Contractor's own cost. Other insurances required by local practice (if any) shall be detailed in the Contract Data and the Contractor shall provide such insurances in compliance with the details given, at the Contractor's own cost.

The Contractor shall provide insurances, as shown below, by an insurance company registered in the Republic of Tanzania.

3.3 Quality Management and Compliance Verification Systems

The Contractor shall prepare and implement the Quality management and verification system to demonstrate the design, materials, plants and workmanship comply with all respects with the contract especially CoC clause 4.9. These documents shall be submitted for Engineer approval within 28 days of the commencement date.

3.4 Topographical Survey

Accurate and detailed topographic information is required for the design of pipelines, locating and design of main and secondary works as water sources, water treatment plants, pumping stations, reservoirs, chambers, inspection pits etc. and associated appurtenances. The contractor will review the existing topographic information and prepare a detailed scope of work necessary to meet the requirements to design all of the facilities to international standards. The Topographic survey methods to be considered in the scope of work are as follows.

The tasks involved in this activity are as follows:

- The contractor shall provide a complete detailed digital terrain model (DTM) and topographical survey of the whole of the project site
- The contractor shall liaise with the Ministry responsible for lands for the existing National Grid/Datum reference beacons and benchmarks in order to establish the permanent control points. The coordinates of all intersection points shall be in Universal Transverse Mercator system and shall be tied to the National Survey Grid, and levels related to the National Benchmarks.
- The contractor shall establish all necessary permanent leveling polygon bench marks at appropriate intervals. The locations shall be on permanent structures that cannot be disturbed by construction, to the extent possible.
- The contractor shall submit detailed topographic maps covering the area of the project including the location of the benchmarks (existing national trigonometric points, and national benchmarks, Contractor's primary and secondary trigonometric points and benchmarks with co-ordinates and elevations). On the map, natural and manmade features shall be shown and labeled in English. Hard copies shall be in scale of 1:500 with 1-meter contour interval.
- All files for the topographic survey shall be submitted to the Engineer in charge in electronic format; copies of the topographic survey data saved in MS Excel and topographic drawings in DXF and DWG format (Auto Cad version 2021); they shall be submitted in CD ROMs for future reference and actions.
- A list of benchmarks showing the coordinates shall be provided to the Engineer.
- The contractor's final topographical survey report shall include detailed topographical map of the sites, Digital Terrain Model (DTM) of the area and benchmarks details. The additional topographical investigations shall be at

no additional cost for the Employer and will not entitle the Contractor to any contractual time extension.

3.5 Geotechnical Survey

Geotechnical drilling is required at the main structures and may be required for other selected sites. Also, geotechnical soil information and results of analysis will be assessed by the contractor with a view to using the geotechnical information to establish design parameters for the project. The contractor shall execute exploratory drilling, analysis, soil and rock mechanics testing, to be supervised by the Engineer in charge, whenever it shall be necessary to meet the requirements of design of the project components to applicable standards and codes.

The work as a minimum will include borings, preparation of the borehole logs, collection of disturbed and undisturbed samples, Standard Penetration Tests, installation of temporary piezometric wells, backfill of boreholes, laboratory tests and preparation of a data analysis report including a summary of the field activities, all field data, including borehole logs, laboratory test results, groundwater level observations. Specification and the requirements for the subsurface investigations are provided at the end of this section. The contractor will give the detailed survey plan in his early report concerning the proposed investigation campaign.

3.5.1 General Scope of Works

The Contractor shall first excavate a test pit at the proposed drilling site to a depth of 1.5 meters below ground to prove the absence of services before commencing any borehole.

The depth of the test drilling shall be min. each 15 m. The depth of the test drillings can be reduced if hard rock is reached/detected. The drilling in rock however shall reach at last 2 m depth.

All drilling is to be carried out in accordance with the EN ISO 22475-1:2006, EN ISO 22475-2:2006 and EN ISO 22475-3:2007.

Continuous soil samples shall be recovered on all the test drillings. The samples shall be placed in to core boxes. In soil layers, Standard Penetration Test (SPT) samples shall be recovered each time a change in soil characteristics is found.

Boring logs shall indicate:

- Description of soil layer by a geotechnical expert surveyor.
- Percentage of rock recovery (Rock Quality Determination, RQD).
- SPT when possible.
- Water table level (if any).

Logging of the boreholes, and soils and rock description and determination of the RQD shall be performed by an experienced / geologist of the Contractor.

Laboratory tests shall be performed in soil samples to determinate the following soil properties:

(i) In soil:

Identification tests:

- Granulometric analysis
- Index of plasticity
- Density
- Moisture content
- Soil classification

Mechanical tests:

- Compression strength
- Undrained direct shear strength
- Oedometer deformability

Chemical analysis:

- Contents on SO₄, NO₃

(ii) In rock:

Identification tests:

- RQD
- Density
- Geological classification

Mechanical tests:

- Compression strength

(iii) Water tests:

- Level of water table
- Chemical analysis

All findings and results shall be presented in a comprehensive report, submitted as a hard copy in English language.

3.5.2 Scope of Geotechnical Works for Foundations of Pipe Bridges and Pipe routes

At least one deep test drilling shall be bored for each Site of pipe bridge. The exact location of the test drilling will be determined by the Engineer. Contractor should identify at least twenty trial pits within the corridor of pipe routes from Intake to WTP and from WTP to storage tanks in order to protect the pipes against cathodic reaction.

3.6 Environmental, Social, Health and Safety Management Site and Action Plan

The Employer has prepared an ESMP on detailed level, which will be updated after having final ESIA. Prior to work commencement Contractor must prepare and submit his update ESMP and ESHS-AP to be in line with NEMC certificate. These documents have to be followed-up during the whole construction period and shall be updated regularly, on a quarterly basis.

4. Design

4.1 General

Within the period stated in the Contractor's Program the following preliminary and detailed drawings and information shall be submitted for approval before start of Construction.

All design documents, prepared by the Contractor for approval by the Engineer and by the authorities (as the case may be), shall be adequate for the purpose, meeting required standards and fully in accordance with national laws and the General Technical Specifications.

Preparation of the design documents and drawings to be submitted by the Contractor shall be conducted by the Contractor's Design Manager. The Design Engineer shall, within 21 days after the Commencement Date, submit the preliminary content of the execution design, including the draft list of drawings for the Engineer's approval.

4.2 Scope of Works - General

The Execution Design Documents shall cover all measures under 20,000m³/d WTP. The future extension of 97,000m³/d shall be covered only insofar that the extension requirements will be attended.

4.3 Scope of Works - Preliminary Design and Information

Within the period stated in the Contractor's Program the following preliminary drawings and information shall be submitted for interim approval, for all sites:

- (a) General layout of the sites, sections and views, clearly showing new structures, plant and M&E installations, etc.
- (b) Process flow diagrams and mass balance diagrams.
- (c) Process control, instrumentation and interconnection diagrams.
- (d) Process description and design calculations.
- (e) Hydraulic profiles of WTP
- (f) Layout for stormwater drainage and sludge handling facilities
- (g) Layout of sludge handling facilities covering 20,000m³/d
- (h) Functional diagrams.
- (i) General pipework layout diagrams/routes and connection details, and the provision of necessary connections for future treatment.
- (j) General electrical wiring and schematic diagrams.
- (k) Cable routing diagrams.
- (l) General layout, showing preliminary electrical cable routes.
- (m) Plant performance data.
- (n) Layout plans and profiles for all transmission mains from the intake to WTP and from WTP to Storage Tanks
- (o) Access road to Storage Tanks

4.4 Scope of Works - Detailed Design and Information

Within the period stated in the Contractor's Program the following detailed drawings and information shall be submitted for approval before start of Construction:

- (a) Update of preliminary design, as commented by the Engineer, in particular P&ID drawings for all scope of works;
- (b) Layout and longitudinal sections of all transmission pipelines;
- (c) Layouts of all Sites, showing arrangement of tanks, pump stations, water pipes, buildings, valve and flow meter chambers, roads, stormwater drainage;
- (d) Layouts of all Sites, showing roads, parking areas, walkways, routes of internal drinking and process water pipelines, valve and flow meter chambers, road drainage channels / pipes, sewers, sewer manholes, sludge handling facilities, septic tank, cables, cable ducts, cable manholes, outdoor lighting;
- (e) Hydraulic design of transmission pipelines: transient analysis of all pipeline sections and sizing of surge and water hammer protection measures;
- (f) Process design: sizing of chlorination units;
- (g) Mechanical design: pump sizing for each pump;
- (h) Mechanical design: machine / pump / valve / pipe appurtenance schedules listing each component with identification number, size, type, connections, rating, figure number and duty;
- (i) Layouts, sections, and details for mechanical / pipe installation for each tank, valve and flow meter chamber, building;
- (j) Geotechnical calculations for shallow foundations, retaining structures, pipe trenches, pipe shoring systems, slopes and embankments, pile and deep foundations, rock anchoring, etc.;
- (k) Typical profiles for installation of pipes in pipe-trench, including bedding details, and for pipe installed on concrete foundations above ground, layout and sections of pipe road crossings, and creek under-crossings;
- (l) Structural calculations and reinforcement drawings / bar bending schedules for all tanks, buildings, chambers, pipe bridges (incl. piles and foundations), pipe road under-crossings, pipe creek under-crossings, pipe foundations / support structure for above ground pipes, thrust blocks, roof trusses, steel work (stairs, platforms, covers, etc.), formwork drawings and details (showing joints and water stops, openings, cast-in parts and pipes, etc.), rock anchor details;
- (m) Steel work execution drawings: layout, sections, details;
- (n) Electrical design: calculation of power consumption and supply sizing, calculation and sizing of MV and LV switch gear and MCCs, transformers, lighting, reactive power correction, UPS, diesel gensets, facilities for discrimination / surge / overvoltage / lightning protection, earthing, elaboration of single line diagrams for each Site, cable sizing calculations

and elaboration of cable schedules, wiring diagrams and panel drawings for each MCC;

- (o) ICA design: design and elaboration of PLC/RTU logic and wiring diagrams for each Site, panel drawings for each PLC;
- (p) HVAC design: calculation of cooling loads, air volume flows, sizing of air conditioners, ventilators for each room / building;
- (q) Electrical / ICA design: schedules listing each power consumer / instrument with identification number, size, type, connections, rating, figure number and duty;
- (r) Architectural layouts, sections, and details (for roof, ceiling, flooring, doors and windows, finishing, and other details), for each building;
- (s) Electrical design: Layouts and sections for electrical / ICA installations (cabinets, distribution boards, luminaries, socket outlets, heaters, ventilation fans and ductwork, air conditioning units, fire detectors and manual call points, intruder alarm switches and detectors, alarm sounders, process and general instruments, etc., and associated switches, thermostats, control panels, etc.) and related cabling routing and routing of cable trays, conduits, and sheaths, for each tank, chamber, building;
- (t) Plumbing layouts, sections, and details (for internal water supply and drainage), for each building, showing location of sanitary equipment, sinks, drains, pipeline routing, etc.;
- (u) Road longitudinal and cross-sections, road drainage profiles, details of septic tank;
- (v) Landscaping and fencing details.

4.5 Scope of Works - Particular Drawings

4.5.1 Codes and Standards

The proposed works to be carried out under phase 1 shall be designed, manufactured, installed and constructed to appropriate International Standards and Codes as well as local national Codes and Regulations as stipulated in the General Technical Specification. In all cases, the more rigorous conditions shall apply. The Contractor shall indicate relevant codes with his submission of proposed design of civil works, plant and equipment.

4.5.2 Specific Design Requirements

The Contractor's design submissions shall include, but not be limited to:

- Detailed calculation sheets, including assumptions, design criteria, relevant codes, software name and version (with licensed copy of proprietary programs where appropriate), applied loads, etc;
- Scaled drawings including site plans, general arrangement and plant layout drawings, process and instrumentation drawings, control diagrams, electrical single line and other drawings, structural drawings, pipeline plan and profiles, special details, alteration drawings, etc;

- Hydraulic profiles;
- Plant and material schedules;
- Manufacturer's shop drawings;
- Manufacturer's specifications, data sheets, dimensional drawings, etc;
- Manufacturer's performance curves;
- Manufacturer's installation / operation / troubleshooting data.

4.6 Hydraulic Profiles of WTP

Process and hydraulic Design and drawings for structures of Water Treatment Plant shall be done to establish the sizing, water levels and/or operating pressures throughout the plant. The drawings shall indicate water levels / pressures under normal operation as well as with individual or combinations of process units out of service. Designs shall ensure adequate free-board under worst-case conditions, as well as non-submergence of weirs and proper operation of overflows. Designs shall indicate cut-in and cut-out water levels and, where necessary shall indicate the cycle times for pumps and other plant.

Designs shall also indicate, velocities, detention times, detention volumes, weir loading rates, overflow capacity, etc. For the purpose of process and hydraulic design and selection of plant and equipment, the Contractor's designs shall also comply with the target water quality requirements described for the individual units. All formulae, coefficients, source references and assumptions made in the calculations shall be detailed.

4.7 Pipeline Layouts and Hydraulic Profiles

The tentative drawings indicate recommended diameters of the proposed pipelines. The Contractor shall confirm the sizes with due consideration to velocities (maximum and minimum), head losses, flow requirements and cost, among others. Where the Contractor proposes a different diameter, he shall provide all necessary calculations to support his proposal. However, unjustified under-sizing of pipes shall not be allowed as it will increase OPEX. The Contractor shall identify such locations, early in the contract, and determine the elevation, gradient, direction, size, material, function, etc of the existing pipe. Horizontal and vertical alignment drawings shall be prepared for each proposed pipeline at suitable scales. The drawings shall also indicate, horizontally and vertically pipelines complete with diameters, chainages, elevations and gradients, location of various type of valves, offtakes as well as other structures.

Detailed drawings shall be provided for pipeline crossings, indicating the separation distance, protective measures if required, and other details.

Ideally, pipelines shall cross one underneath the other with a suitable separation distance. Where this is not possible, the Contractor shall design a suitable crossing structure complete with calculations and operating and maintenance details. In general, pressure pipelines are easier to re-align vertically than open channel drains

laid to gradient. However, in such cases, the designer shall check for changes of grade, including sign reversals of gradient. Structures, houses, electricity poles, trees, and any other item within road reserve limits that is relevant for the pipeline routing (i.e.: has to be demolished, removed, re-located, over-under-crossed, etc.). Pipeline axis, chainage, diameter, pressure class, material, change in slope / direction that needs anchor blocks, buried pipe appurtenances (bends, tees, etc.), chambers for washouts / air valves, sectional valves, branches, pipe bridges, culverts, protections, etc. Existing surface level, new surface level, pipe invert and crown levels, pipe grade, pipe section length, etc.

4.8 Access Roads Layouts and Profiles

Access Roads layouts and longitudinal and cross-sectional profiles shall show the following information, but not limited to:

- Road axis, radii, high points, low points, top and bottom edge of slope (envelope), position of pipe axis relative to road for selected cross-sectional profiles, existing surface level, road surface level,
- Road drainage channel's locations and details,
- Road culverts, locations and details,
- Structures, houses, electricity poles, trees, and any other item within corridor of the access road that is relevant for the road construction (i.e.: has to be demolished, removed, re-located, over-under-crossed, etc.).

4.9 Reinforced Concrete Structural Design

Structural designs shall include all calculations, reinforcement detailing and bar bending schedules. The relevant design standards, including local building codes and regulations, shall be clearly indicated. All loads and load combinations shall be identified and considered to determine worst case loading conditions. Foundation designs shall refer to, and correspond with, the geotechnical investigations.

4.10 M&E - Fabrication and Installation Drawings

Where drawings are required for fabrication or installation of items of plant or equipment provided by the Contractor or his sub-contractors, such drawings shall be provided by the Contractor and submitted to the Engineer for approval.

These drawings shall include, but not be limited to, shop fabrication detail drawings including details of welding, match-marks, erection diagrams, and other details such as connections, all as necessary for proper installation and erection of the equipment and performance of the work.

The drawings of a specific piece of equipment shall identify components with the manufacturer's part number of reference drawings number clearly indicated. If reference drawings are used, the approval date of such drawings shall be included. The drawings shall indicate dimensions and allowable tolerances.

4.11 Temporary Works Designs

If required by the Engineer, the Contractor shall submit for comment full particulars, including drawings, of any of the Temporary Works. If required, the

Contractor shall also submit calculations of the Temporary Works and these calculations shall be accompanied by detailed working drawings to show the Contractor's proposals. Engineer's comments on Contractor's proposals, calculations or drawings, shall not relieve the Contractor of any of his duties or responsibilities under the Contract.

4.12 Additional Drawings for Use by Contractor's Employees

Any additional drawings which the Contractor prepares for the use of his employees, shall be copied to the Engineer, for information.

4.13 Progress Reports / Progress Meetings / Photo Documentation

4.13.1 Progress Reports

During the first week of each month the Contractor shall submit a progress report to the Engineer in the Contract language.

The progress reports shall contain as a minimum the following information:

- tabular description of physical progress for the reporting month and brief on estimated progress for the next reporting month;
- updated work program showing completion of each activity, and critical path, including explanatory notices;
- updated S-curves for physical progress at different sections of the Works;
- brief description of compliance with approved work program, occurred delays and reasoning, proposed measure to make up lost time;
- summary on quality issues, ESHS issues,
- update on invoices, payment, budget remaining and up-dated cash flow;
- photo documentation as attachment (see below);
- Summary of the risk assessment/register as per CoC clause 8.4

4.13.2 Progress Meetings

The Engineer will call for regular progress meetings, in his office or at the Contractor's offices or at the Site. If necessary, the Engineer has the right to call for exceptional meetings at any time. The Contractor's Project Manager and Contractor's experts, as the case may be, shall attend all such meetings.

4.13.3 Photo Documentation

The Contractor shall prepare a photo-panel monthly and additionally as ordered by the Engineer, in order to record progress. All pictures shall be clear, in focus and properly exposed and printed.

Each picture on the panel shall be identified with the date and place of taking and such other information as may be relevant. The Contractor shall also ensure that no unauthorized pictures will be taken on site.

In addition, the Contractor shall take videos, covering the significant parts of the works. The videos shall later be cut, assembled and transformed into a comprehensive project documentary video.

4.14 Method Statements, Material Approval, Shop Drawings

The Contractor shall provide in writing a general description of the arrangements and methods, which the Contractor proposes to adapt for the execution of the Works. An overall Method Statement, describing generally all works, shall be submitted to the Engineer at the same time, as the Contractor's Program for the Works.

Additional detailed method statements shall be submitted in due course of construction, which shall show in detail the methods proposed by the Contractor for carrying out the principal activities of Construction. In particular, the Contractor shall indicate the resources (plant, personnel, materials) to be allocated, their timing and sequencing, emergency/contingency measures, and any other information required to clearly detail the proposed methods. The statement shall be supported by sketches, diagrams or other supportive detail to enable a clear understanding of the method and significance of each step of the operation or work. No work shall be carried out until a method statement has been submitted to and approved by the Engineer.

The Method Statement shall be submitted at least 28 days before commencement of the proposed operation or work.

Any product or substance used by the Contractor or its Subcontractors, which is toxic or hazardous shall be identified to the Engineer by the Contractor's submission of a Material Safety Data Sheet and in accordance with the ESHS proposed management and action plan.

4.15 Functional Design Specification

The Contractor shall elaborate a Functional Design Specification ("FDS") that shall describe for all process units:

- the control functions (verbal and in form of flow charts, with refer to P&ID), and for more complex control functions the mathematical description,
- the interlocks for each motor, drive, device, etc.,
- the input signals and how they are processed for each motor, drive, etc.,
- the output signals of every device, instrument, etc. and for what they are used and to which other device they are routed,
- the set values (min, typical, max) and how they can be changed,
- the status / position of all devices before start-up and after shut down.

The FDS shall serve as a basis for the Contractor and the Engineer to agree on principle and details of PLC programming and furthermore as a basis for testing the PLC functions during testing and commissioning. The Contractor shall execute programming of PLCs in accordance with the approved FDS. Any substantial change during programming shall be documented in the final FDS, to be submitted as part of the O&M Manual.

4.16 Operation & Maintenance Manual

The Contractor shall prepare, and keep up-to-date, a complete set of operation and maintenance manuals for the Works (the "O&M Manuals" in accordance with CoC clause 5.7)

The Contractor shall prepare and submit for the approval of the Engineer "user friendly", Operation and Maintenance Manuals for the Plant, in English, which shall

- a) be in sufficient detail for the Employer to:
 - i. operate, maintain and adjust the Works to ensure that the performance of the Works, Section and/or Plant (as the case may be) continues to comply with the performance criteria and the Schedule of Performance Guarantees; and
 - ii. operate, maintain, dismantle, re-assemble, adjust and repair the Plant; and
- b) include an inventory of spare parts required for the Employer's future operation and maintenance of the Plant.

The Manual shall consist of the Master Manual, which covers the integral operation of the whole plant as well as manufacturers' manuals for individual equipment.

The preparation of the manual shall be carried out by personnel who are trained and experienced in the operation and maintenance of the plant described and are skilled as technical writers to the extent required to communicate essential data and are competent to prepare the required drawings and documentation.

Each volume of the manual shall be identified both on the front and on the spine, with the typed or printed title "OPERATION AND MAINTENANCE INSTRUCTIONS", the title of the project and the identity of the general subject matter covered in the manual. The text and drawings shall be placed in commercial 4 or 5 ring binders with durable and cleanable plastic covers.

Each manual volume shall contain a neatly typewritten table of contents arranged in a systematic order.

It shall include the necessary descriptions, concerning O&M of civil structures, mechanical and electrical equipment, system control, water quality control, safety, record keeping, emergency measures and shut down.

All information in these manuals shall apply specifically to the plant and equipment being supplied and the documentation shall be free from irrelevant matters. Manuals shall relate to as-built conditions and shall include all necessary drawings and diagrams for a proper understanding of the Plant.

The Operation and Maintenance Manual shall be supplemented with a Maintenance Plan, which shall consist at least of the following:

- Spare Parts Replacement Schedule.
- Routine maintenance activities, which shall be performed at intervals. This plan shall show daily, weekly, monthly, and quarterly, half-yearly and yearly maintenance activities regarding each component of the Plant.
- Maintenance Database, with the purpose to enable archiving and QA/QC of all Routine and Emergency Maintenance activities for each component of the Plant.

- All Maintenance Plan documents should be prepared in way, which shall enable flexibility in changes based on Plant operation experience.
- All necessary, schedules, data-sheets, drawings, certificates, diagrams, etc.

The manual shall be updated and complemented continuously, during the Commissioning and Testing, as well as during the DNP. Prior to the end of the DNP the final O&M manual shall be elaborated and submitted to the Engineer for approval.

4.17 Operation during Commissioning Period

The Contractor shall operate the plant during the commissioning period of three months, before handing over to the employer. Simultaneously, the training on the job for Employer's staff shall take place.

4.17.1 Financing Plan for the O&M Period during Commissioning

During commissioning period (three months), the cost for initial operation and maintenance shall be borne by the Contractor (mainly power, consumables, O&M staff, and training costs), which shall be included in the Schedules of Prices.

The operation of the facilities shall be quoted on the basis of one eight hour per day, fully manned shift, 7 days per week. Adequate allowance should be made for night work, time-off, holidays, sick-leave and other absences.

4.17.2 Training Plan

The Contractor will be required to submit a training plan for the newly recruited staff. The respective chapter of the Conceptual Design Report shall provide guidance for the elaboration of the plan. This is to include classroom training, on-the-job training and off-site training on similar facilities, in order to provide theoretical and practical understanding of the required operation of the project components and get familiar with the equipment in use.

Preparation of a training plan shall commence early enough to accomplish in time the following expected components of operation:

- Training of Staff on all aspects of the operation and maintenance of the WTP, pipelines and reservoirs.
- Advice on the setting-up of external supply Contractors for supply of consumable materials and disposal of waste.
- Establishment of stores inventories.
- Set up of maintenance management systems.
- Set up of operational software and administrative systems.
- H&S considerations.
- SCADA

Tests of employees' competency shall be included, backed-up by appropriate certification and training records. Provision shall be made for re-training at regular periods, and re-certification, during the DNP.

4.17.3 Contractor's Staffing Plan for the Plant

The Contractor shall propose a staff plan defining members and types of staff to be employed by the Contractor to operate the plant for 3 months after taking over. This would include:

- Staffing levels;
- Required competences
- Job descriptions;
- Any particular requirements of the Authority responsible for the plant operation.

4.18 As-Built Drawings

The Contractor shall provide a complete and accurate set of all drawings to detail parts of the permanent works, regardless of whether such details were originally prepared by the Engineer or by the Contractor, his sub-contractors, or suppliers. This set of drawings is referred to as "Contractor's As-Built Drawings".

During manufacture, installation, or construction, the original drawings shall be progressively marked up to show any changes, which may become necessary, and to show correct dimensions, sizes, equipment details and the like so as to provide a complete and accurate record of the part of the permanent works covered by such drawings.

All drawings shall be produced by using appropriate computer aided design software preferably, AutoCAD Civil 3D, which will allow undistorted data transfer to the Standard format being used in a GIS system that will be developed by the Employer.

The Contractor's As-Built Drawings shall be completed and submitted to the Engineer within one month of the time that the work detailed on them is completed. Each drawing shall be certified by the Contractor as being a complete and accurate record of the work detailed on it. A copy of the current drawings shall be kept on site, always available for inspection by the Engineer.

The Taking Over Certificate required to be issued under Clause 10.1 of the Conditions of Contract shall not be issued until the Contractor has submitted the completed and approved As-Built Drawings.

4.19 Submission and Approval of Contractor's Documents

All drawings and documents shall be prepared by the Contractor within the time specified in the approved Program of Works and shall be submitted for examination and approval to the Engineer. Within his authorities, the Engineer is responsible for examination and approval/rejection of the Contractor's Documents including the Detailed Design.

4.20 Schedule for submission of Documents

Table 2: Documents to be submitted

No.	Document	No. of copies Hardcopy + Digital	Submission
1.	Work Program + Gen. Method Statements	3+1	Commencement date + 28 days
2.	Review and Approval Period of Contractor Documents by Engineer	3+1	21 days after Receiving Contractor document and Contractor Notice
3.	Bill of Principal Quantities of the Permanent Works (Contract Sum Analysis)	3+1	28 days before commencement date
4.	Insurances	3+1	28 days before commencement date
5.	Quality Assurance Plan	3+1	Commencement date + 28 days
6.	Topographical survey and raw water analysis/assessment	3+1	Commencement date + 28 days
7.	Geotechnical survey	3+1	Commencement date + 56 days
8.	Environmental, Social, health and Safety Action Plan (ESHS-AP) Environmental and Social Management Plan (ESMP)	3+1	Commencement date + 21 days
9.	Notice to the Error in the Employer Requirements	3+1	Commencement date + 42 days
10.	Notice to the Errors in Setting out		Commencement date + 28 days
11.	Approval of all concrete mix designs of all concrete classes		Commencement date + 28 days
12.	Preliminary Design	3+1	Commencement date + 2 months
13.	Detailed Design	3+1	Commencement date + 4 months
14.	Progress Reports	3+1	First week of Month

No.	Document	No. of copies Hardcopy + Digital	Submission
15.	Material approval, Method Statements, Shop drawings	2+1	21 days prior to the execution of the specific activity
16.	Draft FDS	3+1	21 days before the commencement of the Tests on Completion
17.	Provisional O&M Manual	3+1	42 days before the commencement of the Tests on Completion
18.	Plan for the commissioning period	3+1	21 days before the commencement of the Tests on Completion
19.	Plan for tests on completion	3+1	21 days before the commencement of the Tests on Completion
20.	Financing Plan for O&M Period	3+1	21 days before the commencement of the Tests on Completion
21.	Training Plan + Contractor's staffing plan for the plant	3+1	90 days before Taking Over of the Works
22.	Provisional As-Built Drawings	3+1	42 days before the commencement of the Tests on Completion

5. Detailed Scope of Works for Design and Construction

Contractor's design shall include the detailed designs of the water treatment plant with the effective capacity of 20,000m³/d, fit for construction purposes. All facilities, which will be required for the future extension of 97,000m³/d shall be included in the conceptual design, in order to define space requirements for future extensions. Where future units are shown, provision must be made for easy connections. Distribution chambers and similar structures shall be designed to allow for the connection of future units.

The Contractor shall design and construct the WTP and its respective reservoirs to fit within the site boundaries shown on the general arrangement drawings of the Conceptual Design. Within the limit of the WTP and reservoir site boundaries, the Contractor is free to design, position and size the various elements of the project facilities, as the Contractor considers best, but only to the extent that they comply with the requirements set out in these Employer's Requirements and other relevant documents, e.g. the Specifications.

The Contractor's design shall take the process solutions used in the conceptual design as a guideline. If standby equipment has been designed in the Conceptual Design, it is mandatory for the Contractor to reflect this in his design. Also, similar levels of redundancy shall be reached in any alternative design.

The WTP shall be designed and constructed so that individual elements of the WTP such as tanks, chambers, etc. can be isolated from the flows without affecting the overall operation of the works. Furthermore, the WTP shall be designed for maximum flexibility of operation.

5.1 Conceptual Design and Drawings

The designs shown on these drawings are outline/tentative drawings only and are based on information available prior to preparation of the Tender Documents. All levels indicated or proposed are provisional and may need to be revised subject to the results of survey and site investigation carried out by the Contractor. The Contractor shall prepare his own proposals, consistent with his own experience, particulars and guarantees given by him in the schedules.

The dispositions of structures and buildings and of associated systems may be varied from those shown on the outline/tentative drawings to suit the particular requirements of plant and civil works proposed by the Contractor.

Although Employer's design proposals are intended to be recommendations to the Contractor, he shall be fully responsible for his proposals. If the contractor wishes to modify the tentative design and drawings via value engineering, he/she informs the engineer in timely manner. His designs shall reflect his own experience in works of a similar nature and magnitude.

The Contractor shall carry out such surveys, studies and analyses of any kind as necessary for his general and detailed designs and to verify the information shown on the outline drawings and other data furnished by the Employer. The Employer shall not be liable for any claim by the Contractor because of incompleteness or inaccuracy of the outline/tentative drawings and data furnished by the Employer,

irrespective of when the Contractor shall discover such incompleteness or inaccuracy.

5.2 Basis of Contractor's Design & Design Responsibility

The Contractor shall be responsible for all design works in accordance with the Contract. The Plant shall meet the specified performance requirements and safe construction has to be ensured. The Contractor's responsibilities shall also include the selection of a safe operational and maintenance regime in line with his design.

The Conceptual Design forms part of the Contract. The Contractor's design shall follow the principles of the Employer's conceptual design for the Intake, Raw Water transmission main, the WTP, treated water transmission pipes, booster pumping station, the storage tanks, access road, and associated ancillary works, and shall incorporate the requirements of the General Technical Specifications and other parts or sections of the Documents, as relevant.

In particular, the Contractor shall pay attention to the following aspects of the design:

- Water retaining structures shall be cast in-situ reinforced concrete construction. Special attention shall be made to predicting and designing for thermal movement, limiting crack widths and ensuring the structures are watertight under all conditions.
- Calculation of structures shall consider all possible loading conditions during construction, operation and maintenance.
- Structural calculations shall also take account of forces likely to be induced in the event of strong earthquakes of 6.0 magnitude especially for buildings (e.g. pump stations).
- Pipeline routings shall not substantially deviate from the Conceptual design. Local, limited deviations in pipeline routing shall be coordinated with the Employer in order to minimize additional cost for compensation to plot owners / tenants and for acquisition of land / right of way.

5.3 Alternative Designs

Design alternatives shall be constrained by the following requirements:

- The level of reliability of operation achieved shall be at least equal to the tender scheme.
- An equivalent level of standby plant shall be available to that presently proposed.
- The plant shall be able to operate flexible within the specified limits, to meet water demand variations of the system.
- The operational approach shall be similar to existing systems under the control of the MoW.
- The dispositions of structures and buildings and of associated systems may be varied from those shown on the outline drawings to suit the particular requirements of plant and civil works proposed by the Contractor, however, Site locations, Site plot dimensions, and pipeline routings shall not in principle deviate from the Employer's Design. Local, limited deviations in

pipeline routing shall be designed in cooperation with the Engineer/Employer in order to minimize additional cost for compensation to plot owners / tenants and for acquisition of land / right of way.

5.4 Design Regulations & Standards

The Design Works shall be carried out in accordance with the relevant quality standards, test procedures or codes of practice, collectively referred to as Reference Standards, listed in the relevant Part of the General Technical Specifications. The Contractor shall familiarize himself fully with the requirements of such standards.

The applicable Reference Standards shall be the one in force, at the date 90 calendar days preceding the date of submission of the Tender.

Acceptable Technical Standards are Eurocodes, British Standards, American Standards and other EN Standards. The use of other official standards that provide equal or better quality, than the aforementioned standards can be accepted after review by the Engineer and his written approval. In that case, the Contractor shall submit all the necessary information in compliance with the Engineer's instructions. The Engineer shall issue his decision within 21 days after the receipt of that information. If the Engineer decides that the standards and codes proposed by the Contractor do not guarantee equivalent or better quality, the Contractor is obliged to apply the specified standards.

5.5 General Drawings Standards

Drawings submitted by the Contractor shall be clearly printed with black lines on white paper and shall be resistant to fading on exposure to light. If colors are used, only bold and strong types shall be applied, which are clearly visible on white paper. Prints shall be on durable paper of good quality and 80 g/m² minimum weight.

Sheet sizes shall be in accordance with A series to ISO 216 and shall not be larger than A1 size, unless otherwise agreed by the Engineer.

All drawings submitted by the Contractor shall use the English language and SI units.

All drawings shall be clearly and fully cross referenced to each other, to the Employer's Design Specification and to the Engineer's drawings, as relevant.

Where drawings are revised, the revision letter or number shall be incorporated in the title block and the revision shall be clearly indicated on the drawing with the revision letter or number shown in an adjacent triangle.

Drawings shall be drawn to specified scales or to such scales as are appropriate for clearly detailing and conveying the Contractor's proposals and shall include a graphical scale bar. Preferred Scale shall generally be as in the table below

Table 3: Preferred Scales for the Drawings

USE	SCALE
Maps	1:1000000 1:500000 1:200000 1:100000
Block Plan	1:2500 1:2000 1:1250 1:1000
LOCATION DRAWINGS	
Site plans	1:500 1:200
General location Drawings	1:200 1:100 1:50
Range Drawings	1:100 1:50 1:20
COMPONENT DRAWINGS	
Assembly Drawings	1:20 1:10 1:5
DETAIL DRAWINGS	1:10 1:5 1:1

The scale used shall be shown on the drawings.

All layout plans shall show the "north" direction.

The Contractor shall apply a reference numbering system for designs, drawings and documents so that each number used is unique.

All drawings shall bear a stamp, similar to the stamp on the drawings of the Employer's Design. The details of the stamp shall be agreed by the Engineer.

Drawings prepared by subcontractors and manufacturers employed by the Contractor shall be checked by the Contractor for accuracy and compliance with the Contract, as specified, before being forwarded to the Engineer. Working drawings not so checked and noted, will be rejected and returned to the Contractor without further examination.

5.6 Design Principles

5.6.1 General

The design and construction of the Works shall be in accordance with the General Technical Specifications, and good engineering practice. All aspects of the Works shall be fit for purpose and the design shall be sound, safe and robust.

The design of the Works must guarantee 24 hours per day continuous operation of the new facilities, under the given local climatic conditions.

The design shall facilitate operation, inspection, cleaning, lubrication and repair of the various components of the plant and provide easy access to all operation points, lifting devices shall be installed, where necessary.

The design shall ensure adequate ventilation and lighting to all operating areas, controls, proper electrical insulation facilities, noise suppression and insulation, suppression of vibrations, stairs, handrails, covers, etc.

All tanks and channels shall be provided with reasonable freeboard, following state of the design principles.

The design and construction shall ensure that the design life specified in these Employer's Requirements will be attained, and that such components operate satisfactorily under all service conditions.

All alarms and plant status shall be indicated on the local control panels and the central LED touch screen panel.

For all pump stations, it has to be determined if surge (water hammer) protection measures are applicable and implemented, if necessary.

Preference shall be given to treatment processes, for which MoW has gained positive operation experiences in the last years;

Design shall ensure cost-efficiency in terms of investment and operation costs;

Availability (safe supply) of proposed chemicals is an asset.

The proposed treatment process is based upon the raw water quality of the Momba River. However, the Contractor's proposed water treatment process must ensure to treat also deteriorated raw water quality to allowable limits.

Design shall be flexible in coping with varying raw water conditions and to allow for staged WTP extension. Consequently, certain degree of space reserves must be foreseen, for future installation of additional treatment units, in order to cope with deteriorating raw water quality and/or other non-favorable operational conditions.

5.6.2 Reliability and Redundancy

Redundancy means the duplication of critical components or functions of a system with the intention of increasing reliability of the system, usually in the form of backup or fail-safe measures.

The Facilities shall be designed with a high grade of flexibility and reliability of the various parts. Therefore, all process and control items shall be designed with a sufficiently high degree of redundancy (by means of installed spare capacity and/or

spare units of equipment), enabling to overcome all maintenance situations and/or local system malfunctions.

The Redundancy Concept set out in the Conceptual Design shall serve as a principle guideline for Contractor's design. However, in accordance with the FIDIC yellow book conditions the Contractor may modify the redundancy concept, in line with his envisaged design solution.

5.6.3 Standardization

Whenever possible, the Contractor shall provide equipment of a similar nature from the same manufacturer, e.g., electric motors, in order to facilitate the spare part regime. Standardization of types of equipment shall also be considered.

The Contractor shall note that particular attention will be paid to these requirements. In cases where the proposed equipment, which could be standardized, has not been standardized regarding manufacturer and type, the Contractor shall be required to provide conclusive technical justification; considerations of price alone will not be accepted.

5.7 General Design Requirements

5.7.1 Design Life

Civil, structural and building works shall be designed on the basis of a 50-year service life.

Mechanical and electrical equipment shall be suitable for 24 hours per day continuous operation also under discontinuous operation and under all local climatic conditions.

Wearing parts, other than consumable items, shall have a design life of at least 5 years assuming continuous operation. All mechanical and electrical equipment to be supplied shall, wherever possible, have a proven reliability record in similar works. Design life for the components is shown in the Table below:

Table 4: Minimum Design Lives

Description	Design Life Years
Civil Works and Buildings	
Process Structures	50
Underground Pipework	50
Buildings	50
HVAC equipment	20
Building Services	20
Pipelines	50

Description	Design Life Years
Mechanical and Electrical Plant	
Process Plant – water supply	25
Process Plant - sludge	20
Pumps – water supply and sludge	15
Valves and Actuators– water supply and sludge	20
HV switchgear and transformers	40
LV Motor Control Centers	20
Electrical Installations	20
Instrumentation, Control and Automation equipment	15
SCADA and PLC systems	15
Computers	5
Flow meters	20
Instrumentation	15

5.7.2 Maximum Noise Levels

The following noise levels, measured at approximately 1 m distance from mechanical and electrical facilities, are permitted:

- Hydraulic and pneumatic equipment max. 80 dB
- Gears and drives max. 80 dB
- Water pumps max. 80 dB
- Blowers max. 85 dB
- Gen-sets max. 85 dB

5.8 Particular Scope of Works and Requirements for each Site

5.8.1 Raw Water Quality

The water quality at this raw water abstraction point can be summarized as follows:

Main positive water characteristics:

- rather low to moderate raw water turbidity (acceptable for surface water sources)
- low raw water color

- very good dissolved oxygen level
- rather low nutrients concentrations (nitrogen and phosphorous)
- very low inorganic contaminant contents (chlorides, sulphates, etc.)
- very low heavy metals contents (mercury, nickel, lead)
- rather low algae presence (see concentrations of parameter "chlorophyll")

Main negative water characteristics (thus to be carefully considered in the WTP concept/design):

- low total hardness and low alkalinity, thus representing a very soft water with low buffer capacity, nearly non-existing permanent hardness and thus resulting in a strong water aggressiveness/corrosiveness,
- increased total iron concentrations (in the range of 0.3 mg/l being the former WHO guideline value and still being recommended the maximum drinking water iron concentration),
- moderate bacteriological water pollution (coliforms),

The concentrations of most of the water quality parameters remain well within WHO guidelines for drinking water quality (see section below), the raw water quality is thus considered as being adequate for drinking purpose following typical surface water treatment.

Raw water sampling was done in 2022 - sampling date: 27.05.2015, at sampling point Momba river, 200 m distance to shore, 3 m water depth.

5.8.2 Drinking Water Quality

It is required that the final quality of treated water, produced by the WTP, meets the WHO and Tanzanian Drinking water quality standards.

The two relevant standards are thus

- Tanzania Standard: "US EAS 12: 2014 – Potable Water – Specification", 1st edition 2014
- WHO: "Guidelines for Drinking-water Quality", 4th edition, 2011

Where two given parameters in the two standards/guidelines are in conflict, the more superior parameter shall prevail.

The following table summarizes the corresponding drinking water standards for the main relevant raw water parameters:

Table 5: Raw Water Quality characteristics – March 2023

PARAMETERS	UNITS	MEASURED VALUE	RECOMMENDED VALUE	
			TZS 789: 2019/ EAS 12: 2018	WHO, 2022
Temperature	°C	24.7	nm	nm
Colour	TCU	2060	<50	<15
Turbidity	NTU	466	<5	<1
Total Suspended Solids (TSS)	mg/L	253	nm	nm
Total Dissolved Solids (TDS)	mg/L	86.2	<1500	<600
Alkalinity (as CaCO ₃)	mg/L	75.88	nm	nm
Hardness (as CaCO ₃)	mg/L	61.39	<600	100 – 300
Iron (Fe)	mg/L	1.04	<0.3	<0.3
Manganese (Mn)	mg/L	0.132	<0.1	<0.08
Total coliform	CFU/100mL	152	nm	0
Fecal coliform	CFU/100mL	112	nm	0
E-coli	CFU/100mL	56	nm	0

Table 6: Treated water (effluent) quality.

PARAMETERS	UNITS	TBS	WHO 2011
pH		<5.5 – 9.5	6.5 – 8.5
EC (Conductivity)	(µS/cm)	<2500	nm
TDS (Total Dissolve Solids)	(mg/l)	<1500	<600
Temperature	(°C)	nm	nm
Colour	TCU	<50	<15
Turbidity	(NTU)	<5	<1
Total Suspended Solids (TSS)	(mg/l)	nm	nm
Chloride	(mg/l)	<250	<150 (C)
Nitrate-NO ₃	(mg/l)	<45	<50
Nitrate Nitrogen	(mg/l)	nm	nm
Nitrite-NO ₂	(mg/l)	nm	<3
Nitrite Nitrogen	(mg/l)	<0.9	nm

Sodium	(mg/l)	<200	<200
Potassium	(mg/l)	<50	nm
Fluoride	(mg/l)	<1.5	<1.5
Alkalinity (as CaCO ₃)	(mg/l)	nm	nm
Total Hardness	(mg/l)	<600	100 - 300
Calcium	(mg/l)	<150	nm
Magnesium	(mg/l)	<100	nm
Iron	(mg/l)	<0.3	<0.3
Manganese	(mg/l)	<0.1	<0.08 (P)
Phosphates	(mg/l)	<2.2	nm
Sulphate	(mg/l)	<400	<250
Aluminium	(mg/l)	<0.2	<0.2
Barium	(mg/l)	<0.7	<1.3
Boron	(mg/l)	<2.4	<2.4
Dissolved oxygen	(mg/l)	≥6.0	nm
BOD ₅	(mg/l)	<30	<1-2
COD	(mg/l)	<60	nm
Oil and Grease	(mg/l)	<20	nm
Metalaxyl	(mg/l)	nm	<0.01
4,4-DDT	(mg/l)	<0.001	<0.001
Tebuconazole	(mg/l)	nm	nm
2,4-Dichlorophenoxyacetic acid	(mg/l)	<0.03	<0.03
Cypermethrin	(mg/l)	nm	<0.003
Cadmium	(mg/l)	<0.003	<0.003
Lead	(mg/l)	<0.01	<0.01 (A, T)
Zinc	(mg/l)	<0.01	<4
Arsenic	(mg/l)	<0.01	<0.01 (A, T)
Copper	(mg/l)	<1	<2
Cyanide	(mg/l)	<0.01	nm
Selenium	(mg/l)	<0.01	<0.04 (P)
Mercury	(mg/l)	<0.001	<0.006
Total Chromium (as Cr)	(mg/l)	<0.05	<0.05
Cyanobacteria (<i>Microcystis aeruginosa</i>)	Cells/10mL	nm	<10

Total coli/100mls (cfu)	CFU/100mL	absent	0
Fecal coli/100mls (cfu)	CFU/100mL	absent	0
E. coli (cfu)	CFU/100mL	absent	0

nm: Not mentioned.

According to WHO Drinking-water Quality 4th edition (2011):

A = Provisional guideline value because calculated guideline value is below the achievable quantification level.

C = Concentrations of the substance at or below the health-based guideline value may affect the appearance, taste, or odor of the water, leading to consumer complaints.

D = Provisional guideline value because disinfection is likely to result in the guideline value being exceeded.

P = Provisional guideline value because of uncertainties in the health database.

T = Provisional guideline value because calculated guideline value is below the level that can be achieved through practical treatment methods, source protection, etc.

Toxins produced by cyanobacteria that is, "cyanotoxins"—are the only organic substances for which guideline values have been derived. Cyanobacteria occur widely in lakes, reservoirs, ponds, and slow-flowing rivers. They produce a wide range of compounds. Four key groups—anatoxin-a variants, cylindrospermopsins, microcystins and saxitoxins account for much, but not all, of the toxicity to humans (and other vertebrates) and are frequently reported. Whereas anatoxins, microcystins and saxitoxins are primarily found within cells, major fractions of cylindrospermopsins are released into water. The toxicity of different structural variants within a cyanotoxin group varies widely, and further cyanotoxins may yet be recognized. Mass developments of cyanobacteria ("blooms") also cause a high organic load that challenges treatment. Therefore, preventing blooms is the preferred control option.

The Bidders are recommended to confirm the pH value during their Tender preparation.

5.8.3 Process and hydraulic design Criteria

The WTP process design shall be based on internationally acknowledged process design criteria. The following Tables shall be considered as a guideline:

Table 7: Main Design Criteria for pipework's (Gravity and Pumping Main)

Main Design Criteria		
Criteria	Unit	Value
○ Minimum pressure at the remotest end of the system	m	5
○ Maximum velocity of flow in pipes-Transmission Lines	m/s	3

Main Design Criteria		
Criteria	Unit	Value
○ Maximum velocity of flow in pipes in Distribution Lines	m/s	1.5
○ Maximum velocity of flow in pipes	m/s	0.4
○ Maximum value of Demand Factor	-	0.3
○ Maximum value of Demand Factor – Peak Demand	-	3
○ Minimum allowable head loss in pipes	m/km	0.005
○ Maximum allowable head loss in pipes	m/km	0.01
○ Minimum allowable pressure	m	5
○ Maximum allowable pressure	m	60
○ Minimum Demand	factor	0.3 ADD*
○ Maximum Day Demand	factor	1.3 ADD
○ Peak Hour Demand for served population <1000	m ³	3 ADD
○ Peak Hour Demand for served population >1000	m ³	2.5 ADD
○ Allowable Non-Revenue Water for a new system (NRW)	%	15
○ Household per water point	Households	50

ADD = Average Daily Demand

Table 8: Main process design criteria for WTP

Description	Unit	Range of values	Adopted Design value
1. Design flow			
Max. WTP production rate	m ³ /day		20,000
Peak day factor		1 – 1.5	1.2
Average WTP production rate	m ³ /day		16,667
Intake to WTP losses	%	1 - 3	2
WTP losses	%	4 - 6	5
Total losses	%		7

Raw water flow rate	m ³ /day		21,506
Operation time per day	hours/d ay		22
Transmission main (Intake to WTP)			
Size	mm		Steel DN600
Length	m		1177
Flow velocity	m/s		0.97
2. Intake Coarse screen			
Bar diameter	mm	13 - 19	15
Bar spacing (c/c)	mm	50 - 75	70
Velocity through the screen	m/s	≤0.6	0.4
3. Intake Fine screen			
Screen opening size	mm	5 - 9.5	8
Velocity through the screen	m/s	≤0.6	0.3
4. Intake Grit chambers			
Number of units	u	≥2	2
Mean flow velocity	m/min.	3 - 4.5	4
Hydraulic detention time	Min.	6 - 15	10
Free board	m	≥0.5	0.5
Surface loading rate	m/h	10 - 25	15.27
Length to width ratio (L: W)		≥4:1	4:1
Length to Depth ratio (L:H)		≥6:1	6.15:1
5. Aeration Unit			
Type of aerator			Cascade aerator
Shape of aerator			Circular
Surface loading rate	m ² /m ³ /s	90 - 120	100
Flow velocity on the cascade	m/s	0.3	0.3
Number of drops/steps		4 - 6	4
Height of drops	cm	15 - 60	50
Overflow rate	m ³ /s over m	0.01	0.01

	width of step		
Height of aerator	m	0.9 - 3	1.8
Outlet velocity	m/s	0.6 - 1	0.6
Efficiency of raising O ₂	%	60 - 70	65
6. pH Adjustment tank			
Detention Time, T	Min.	≥2	3
7. Coagulation tank			
Type of mixing			Mechanical mixing using vertical shaft impellers
Detention Time, T	s	10 - 40	40
G value	s ⁻¹	600 - 1000	600
8. Clariflocculator			
Type of feed system		Center feed and peripheral (rim) feed	Center feed
Velocity gradient, G	s ⁻¹	20 - 75	45
Maximum Velocity gradient, G	s ⁻¹	50	45
Detention Time	min.	10 - 40	30
Side water depth (SWD)	m	4.5	4.5
Walkway width	m	1.2	1.2
Number of units	u	≥2	2
Paddle area/tank area	%	<25	23
Number of tanks	u	≥2	2
Launder weir loading	m ³ /d.m	<300	280
Side Water Depth	m	3.0 - 5.0	3.5

Sludge collector speed	m/min	0.3 – 0.9	0.6
Outlet velocity	m/s	< 0.4	0.3
Floor slope		1:12	1:10
9. Filters			
Type of filters		Slow sand filters and Rapid Sand filters	Rapid sand filter
Influent location		Top	Top
Effluent location		Bottom	Bottom
Length to width ratio L:W		2:1 – 4:1	2
Filter cell area	m ²	25 - 100	50
Depth of filter box	m	4 - 8	5.5
Width of filter cell	m	<6	5
Depth of water over sand	m	≥1.8	1.8
Media depth: Effective size		>1000	2400
Bed depth	m	>0.25	0.45
Effective diameter	mm	0.8	0.8
Uniformity coefficient		1.7	1.7
Max. filtration rate	m/h	12	7
Control System		Modulating control valve system, influent weir splitting, and declining- rate filtration.	Influent weir splitting
Under Drain System		manifold pipe system, blocks, screens, false bottom with nozzles, and porous bottom.	False floor (bottom) underdr

			ain with nozzles
Type of Media		Quartz and Silica	Quartz Sand
Media arrangement		Single media, Dual media and Multimedia	Single medium
Effective Size	mm	0.45 – 0.55	0.5
Specific gravity		2.6 - 2.7	2.65
Uniformity Coefficient		1.3 - 1.8	1.5
Filter media Depth	mm	1000 - 1200	1200
Backwash type		(1) Water-only backwash with a surface wash (2) Water and air backwash (air scour).	Air and water backwash
Total Backwash duration	Min	10-20	18
Number of nozzles	Nozzles/ m ²	≥ 35	42
Nozzles slot size	mm	0.3 - 0.5	0.3
Filter offline duration	Min.	≥30	45
Backwash cycle	hour	24 - 72	24 - 72
Air backwash rate	m ³ /h.m ² of filter bed	0.9 – 1.5	1.2
Water backwash rate	m/h	37 - 45	37

Water backwash duration		Min.	10 - 20	16
Air backwash (alone) duration		Min.	2 - 3	2
% Bed expansion		%	37	37
10. Disinfection (Contact) tank				
Contact time		min	≥30	30
Minimum chlorine storage time		days		90
Baffling factor			0.1 - 1	0.7
Log inactivation	For virus		Depends on pH and temperature	4
	For giardia		Depends on pH and temperature	3
Chlorine residual		mg/l	0.5 – 0.8	0.6
11. Treated Water Tank				
Storage Time		h	3 - 8	4
12. Gravity Thickeners				
Solids Loading Rate		kg/m ² /d	20 - 80	50
% Solids of sludge		%	0.1 - 4	2
Specific gravity of wet sludge			1.025 – 1.1	1.05
Specific gravity of dry solids			1.2 – 1.5	1.3
Initial sludge settling velocity		cm/s	0.06 – 0.15	0.1
Solids capture Efficiency		%	80 - 90	85
Solid's concentration of Thickened sludge		%	2 - 4	4
Safety factor for solids flux			0.667	0.667
13. Sludge drying beds				
Sludge layer		mm	300 - 400	350
Sand layer depth		mm	100 - 300	300
Sand effective size		mm	0.3 – 1.2	0.9
Sand uniformity coefficient			3.5 - 5	4.5

Gravel layer depth	mm	300 - 460	400
Gravel effective size	mm	3 - 35	20
Sludge cake Solid's content	%	15 - 30	25

Table 9: Design Summary

ITEM	DESCRIPTION
General site Layout Plan for WTP	The proposed Momba Water treatment plant is designed to operate at maximum raw water flow rate of 21,506 m ³ /day from a proposed Momba raw water intake and produce a maximum flow rate of 20,000 m ³ /day. A raw Water inlet pipe (DN600-Steel) from the intake enters the circular cascade aerator at an approach velocity of 0.97 m/s. The proposed WTP layout comprises of the following water treatment structures. Cascade aerator, pH Adjustment and coagulation units (3), Clariflocculators (2), Rapid sand gravity filters (5), Contact (disinfection) tank, and 4,000 m ³ clear water tank. The sludge treatment structures are as follows: Primary and secondary sludge tanks (2), Primary and secondary sludge gravity thickeners (2), Mixed sludge tank, sludge drying beds (5), Decantation lagoons (2). Operation buildings are; Blower house, Chemical building, Power house, Workshop, Administration building, Plant manager's house, staff houses (4), and Guard house. All access roads are with total carriage way of 8 m (i.e., 2.5m lane width, 0.5m shoulder and 1m side ditch)
Pre-treatment Unit (Grit Chambers)	Two grit chambers are provided at the proposed Momba raw water intake to aid the removal of grits greater than 0.1 mm size. Each unit has the ability to carry maximum flow rate. The criteria used to design the grit chambers are detention time, surface loading rate, Length to width ratio and depth of water in the chamber. The effective detention time chosen is 10 minutes which will ensure a surface loading rate of 15 m/h. These chambers are provided ahead of fine and coarse (bar) screen.
Cascade Aerator	A circular cascade aerator is designed to aid the removal of iron and manganese through aeration process as water drops down the four steps. The total height of cascade aerator is 1.8m with a surface area of 27.6 m ² . A collection channel at the bottom of the cascade aerator is provided to collect and convey water to the DN600 steel pipes outlet pipes towards the pH adjustment and coagulation units. The removal of iron and manganese are efficient up to 70% with the cascade aerator.

pH Adjustment and Coagulation Units	Three (3) units designed to convey full flow at peak flow rates are provided. The pH Adjustment chamber is essential for dosing chemicals which can adjust the pH range either by raising or lowering to the recommended coagulation pH range. The chemicals are Milky of Lime (for raising pH) and Carbon dioxide (for lowering pH). The coagulation tank is for coagulation process using Aluminium sulphate, Vertical shaft impellers are used for mixing and allowed to take place for 30 seconds at a velocity gradient of 600 s^{-1} .
Clariflocculator	A circular Clariflocculator is used to aid the flocculator and sedimentation processes. The inner structure is a flocculation unit with an internal diameter of 9m and side water depth of 4.5 m designed at a slow mixing time of 30 minutes. The slow mixing process is a by mechanical means using vertical shaft paddles. The outer structure is a clarifier used to allow flocs to settle down towards the sludge hopper. The side water depth of a clarifier is 3.5 m with an inner diameter of 24m.
Rapid sand gravity filter	The designed filter cells comprise of deep bed homogeneous sand where the air and water applied simultaneously, air is first introduced about 1.5-2 minutes before the water to allow the air flow to become established, after which the combined air-water wash proceeds for about 6-8 minutes. The air flow is then stopped and the water flow continues to rinse the bed for another 8-10 minutes. This regime is also referred to as 'collapsed pulse' (Twort, 2017). A total of 5 filter cells are designed to operate at an average flow rate where 2 cells can be taken out of service during backwashing and maintenance. One filter cell has a plan area of 50 m^2 (10x5) with a nozzle density of 42 Nozzles/m^2. The backwash system is by continuous application of air and water at 60 m/h and 45 m/h respectively. All filter cells are designed to operate at a filtration rate ranging from 4 m/h – 7 m/h. The filters are backwashed at a duration of 18 minutes and repeatedly at a minimum cycle of 24 hours during high turbidities to discourage bacterial growth in the filter bed. The total number of filter cells backwashed per day should not exceed 2 cells.
Contact (Disinfection tank)	A two-compartment contact tank conveying full flow at maximum flow rate is proposed. A tank is designed to provide a minimum actual contact time of 30 minutes. The disinfection is to meet a desired log inactivation for virus and giardia of 4 and 3 respectively. The disinfection process is by calcium hypochlorite providing a minimum chlorine residual of 0.6 mg/L.
4,000 m^3 Clear Water Tank	A single unit tank is designed to store water for a minimum storage time of 4 hours. The stored water is for main transmission to Ikana GBR by pumping, Backwash water for backwashing Filters and service water at WTP.

Clear Water Pump House	A two-floor pump house with a basement and ground floor is provided. The pumps are installed at the basement floor and the ground floor used as a control room. A total of four (4) pumps (3W+1S) are provided to convey 304 m ³ /h for each pump through a head of 330 m at flow velocity of 0.9 m/s to Ikana GBR.
Backwash water storage tank	A 1,000 m ³ circular Elevated backwash water storage tank is provided to store water for two backwashes and service water. Water should be fed from the Clear water tank by pumping at a head of 15 m.
Blower House	A blower house with 2 pumps (1W + 1S) is provided to aid air scour backwashing through a 300 mm pipe to the filter cells. The air scour should last for 2 to 3 minutes before introducing water backwash. The simultaneous application air and water should last for not less than 8 minutes.
Primary Sludge Tank	A baffled primary sludge tank receiving sludge from primary treatment units (Cascade aerator to clarifier) by gravity is provided. Sludge from primary sludge tank is conveyed by pumps to the sludge gravity thickener due to elevation difficulties.
.Primary sludge gravity thickener	A circular sludge gravity thickener is provided to thicken daily sludge produced from a primary sludge tank at a loading rate of 50 kg/m ² .day. A thickened sludge solid content should be 4% or more. The diameter of thickener is 20 m with side water depth of 4.
.Secondary sludge (dirty backwash water) tank	A rectangular secondary sludge tank receiving sludge from filter backwashes is placed at the bottom of rapid sand gravity filters. The designed tank has the capacity of holding two backwashes per day. The total capacity of a dirty backwash water tank is 1,000 m ³ which is pumped to the secondary sludge gravity sludge thickener.
.Secondary sludge gravity thickener	One sludge gravity thickener is provided to thicken daily sludge produced at a loading rate of 50 kg/m ² .day. A thickened sludge solid content should be 4% or more. The diameter of each thickener is 10 m with side water depth of 4.
.Mixed Sludge Tank	This tank receives thickened sludge from primary sludge gravity thickeners and secondary sludge gravity thickener. Acts as an equalization basin before discharging sludge into sludge drying beds.
.Sludge Drying Beds	Five (5) drying beds are provided to enhance drying of sludge from the mixed sludge tank. The sludge is applied at initial sludge depth equals to 1 m. The Initial solids content of wet sludge, P_{s-wet} is 5% and sludge is allowed to dewater to a solid concentration, $P_{s-dewatered}$ of 10% thereafter left to dry to final

	cake solids concentration, $P_{s\text{-cake}} = 25\%$. The yearly sludge cake production is 304,658.4 m³/year. More drying bed details are as tabulated bellow; <table><tr><th>Detail</th><th>Unit</th><th>Value</th></tr><tr><td>Feeding duration</td><td>days</td><td>4</td></tr><tr><td>Drying (dewatering) duration</td><td>days</td><td>4</td></tr><tr><td>Emptying time</td><td>days</td><td>2</td></tr><tr><td>Total cycle duration</td><td>days</td><td>10</td></tr><tr><td>Free board</td><td>m</td><td>0.5</td></tr><tr><td>Sand layer depth</td><td>m</td><td>0.3</td></tr><tr><td>Gravel layer depth</td><td>m</td><td>0.4</td></tr><tr><td>Slab thickness</td><td>m</td><td>0.15</td></tr><tr><td>Slab slope towards the drain pipe</td><td>%</td><td>1</td></tr><tr><td>Slab depth at the slope end</td><td>mm</td><td>361</td></tr><tr><td>% Decrease in volume due to percolation</td><td>%</td><td>50</td></tr></table>	Detail	Unit	Value	Feeding duration	days	4	Drying (dewatering) duration	days	4	Emptying time	days	2	Total cycle duration	days	10	Free board	m	0.5	Sand layer depth	m	0.3	Gravel layer depth	m	0.4	Slab thickness	m	0.15	Slab slope towards the drain pipe	%	1	Slab depth at the slope end	mm	361	% Decrease in volume due to percolation	%	50
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.Decantation Lagoon	Lagoons are generally operated in a cyclic fashion: fill, settle, decant. This cycle is repeated until the lagoon is full or the decant can no longer meet discharge limitations for about three months. In this design two lagoons are provided to facilitate decantation process interchangeably throughout the year. Lagoons with 3:1 horizontal to vertical slope is provided to facilitate a 1.2 m initial depth of dewatering layer. A 3 m wide ramp is provided in each lagoon to provided vehicle access during removal of settled sediments.																																				
.Operational Buildings	Other structures on the proposed WTP site are operational buildings as follows; <table><tr><td>A. Chemical building</td></tr><tr><td>B. Power House</td></tr><tr><td>C. Workshop building</td></tr><tr><td>D. Administration building</td></tr><tr><td>E. Plant managers House</td></tr><tr><td>F. Staff Houses</td></tr><tr><td>G. Guard House</td></tr></table>	A. Chemical building	B. Power House	C. Workshop building	D. Administration building	E. Plant managers House	F. Staff Houses	G. Guard House																													
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6. Chemical Dosages

Chemical Name	Uses	Min. Dosage rate mg/L	Avr. Dosage rate mg/L	Max. Dosage rate mg/L
Aluminium Sulphate (Alum)	Coagulant	12	20	25
Potassium permanganate	Oxidizing agent	0.3	0.5	1
Calcium hypochlorite	Disinfectant	2	4	5
Lime water	Final remineralization	8	10	14
Milk of Lime	pH Adjustment (Raising pH)	10	12	16
Carbon dioxide gas	pH Adjustment (Lowering pH)	8	10	15
Polymer*	Flocculant aid	0.1	0.25	0.5
Powder activated carbon (PAC)	Taste and Odor (T&O) control	2	6	10
Chemical Name	Uses	Max. Dosage rate g/kgDS	Average Dosage rate g/kgDS	Min. Dosage rate g/kgDS
Polymer**	Sludge thickening Aid	2	2.5	4

*Used in water treatment

** Used in sludge treatment

Capacities to be designed

The following capacities shall be designed and constructed:

Table 10: Summary of the capacities for Intake, WTP, Pumps and Storage Tanks

LOCATION	DESIGN RESULTS	COORDINATES
	PROJECTION 2035	
Momba Intake		E = 419787.905, N = 9020987.744 Z = 1363.04m
✓ Intake capacity	Q = 21,506 m³/day	
✓ Grit Water Tank	978 m³/hr	
Intake pump		
✓ Low lift pump	Q = 978 m³/hr, H = 10m (1W+1S)	
✓ High lift pump	Q = 978 m³/hr, H = 25m (1W+1S)	
Momba (Ruwasho) WTP		E = 420870.671, N = 9020760.594 Z = 1372 m
✓ WTP Capacity	Q = 20,000 m³/day	
✓ Pumps	Q = 304 m³/hr, H = 330 m (3W+1S)	
✓ Clear Water Tank	4,000 m³	
Ikana GBR		E = 434507.551, N = 9008667.396 Z = 1663 m
✓ Tank Capacity	3,000 m³	
Nkangamo BPS		
✓ Pumps to Tunduma GBR	Q = 455 m³/hr, H = 221 m (2W+1S)	E = 457582.688, N = 8980505.059 Z = 1555.5 m
✓ Sump Capacity	2,000 m³	
Tunduma Tanks		
Proposed new tanks		E = 474261.639, N = 8972291.901 Z = 1697.4 m
✓ Uhuru Park GBR	5,000 m³	
Existing tanks		
✓ Existing ESR	120 m³	E = 474027.605, N = 8972364.379, Z = 1697.4 m
✓ Existing Uhuru GSR	500 m³	E= 474244.828, N = 8972291.901, Z = 1697.4 m
✓ Existing Ipito GSR	200 m³	E = 475935.922, N= 8970122.425, Z = 1682.7 m
✓ Existing Makambini GSR	200 m³	E = 475884.205, N = 8972108.747, Z = 1655.7 m
✓ Existing Chapwa GSR	200 m³	E= 477321.942, N = 8972978.587, Z = 1522.2 m

Pipe size and Length summary

S/N	TUNDUMA WATER SUPPLY PROJECT DESIGN PROJECTION YEAR 2035	
	PIPE SELECTION	LENGTH (m)
1	ABSTRACTION PIPE	
	STEEL-DN600-PN10	100
2	RAW WATER PIPE	
	STEEL-DN600-PN10	1,177
3	RUWASHO_WTP TO IKANA GBR	
	STEEL-DN600-PN10	305
	STEEL-DN600-PN16	119
	STEEL-DN600-PN25	4,705
	STEEL-DN600-PN30	10,798
	STEEL-DN600-PN40	5,067
	SUB-TOTAL-02	20,994
4	FROM IKANA GBR TO NKANGAMO BPS	
	STEEL-DN600-PN10	2,877
	STEEL-DN600-PN16	13,247
	STEEL-DN600-PN25	22,380
	SUB-TOTAL-03	38,504
5	NKANGAMO BPS TO UHURU PARK TUNDUMA	
	STEEL-DN600-PN10	10,310
	STEEL-DN600-PN16	7,177
	STEEL-DN600-PN25	4,185
	SUB-TOTAL-04	21,672
6	FROM J-0001N TO EXISTING TUNDUMA_ESR	
	STEEL-DN100-PN10	38
7	FROM J-0002N TO EXISTING TUNDUMA_GSR 01	
	STEEL-DN150-PN16	40
8	FROM TUNDUMA GSR TO EXISTING TUNDUMA_IPITO GSR 02	
	STEEL-DN150-PN10	4,285
9	FROM TUNDUMA UHURU PARK TO JUNCTION J-0004N	
	STEEL-DN150-PN10	1,555
10	FROM J-0003N TO EXISTING TUNDUMA_MAKAMBINI GSR 03	
	STEEL-DN100-PN10	494
11	FROM J-0004N TO EXISTING TUNDUMA_CHAPWA GSR 04	
	STEEL-DN100-PN25	2,664
	SUB-TOTAL FROM ALL JOINTS	9,076
	GRAND-TOTAL	91,523

6.1 The design of a Conventional Water Treatment Plant

The water and sludge conventional treatment units constituting the base solution are summarized in the simplified Process Diagram presented in the following figures, which are to be considered as concept diagrams.

These treatment units include the following steps:

6.1.1 Water treatment line:

- Cascade aeration
- Distribution Chamber
- Coagulation
- Flocculation
- Conventional Sedimentation and remineralization
- Sand Filtration
- Recycling of supernatant;
- Back-wash facilities;
- pH adjustment if required
- Disinfection
- Clear water reservoir
- Service water tank
- Fencing and gates, guard houses;
- Utilities – internal water supply, power, internet, sewer, waste;
- Internal road system;
- Administration building
- Chemical and storage building
- Staff houses;
- Street lighting;
- Access roads;
- Transmissions – pumped mains:
- Storage tanks facilities and infrastructures:
- Transmissions – Gravity lines and pumped one
 - ✓ Intake to WTP and WTP to other storage facilities

6.1.2 Sludge line:

- Thickener for dirty backwash water from sand filters
- Thickener for sludge from sedimentation tank (this stage can be integrated into the sedimentation tank)
- Mixed storage tank
- Sludge drying beds
- Decantation lagoon (overflow from thickeners and occasionally rain water from WTP platform)

Plant shall operate satisfactorily for the design flow rate, but also for lower flow rates down to 80% of the design flow rate.

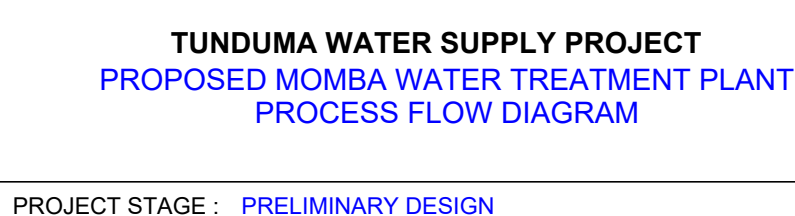
The retention times mentioned in the diagram must be considered by the Contractor as minimum values, and be increased if needed to guarantee the treated water quality as required.

The Contractor shall determine the nature and dosage of chemicals to be used at each step of the treatment, taking into account OPEX optimisation.

Figure 4: Simplified Process Diagram

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NOTES:

1. All dimensions shown are in millimeters unless otherwise shown.
2. Levels shown to this drawing are the levels in meters above the mean sea level.

6.1.3 Raw Water Pipeline and Intake

A new raw water pipeline, steel - DN 600mm, shall be constructed from the intake to the WTP. The pipe shall cope with a capacity of 21,000m³/d raw water (including internal water losses).

In order to avoid floating and movement, the pipe shall be fixed, secured and protected against physical damage. The pipe shall be equipped with all necessary anchorages, according to Contractor's structural calculations, to safeguard it against buoyancy.

6.1.4 WTP - General

The following treatment units of the WTP (where not otherwise mentioned) shall be designed and constructed. Scope of works for construction is for the WTP with the capacity of 20,000m³/d. Based on the results of the raw water quality assessment, topographical survey and geotechnical investigation, the contractor shall carry out detailed design of the required treatment process, as well as the required hydraulic elements of the treatment process. The design shall be as per the applicable standards. The treatment system shall be designed to remove all pollutants/contaminants from the raw water and the foreseen emerging pollutants. Water treatment process structures and water storage reservoirs shall be designed in reinforced concrete. **However, sedimentation and filtration are compulsory treatment units.**

The design shall be based on respective required capacity and quality and will include but not limited to:

- General layout for the plant
- Detailed design and plans for Aeration system
- Detailed design and plans for flocculation system,
- Detailed design and plans for lamellar sedimentation tanks
- Detailed design and plans for filters
- Detailed design and plans of disinfection system
- Characteristics and specifications of filtering media (anthracite or other approved filtering materials)
- Detailed design and plans for hydraulic installations
- Detailed design, plans and specifications of backwash system
- Detailed design, plans and specifications for laboratory and equipment
- Detailed design and plans for office facilities and guard rooms
- Details and plans of electrical, electronic and electromechanical installations
- Detailed design, plans and specifications for instrumentation, control, automation and telemetry
- Detailed design, plans and specifications of pumping stations where required
- Detailed architectural plans and drawings
- Detailed bills of quantities

- technical specifications for works, materials, equipment, installations, workmanship, and all related items and activities
- Clear water storage reservoir
- Operation manual and maintenance manual and schedule

6.1.5 Inlet flow rate metering and regulation

A flow metering chamber shall be installed upstream of the treatment works. The raw water flow rate shall be measured by an electromagnetic flow meter installed on the raw water transmission pipeline. Turbidity and pH shall be also measured in the inlet valve/measurement room.

6.1.6 Aeration and distribution chamber

In the design of the aeration structure unit to treat iron (dissolved form), the following should be provided:

- Cascade type aeration to increase the dissolved oxygen concentration up to minimum 60% saturation levels, taking into account the % of dissolved oxygen in raw water as indicated in raw water analysis
- Maximum 2.0 m head loss between upper water levels between inlet and outlet of the aeration unit;
- Equal distribution of flow over the entire width of the cascade;
- By-pass of the aeration cascades for cleaning purposes;
- Safety access to cascades provided for cleaning purposes.

The splitting chamber shall be designed to enable an equal distribution between the treatment units. The distribution structure should preferably be of the overflow type. This must be adjustable and easily accessible. Each treatment unit, whose number will be determined by the Contractor, shall be easily isolated for maintenance purpose.

6.1.7 Coagulation

The coagulation step shall be achieved by injecting metallic salts under rapid mixing conditions. The coagulation reagent shall locally available.

The Contractor shall specify and justify the following in his design document:

- Type of reagent,
- Location of injection point,
- Treatment rate / consumption,
- Regulation.

The minimum sizing for coagulation tanks shall be in accordance the main process design criteria for WTP in table 7.

The coagulation tanks shall be designed to avoid hydraulic short-circuits and enable uniform energy dissipation in the water. In order to ensure an efficient dispersion of the coagulant, the coagulation tanks will be provided with rapid mixing devices.

If necessary, the Contractor shall provide facilities for pH adjustment. In that case, the Contractor shall specify in his design documents the following:

- type of reagent,
- location of injection point,
- treatment rate / consumption,
- regulation.

6.1.8 Flocculation

The flocculation step shall be achieved into slow mixing tanks. The minimum sizing for flocculation tanks shall be in accordance the main process design criteria for WTP in table 7.

The flocculation tanks shall be designed to avoid hydraulic short-circuits and enable uniform energy dissipation in the water. In order to enhance the aggregation of the flocs, the flocculation tanks will be provided with soft mixing devices.

In case the efficiency of flocculation needs to be improved, adjunction of a flocculating agent (polymer) shall be determined by the Contractor. In that case, the Contractor shall specify in his design documents:

- Type of reagent;
- Location of injection point;
- Treatment rate / consumption;
- Regulation.

Polymer solution shall be prepared and dosed automatically in the chemical building.

6.1.9 Clarifiers

Design and installation of clarifiers

The contractor will propose the type of sedimentation tanks.

The description below applies to horizontal sedimentation tanks equipped with mechanical scrapers; however, the Contractor will determine either horizontal sedimentation or up flow clarifiers will be used. After coagulation and flocculation, water should be conveyed to clarifiers with optimum delay and without disturbing the flocs formed.

Maximum surface load will be in accordance the main process design criteria for WTP in table 4.

6.1.10 General

Clarification tanks and installation of the common equipment shall contain the following items:

- Pipe or channel connection to flocculation unit.
- Easy and safe access for operation and maintenance.
- Easy access for energy cables, chemical materials lines and other services.

- When one of the tanks is out of service, other(s) shall provide the required efficiency in operation.
- Travelling bridges spanning the tanks from centre supports to the perimeter walls for circular clarifiers (if circular, rectangular tanks can also be provided as shown in drawings).
- Water content of the drawn sludge will be minimized when discharging automatically by mechanical or hydraulic methods.
- Sampling points.
- Protection of the mechanical and electrical equipment that may be affected by climatic conditions.
- Hydraulic isolation opportunity of each tank for repair and maintenance.
- Safe and easy transportation for inside and on each tank for control and maintenance.
- Adjustments (if needed) in orifice, weirs, etc. levels of the collection channels.
- Connection to filters with clarified water channel system.
- By-pass line.

6.1.11 Entrance system for clarifiers

Each clarifier shall have equal flow rates not differing more than 10% of the average flow to each clarifier. Clarifiers shall have a by-pass arrangement. By-pass valves on clarifier tanks shall be operated manually.

6.1.12 Sludge discharge

The settling tanks will be concrete construction. Settled sludge will be collected into two sludge pits in the bottom of each tank. The walls of sludge pit will have a sufficient slope to facilitate sludge collection. Contractor shall indicate the minimum slope taken into consideration in its proposal. Sludge valves shall be controlled with a motor or pneumatic control. Valves shall be controlled with the signals of sludge control devices. Each valve can also be operated manually.

Sludge extraction system shall be designed as "failure-safe" (meaning that the valves will be closed at a power or mechanical failure) and will activate a local alarm and also the alarm at central control room. Drain pipes shall be installed for drainage of the water and sludge from the tank. Necessary pumps, valves and fittings shall be designed so that the tank can be emptied.

6.1.13 Sludge samples

Local sample taps shall be installed to take sample from three separate levels at each clarifier. Sampling points will be set to make adjusting available in vertical and horizontal platform.

6.1.14 Remineralization

A Remineralization will be performed in the sedimentation tanks and/or at the end of the treatment line, subject to the raw water characteristics

The Contractor shall specify in his design document:

- Type of reagents,
- Location of injection points,
- Treatment rate / consumption,
- Regulation.

6.1.15 Manganese removal

This specific treatment will be forecasted due to of high concentrations of dissolved Manganese in raw water.

The Contractor shall specify in his design documents:

- Type of reagent,
- Location of injection point,
- Treatment rate / consumption,
- Regulation.

6.1.16 Filtration

The filtration shall be carried out by open gravity sand filters.

The sand filters shall be designed to be able to treat the maximal design flow, and maintain the filtered water quality, even when one filter is being backwashed.

At the maximal water flow of the plant, the filtration speed shall not exceed 6.5 m/h when all the filters are in filtration sequence and 10 m/h with one filter is in backwash and the other ones in filtration sequence. The minimum number of filters take into account the need that the total minimum flow) must be treated through $N-2$ filters, if N is the total number of filters, to avoid reducing the flow when one filter is under backwashing. The minimum number of filters is consistent with the filtration maximum velocities

For the calculation of surface area of the filters, the Contractor shall consider that one filter will be out of operation.

Internal dimensions of all filters shall be the same, and shall be designed for cleaning by both air and clean water.

Filters shall be installed such that filter backwash time frequencies will not be less than 24 hours in all conditions.

Hydraulic isolation of each filter unit shall be possible.

Filters shall have a by-pass arrangement.

The design of the floors and nozzles shall ensure uniform distribution of the incoming water, air and backwash water over the complete filter bed area.

Each filter will be equipped with a regulation device to keep constant the water level on the filter media and to allow the detection of clogging. This functioning of this device will be based on compensation of water head losses resulting from clogging, through a decreasing of water head losses at the outlet of the filter.

6.2 Installation and designing principles for filters

Installation of the filters shall comply with the following principles:

- Easy and safe access for operation and maintenance.
- Easy and safe access to go underneath the concrete floor of the sand filters.
- Necessary platforms for energy cables, chemical materials lines and other services.
- Protection of the electrical and mechanical equipment to be affected by climatic conditions shall be provided.
- Each filter can be isolated from others for maintenance and repair purpose.
- Inlet channels of filters will be designed as to distribute equal flow to all filters.
- Height of flood weir of the filter back wash water shall prevent filter media loss during backwash cycles with air and water.
- The filter control gallery will be covered.

6.3 Filter media

The Contractor will define the filter media, including the number of media.

Unless strong justification by the Contractor that other specifications are required, the sand will be according to the minimum following characteristics identified in table:

The Contractor shall be responsible for the supply, transport, storage and protection of the filter materials to the plant. Therefore, all temporary required facilities that can be accepted by the Engineer shall be supplied by the Contractor. The quality of the sand shall be checked with the Gravimetric sieve analysis Method. The analysis shall be conducted on the sand from the source quarry and a second on-site examination shall be conducted by the Contractor. The Contractor will be required to approve the quality of the sand and to ensure that the required granular proportions of the filter media have been met prior to the sand being utilised in the filter as filter media.

Gravimetric Sieve Analysis and the depth of the sand and gravels shall be stated by the Contractor. The total depth of the sand, excluding any bearing or distribution layers, in the filter media shall not be less than 1.0 m.

The Contractor shall be responsible for supply installation and maintenance of the filter material during, construction, installation, commissioning and acceptance without any further charges.

6.4 Lower drain system

Porous types bottoms that are made up of concrete or similar material shall not be accepted. Number and type of nozzles proposed installed shall be submitted for the approval of the Engineer before being installed. The distribution of nozzles shall be at least 50 nozzles/m². The Contractor shall supply 20% of the required nozzles as spares. The Contractor shall take all necessary precautions for partially or totally blocking of the nozzles and the additional installations including air/water transferring pipe and channel system, nozzles or orifice.

Filter nozzles:

- The distance between the nozzles and the orifices shall be arranged so that it will not adversely affect the operation of the filters if the sand layers or gravel located near the drain change their location.
- Nozzles shall contain separate inlets for air and water
- Installation to the required level shall be possible.
- Change of the constant impermeability rings without disturbance shall be possible.
- It shall prevent seepage of the sand.

The Contractor shall apply air pressure test at each filter drain before settling any material is allowed where air washing is used, see below.

6.5 Filter valves and pipes

The filter valves to be used for the washing periods shall be on/off commanded engine types or pneumatic types. Filter exit valves shall be closed automatically during energy cut off.

Air compressor with spares shall be supplied for the pneumatic valves. The velocity and capacity of each compressor group shall be designed sufficiently so that, for all filters, it will allow operating of at least three valves at the same time.

Air pipes shall be resistant to high pressure operation conditions.

6.6 Control consoles for filters

The filter cleaning shall be set manually or automatically by head loss measure into the filter bed. Time mode will be also provided in case of failure in the head loss measurement.

The labels on the control console shall be in English. Labels shall be submitted to the Engineer for approval.

6.7 Filter flow control mechanisms

A flowrate control mechanism shall be established at each filter's outlet. The control system with the control mechanisms of the flow rates of each filter shall work in a way that keeps the average water level in the inlet channels and on each filter accurate. The flowrate control mechanism shall not be affected by small deviations in the water level.

To measure real time flow passing through each filter, the system shall be aided with the use of portable flow meters.

The pipes between the filter and the clear water reservoir shall be arranged to prevent that the filter water level drops below the sand level. The lowest water level in the filter shall be 100mm above the sand level.

6.8 Level indicators

There shall be a "level" indicator on the control console of each filter. The warning and backwash system shall start to operate when the water level at each filter reaches the predetermined value and the exit valve open completely.

6.9 Backwash

6.9.1 Backwash velocity with air

Velocity of the free air coming into each filter unit shall be sufficient to loosen filter environment and to provide a homogenous distribution on the filter surface. The Contractor shall state the air washing velocity as m^3 free air/ m^2 filter area/hour. If the Engineer finds that the air velocity is not sufficient, then the Contractor shall prove this and all relevant costs will be charged to him/her. The Contractor shall take the required corrective measures if necessary.

6.9.2 Backwash water quantity and velocity

The backwash velocity of the filter shall be sufficient to provide an efficient washing of the filter beds.

Height of the backwash water discharge weir shall be designed such that it will not allow any sand loss to occur during a backwash cycle. The Contractor shall state proposed level of the discharge weir and normal height on the filter bed.

During backwash, no water shall enter the filters except for washing water.

The backwash water hydraulic system shall be designed giving sufficient cleaning of filter material with such quantity of backwash water.

Backwash water shall be taken from the backwash water tank (water not chlorinated). The capacity of the tank shall enable at least two filter washings. The backwash water will be directed to the filters by pumps directly to provide the washing requirements.

The available depth of water in the backwash water tank shall be measured automatically with a display to be installed in the plant control room.

The Contractor shall specify in his design documents the following:

- Backwash sequence (air and water velocity, duration...),
- Maximum water losses,
- Maximum TSS content in backwash water.

6.10 Discharge of the filter backwash water

Backwash water shall be discharged away from the filter environment by washed water channels and washing water conduits.

- Dirty backwash water shall be transferred to a specific thickening step, before to passing to the drying beds with sludge extracted from sedimentation tanks.
- The loss of filter material shall be prevented.
- Materials remaining in the filter bed sand after backwashing shall be reduced to a minimum. At least 95% of the material should be removed.

The layout of the system shall ensure that used backwash water will flow to the backwash water reuse tanks by gravity.

6.11 Piezometer connections

Piezometer connections shall be installed to measure the head loss gradient of the filter beds determined by the Engineer.

6.11.1 Disinfection

Disinfection shall be achieved using a form of chlorine locally available (gas, calcium hypochlorite or another one) which will be proposed by the Contractor based on efficiency and cost criteria. Chlorine will be injected in the contact tank. Contact tank shall be designed to improve water circulation and ensure a sufficient contact time, for example with baffle walls or similar in accordance with design criteria in table 4.

Proper dosing of chlorine for water treatment is critical in order to provide enough free chlorine to maintain a residual during storage and use. Recommendations are to dose with free chlorine at about 2 mg/l to clear water (< 10 nephelometric turbidity units [NTU]) and twice that (4 mg/l) to turbid water (> 10 NTU). Although these free chlorine doses may lead to chlorine residuals that exceed the recommended chlorine residual for water that is centrally treated at the point of delivery, 0.2-0.6 mg/l, these doses are considered suitable for water treatment to maintain a free chlorine residual of 0.2 mg/l in stored household water treated by chlorination. In order to inactivate microbes present in raw water contractor shall make sure that the baffling classification T_{10}/T is more or equal 0.7 in order to avoid short circuiting. For more insight Chlorination CT and inactivation of microbial contractor shall familiarize himself/herself with EPA Guidance Manual Disinfection Profiling and Benchmarking.

6.11.2 CT values

Chlorine	Viruses	2 (Ct ₉₉ 2-30 min-mg/l; 0-10°C; pH 7-9)	Free chlorine x contact time predicts efficacy; not effective against Cryptosporidium oocysts. Turbidity and chlorine-demanding solutes inhibit this process; hence, turbidity should be kept below 1 NTU to support effective disinfection. Where this is not practical, turbidities should be kept below 5 NTU with higher chlorine doses or contact times. In addition to initial disinfection, the benefits of maintaining free chlorine residuals throughout distribution systems at or above 0.2mg/l should be considered
	Bacteri	2 (Ct ₉₉ 0.04-0.08 min-mg/l; 5 °C; pH 6-7)	
	Protozoa	2 (Ct ₉₉ 25-245 min-mg/l; 0-25 °C; pH 7-8; mainly Giardia	

The chlorine dosing rate will be adapted with local regulation and no less than 0.6 mg/l at the outlet of the contact tank.

The contact tank shall be divided into two equal capacity compartments. Each compartment can be isolated and emptied for maintenance or cleaning, although the other one remains in operation.

The retention time of water in the contact tank shall not be less than 30 minutes at the maximum flow.

The Contractor shall specify and justify the following in his design documents:

- Location of chlorine injection point,
- Treatment rate / consumption,
- Contact time,
- Regulation.

6.11.3 Treat water storage tank

A clear water reservoir shall be constructed. Its capacity will be 10,000 m³, representing approximately four hours of average demand (one hour of average demand represents a twenty-second of the maximum capacity).

The reservoir shall be divided into two equal capacity compartments. Each compartment shall be isolated and emptied for maintenance or cleaning, although the other one remains in operation.

Each compartment shall include:

- Baffles to avoid preferential streams;
- An access shaft with a security closing;
- Covered air shafts with an anti- insect grid (at least two units);
- Aluminium or stainless-steel ladders;

The bottom of the reservoir will have a slope and will include a sump in order to allow a total emptying of each compartment when needed. The contractor shall supply a mobile sump pump. The reservoir will be equipped of two overflows, one for each compartment, discharging to the storm water drainage network of the plant.

6.12 Plant requirements – Sludge units

6.12.1 Lime Dosing (pH Correction)

The sludge treatment line shall be design to treat the sludge produced in the clarification step and the dirty backwash water from sand filtration, and shall comprise the following treatment units as a minimum:

- Sludge thickening,
- Dirty backwash water thickening,
- Sludge buffer tank,

- Sludge drying beds.

Minimal performances required and process guarantees provided in "Schedule of Guarantees" have to be respected, in particular the maximum water losses and the minimum dryness of the sludge to be evacuated. The liquid waste produced during the treatment (water extracted from sludge thickening) shall pass through wetland before being discharged to the nearby stream. This requirement is a minimum.

The Contractor shall take into consideration the environmental requirements and mitigation measures proposed in the ESIA.

6.12.2 Sludge and waste water production

The Contractor shall specify in his bid the calculation of sludge and liquid waste production including:

- Type of liquid waste,
- Medium and maximal flow and volume,
- Maximum TSS content.

6.12.3 Sludge thickening

The liquid sludge and dirty backwash will be transferred to a thickening step in order to increase their TSS concentration.

In order to optimize thickening design and efficiency, each flow will be thickened separately.

The process of liquid sludge thickening will be a conventional static thickener with a mechanical scraper.

In case the efficiency of thickening needs to be improved, adjunction of polymer shall be forecasted by the Contractor. The Contractor shall specify in his bid the following:

- Superficial mass load,
- Minimal height of the tank,
- Retention time,
- Thickened sludge concentration,
- type of reagent,
- location of injection point,
- treatment rate / consumption,
- regulation.

Polymer solution shall be prepared automatically in the chemical building.

The supernatant water or drainage water shall be discharged into a decantation lagoon before being rejected into the nearby stream.

6.12.4 Sludge buffer tank

A mixed buffer tank shall be designed to collect the thickened sludge and the thickened backwash water. The sludge buffer tank shall have a sufficient slope in

the bottom to avoid sludge deposit and facilitate the cleaning operation. The sludge buffer tank shall be designed for a minimum retention time of 2 hours. The sludge extraction from clarification and the buffer tank filling shall be controlled by automatic valves.

6.12.5 Sludge drying beds

The minimum sizing requirements for the drying beds are the following:

- Minimal sludge dryness: 20%

The Contractor shall specify in his bid the following to reach such a dryness:

- Feed period duration,
- Rest period duration,
- Number of beds,
- Bed composition (type and height of each layer),
- Height of sludge on the bed,
- Density of sludge after dewatering,
- Quantity of sludge to evacuate after dewatering.

The sludge beds shall be placed for easy collection of the sludge by Tunduma WSSA for final disposal, Access roads must be provided around the beds. The Backhoe Loader with 74Hp JCB 3CX should be supplied for dredging the sludges.

7. Plant requirements – Chemical

7.1 General

The chemicals used in the water and sludge treatment could be the following:

- Calcium Hypochlorite,
- Aluminium sulphate as a coagulant,
- PAC
- Polymer as a flocculant (water line),
- Magnesium Carbonate or equivalent for remineralization,
- Potassium Permanganate for manganese removal
- Sulfuric acid and/or soda, if pH adjustment is needed,
- Polymer (sludge line).

The Contractor is responsible for the choice of chemicals that shall respect the minimum requirement expressed in the following paragraphs.

7.2 Transmission Mains and Secondary Gravity Supply Mains

7.2.1 General

The following applies for all transmission mains:

1. Pipelines design flow velocities shall be designed in reasonable engineering range. Preference is given to the least life cycle cost of related assets (CAPEX for pumps and pipeline and OPEX - energy cost).
2. For buried pipelines, the Contractor shall conduct structurally dimensioning of trust blocks, which shall compensate the forces, and safeguard every bend of the transmission mains and secondary gravity supply mains. Alternatively, force restraining joints may be selected. For above ground pipelines, only force restraining connections shall be chosen.
3. The surge protection for the water transmission main shall be through surge control measures at the pump station (e.g. surge vessel incl. compressor, soft start/stop systems, etc.) and the line (e.g. air valves). The surge control should meet the criteria as detailed below:
 - Maximum Positive Surge = 20 % above the maximum working head pressure
 - Minimum Surge = Atmospheric pressure minus 2 m

7.2.2 Pipeline Data

The Schedule of Prices contains details on pipe diameters and length (sub-divided into above ground and below ground).

7.2.3 Pipeline below / above Ground

In principle all pipelines shall be designed and constructed below ground.

Exceptions may be

- pipe bridges or
- laying of pipeline ascending or descending to hill areas, of which geology can be characterized as solid rock. In this case the Contractor shall design and construct proper RC foundations for the pipes to be laid, either anchored to rock or to set in hard soil. The pipeline shall be anchored to the RC foundations considering all occurring forces.

Pipes laid above ground shall be executed in any case with restrained and form-locking connections.

7.2.4 Pipeline Road Crossings TANROAD AND TARURA

The Contractor shall design and build pipes road crossings between different section of the right of way. Trenchless technology should be applied when crossing trunk road under TANROAD.

7.2.5 Pipeline Road Crossings of Secondary or Unpaved Roads

At every crossing of secondary or unpaved roads, the Contractor shall design and construct steel pipe conduits in sufficient depth, diameter and wall thickness into which the media pipe shall be pushed after the conduit pipe has been laid. The conduit pipe may be laid in open trench procedure. The original road surface shall be reinstated after the conduit pipe has been laid.

Table 05: Below are sections which will entail TANROADS AND TARURA on road crossings

TARMAC AND GRAVEL ROAD CROSSING FOR TRUNK MAIN					
TRUNK MAIN	ROAD CROSSING TYPE	MATERIAL	PIPE SIZE-DN (mm)	START CH.(m)	END CH.(m)
Rising main from Momba WTP to Ikana BR/PR	Gravel	STEEL	600	00+270	00+280
Rising main from Momba WTP to Ikana BR/PR	Gravel	STEEL	600	00+990	01+000
Rising main from Momba WTP to Ikana BR/PR	Tarmac	STEEL	600	03+240	03+260
Rising main from Momba WTP to Ikana BR/PR	Gravel	STEEL	600	17+610	17+625
Gravity main from Ikana BR/PR to Nkangamo BPS	Gravel	STEEL	600	01+260	01+275
Gravity main from Ikana BR/PR to Nkangamo BPS	Tarmac	STEEL	600	02+880	02+900
Gravity main from Ikana BR/PR to Nkangamo BPS	Gravel	STEEL	600	14+550	14+575
Gravity main from Ikana BR/PR to Nkangamo BPS	Tarmac	STEEL	600	18+750	18+770
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	00+430	00+450
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	01+390	01+400
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	05+380	05+395
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	07+000	07+010

Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	10+957	11+010
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	11+505	11+520
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	12+355	12+370
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	13+740	13+755
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	14+540	14+560
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	15+450	15+460
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	15+885	15+900
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	15+930	15+945
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	16+035	16+045
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	16+285	16+300
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	16+560	16+575
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	16+830	16+845
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	17+130	17+145

Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	17+370	17+385
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	18+030	18+045
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	18+210	18+225
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	19+110	19+125
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	19+320	19+335
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	19+545	19+560
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	19+935	19+950
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	20+340	20+355
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	20+610	20+625
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	20+820	20+835
Rising main from Nkangamo BPS to Uhuru Park Tunduma GSR	Gravel	STEEL	600	21+495	21+510
Gravity main from Uhuru Park Tunduma to Junction J-0004	Gravel	STEEL	150	00+120	00+130
Gravity main from Uhuru Park Tunduma to Junction J-0004	Gravel	STEEL	150	00+425	00+440

Gravity main from Uhuru Park Tunduma to Junction J-0004	Gravel	STEEL	150	00+660	00+675
Gravity main from Uhuru Park Tunduma to Junction J-0004	Gravel	STEEL	150	00+930	00+945
Gravity main from Uhuru Park Tunduma to Junction J-0004	Gravel	STEEL	150	01+290	01+305
Gravity main from Uhuru Park Tunduma to Junction J-0004	Gravel	STEEL	150	01+440	01+455
Gravity Main from Tunduma Junction J-004 to Chapwa GSR04	Gravel	STEEL	100	00+450	00+460
Gravity Main from Tunduma Junction J-004 to Chapwa GSR04	Gravel	STEEL	100	01+290	01+305
Gravity Main from Tunduma Junction J-004 to Chapwa GSR04	Gravel	STEEL	100	01+790	01+800
Gravity Main from Tunduma Junction J-004 to Chapwa GSR04	Gravel	STEEL	100	02+040	02+060
Rising main from Tunduma GSR/GBR to Ipito GSR	Gravel	STEEL	150	00+120	00+150
Rising main from Tunduma GSR/GBR to Ipito GSR	Gravel	STEEL	150	00+620	00+630
Rising main from Tunduma GSR/GBR to Ipito GSR	Tarmac	STEEL	150	00+720	00+731
Rising main from Tunduma GSR/GBR to Ipito GSR	Tarmac	STEEL	150	01+330	01+340
Rising main from Tunduma GSR/GBR to Ipito GSR	Tarmac	STEEL	150	01+600	01+623
Rising main from Tunduma GSR/GBR to Ipito GSR	Tarmac	STEEL	150	02+057	02+086
Rising main from Tunduma GSR/GBR to Ipito GSR	Gravel	STEEL	150	2+595	2+610
Rising main from Tunduma GSR/GBR to Ipito GSR	Tarmac	STEEL	150	03+275	03+287

Rising main from Tunduma GSR/GBR to Ipito GSR	Railway Crossing	STEEL	150	03+618	03+643
Rising main from Tunduma GSR/GBR to Ipito GSR	Tarmac	STEEL	150	03+793	03+805

7.2.6 Pipeline on rivers and stream

The Contractor shall design and build several under-crossings of smaller streams. The conceptual design foresaw the following number and total length of under-crossings (the information is not warranted by the Employer):

The conceptual design foresaw to construct an RC U-profile upside-down on top of the pipelines to protect the pipeline and bedding material at creek under-crossings against scour and erosion and at under-crossings of secondary roads against additional loads from traffic (refer to standard detail drawing). At stream under-crossings, the top of the RC U-profile should be protected further by stone packing.

The Contractor shall adopt his own design for the pipeline protection considering the following requirements:

- Sufficient scour / erosion protection of laid pipeline at creek crossings,
- Protection of pipe against non-permissible traffic loads.

Table 06: Below are the sections which passes on stream where pipes needed to be encased on concrete, sleeved or supported as per drawing

RIVER CROSSING DETAILS					
TRUNK MAIN	RIVER CROSSING TYPE	MATERIAL	PIPE SIZE-DN (mm)	START CH.(m)	END CH.(m)
Rising main from Momba WTP to Ikana BR/PR	Stream	STEEL	1200	01+060	01+080
Rising main from Momba WTP to Ikana BR/PR	Stream	STEEL	1200	12+577	12+636
Rising main from Momba WTP to Ikana BR/PR	Stream	STEEL	1200	15+513	15+533
Gravity main from Ikana BR/PR to Nkangamo BPS	Stream	STEEL	900	11+576	11+596
Gravity main from Ikana BR/PR to Nkangamo BPS	Stream	STEEL	900	13+140	13+260
Gravity main from Ikana BR/PR to Nkangamo BPS	Stream	STEEL	900	15+513	15+533

Gravity main from Ikana BR/PR to Nkangamo BPS	Stream	STEEL	900	21+705	21+830
Gravity main from Ikana BR/PR to Nkangamo BPS	Stream	STEEL	900	27+400	27+500
Gravity main from Ikana BR/PR to Nkangamo BPS	Stream	STEEL	900	30+379	30+470
Rising main from Nkangamo BPS to Tunduma GSR	Stream	STEEL	1000	06+419	06+585
Rising main from Nkangamo BPS to Tunduma GSR	Stream	STEEL	1000	09+920	10+019
Rising main from Tunduma GSR/GBR to Ipito GSR	Stream	STEEL	500	02+565	02+630

7.2.7 Pipeline Wash-Outs, Air Valves, and Sectional Valves

The Contractor shall design and build all needed pipeline wash-out chambers at low-points of the pipeline.

The Contractor shall furthermore design and build all needed air valves and related chambers at high-points of the pipeline and at points where – according to general design rules for positioning of air valves and according to the transient analysis carried out by the Contractor – air valves (= vacuum valves, air release valves, or combined vacuum and air release valves) would be necessary to limit max and min pressure caused by water hammer and surge to the max and min pressure allowable for the chosen pipe material and pressure class.

The conceptual design foresaw No. of wash-outs (WO) and air valves (AV) as follows (the information is not warranted by the Employer)

Table 07: AIR VALVES (ARV)

S/ N	TRUNK MAIN	PIPE DIAMETER (mm)	VALVE DIAMETER (mm)	VALVE PRESSURE UPSTREAM (bar)	VALVE PRESSURE DOWNSTRE AM (bar)	CHAINAGE (km+m)
TRUNK MAIN 01						
1	Rising main from Momba wtp to Ikana GSR	600	100	40	40	03+030
2	Rising main from Momba wtp to Ikana GSR	600	100	40	40	03+660
3	Rising main from Momba wtp to Ikana GSR	600	100	30	30	05+620
4	Rising main from Momba wtp to Ikana GSR	600	100	30	30	07+020
5	Rising main from Momba wtp to Ikana GSR	600	100	30	30	08+000
6	Rising main from Momba wtp to Ikana GSR	600	100	25	25	09+270
7	Rising main from Momba wtp to Ikana GSR	600	100	30	30	10+110
8	Rising main from Momba wtp to Ikana GSR	600	100	30	30	14+488
9	Rising main from Momba wtp to Ikana GSR	600	100	25	25	18+630

10	Rising main from Momba wtp to Ikana GSR	600	100	25	25	18+950
11	Rising main from Momba wtp to Ikana GSR	600	100	25	25	19+500
12	Rising main from Momba wtp to Ikana GSR	600	100	10	10	20+818
TRUNK MAIN 02						
1	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	03+005
2	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	04+678
3	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	06+620
4	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	07+440
5	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	08+240
6	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	09+840
7	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	11+190
8	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	12+420
9	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	13+880

10	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	16+080
11	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	17+020
12	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	18+840
13	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	19+870
14	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	21+933
15	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	22+940
16	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	24+269
17	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	25+119
18	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	25+738
19	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	26+461
20	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	29+080
21	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	31+830
22	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	32+748

23	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	35+180
24	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	10	10	36+620
TRUNK MAIN 03						
1	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	25	25	00+300
2	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	04+850
3	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	06+330
4	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	07+740
5	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	08+789
6	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	10+060
7	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	12+190
8	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	13+548
9	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	14+690

10	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	16+259
11	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	18+753
12	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	19+950
TRUNK MAIN 04						
1	Gravity main Tunduma Junction J-004 to existing Chapwa GSR	100	80	10	16	00+135
2	Gravity main Tunduma Junction J-004 to existing Chapwa GSR	100	80	16	16	01+680
3	Gravity main Tunduma Junction J-004 to existing Chapwa GSR	100	80	16	16	02+200
TRUNK MAIN 05						
1	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	80	16	16	00+885
2	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	150	10	10	01+440
3	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	150	10	10	03+015

Table 08: WASHOUT VALVES (WO/SCV)

S/ N	TRUNK MAIN	PIPE DIAMETER (mm)	VALVE DIAMETER (mm)	VALVE PRESSURE UPSTREAM (bar)	VALVE PRESSURE DOWNSTREAM (bar)	CHAIN AGE (Km+m)
TRUNK MAIN 01						
1	Rising main from Momba WTP to Ikana GBR	600	100	40	40	00+420
2	Rising main from Momba WTP to Ikana GBR	600	100	40	40	03+400
3	Rising main from Momba WTP to Ikana GBR	600	100	40	40	03+800
4	Rising main from Momba WTP to Ikana GBR	600	100	30	30	05+960
5	Rising main from Momba WTP to Ikana GBR	600	100	30	30	07+330
6	Rising main from Momba WTP to Ikana GBR	600	100	30	30	08+360
7	Rising main from Momba WTP to Ikana GBR	600	100	25	25	09+552
8	Rising main from Momba WTP to Ikana GBR	600	100	30	30	12+460
9	Rising main from Momba WTP to Ikana GBR	600	100	30	30	15+410
10	Rising main from Momba WTP to Ikana GBR	600	100	25	25	18+870
11	Rising main from Momba WTP to Ikana GBR	600	100	25	25	19+410
12	Rising main from Momba WTP to Ikana GBR	600	100	25	25	20+010
13	Rising main from Momba WTP to Ikana GBR	600	100	10	10	20+870
TRUNK MAIN 02						
1	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	10	10	00+090

2	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	02+070
3	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	03+110
4	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	05+220
5	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	07+788
6	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	09+425
7	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	10+655
8	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	11+560
9	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	13+102
10	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	16	14+150
11	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	16+410
12	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	17+800
13	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	19+560
14	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	21+650
15	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	22+816
16	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	23+790
17	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	24+900
18	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	25+360
19	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	26+160
20	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	27+266
21	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	30+338
22	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	32+232
23	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	33+186

24	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	35+216
25	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	37+620
TRUNK MAIN 04						
1	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	25	25	01+202
2	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	05+585
3	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	06+530
4	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	08+050
5	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	08+789
6	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	10+329
7	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	12+902
8	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	14+440
9	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	14+890
10	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	18+078
11	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	18+962
12	Rising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	10	10	20+760
TRUNK MAIN 07						
1	Gravity main Tunduma Junction J-004 to existing Chapwa GSR	100	80	10	16	01+260
2	Gravity main Tunduma Junction J-004 to existing Chapwa GSR	100	80	16	16	01+945
TRUNK MAIN 08						
1	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	150	10	10	00+720
2	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	150	10	10	01+100

3	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	150	10	10	02+790
1	Rising main Tunduma Junction J-002 to existing Ipito GSR	150	80	16	16	03+210 0

Table 09: Isolation Valves (ISV)

S/N	TRUNK MAIN	PIPE DIAMETER (mm)	VALVE DIAMETER (mm)	VALVE PRESSURE UPSTREAM (bar)	VALVE PRESSURE DOWNSTREAM (bar)	CHAINAGE (Km+m)
TRUNK MAIN 02						
1	Rising main from Momba wtp to Ikana GSR	600	100	40	40	05+000
2	Rising main from Momba wtp to Ikana GSR	600	100	30	30	10+100
3	Rising main from Momba wtp to Ikana GSR	600	100	30	30	15+00
TRUNK MAIN 03						
1	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	05+000
2	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	10+000
3	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	15+000
4	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	20+000
5	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	25+050

6	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	25	25	30+000
7	Gravity Main from Ikana GBR to Nkangamo BPS	600	100	16	16	35+010
TRUNK MAIN 04						
1	Raising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	05+000
2	Raising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	10+000
3	Raising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	15+000
4	Raising Main from Nkangamo BPS to Uhuru Park Tunduma	600	100	16	16	20+000

7.3 Storage Tank Sites

7.3.1 General

The dimensions of the storage tanks shall be selected to fit into the storage tank areas, leaving the space requirements for the phased extensions. However, the water level height in the storage tanks (except at WTP) shall be limited to 5.00 m. The storage tanks shall be equipped with all necessary hydraulic inlet and outlet equipment, mainly installed in valve chambers, adjacent to the reservoirs.

For safety reasons, all storage tanks shall be designed to receive chlorine dosing facilities, to provide safety chlorination of 2 mg/l. Construction and installation of chlorine dosing facilities is optional.

All sites will accommodate:

- The water tank(s)
- Valve chambers and flow meter chambers
- Operational building
- Septic tank

All primary reservoirs sites will accommodate additionally:

- Pole mounted MV switch gear and transformers
- Electrical building with LV switch gear
- Chlorination building including internal drinking water and chlorine motive water pump sets.

Access roads to the reservoir Sites shall be constructed under phase 1. Appropriate pipe, cabling and road infrastructure arrangements inside the reservoir Sites shall be constructed phase 1.

The overflow of the reservoirs shall discharge into overflow discharge channels, with appropriate capacities, rough surface, with cascades, which conveys the water to the next drainage system.

Level measurements as well as level indicators shall be provided at the reservoirs. The overflow protection mechanism (controlled by level sensors) shall be executed redundantly (i.e. two independent level sensors, one ultrasonic and one conductive type level switch in combination with mechanical overflow protection).

Information concerning water levels and flows shall be transferred, via radio to SCADA, to the main control room at the WTP.

7.3.2 Flow meters

To measure the pumped and gravity water flow, magnetic inductive flow meters (for pipe nominal diameters up to ND 400 mm) or ultrasonic flow-meters (for pipe nominal diameters greater than ND 400 mm) shall be installed

- in the transmission mains behind the raw and drinking water pump stations.
- at outflow of each storage tank

Flow meters shall be installed either inside pump stations (where sufficient length is available) or in separate shafts outside the pump stations / reservoirs.

Nominal diameter of the flow meters can be reduced compared to the pipeline nominal diameter. In this case the cost of additional electricity consumption due to higher friction losses over 20 years shall be balanced against lower investment cost for flow meter and shaft structure.

Flow meters do not need by-pass in case an additional FF-piece for each flow meter in the length and of the nominal diameter of the flow meter will be delivered to site.

Flow meter readings shall be transferred to the central SCADA system via PLC / RTU.

7.3.3 Buildings Design Requirements

Details concerning layout, structural properties, room-program, areas, etc. of the various buildings can be obtained from the Conceptual Design. All buildings shall be equipped with water supply, sewerage, storm water drainage, power, lighting, lightning protection, telephone network, air conditioning, and furniture, etc. Furniture and facility list is attached in an annex.

The following buildings shall be designed and constructed within the scope of 97,000m³/d:

- Administration and laboratory building
- Workshop / Store
- Guard houses for ST and WTP
- 2 duplex staff house at WTP

Please note that staff houses shall be ready for use, 6 months before the overall completion date. This will enable the Employer to recruit and settle-in the O&M staff at the site, at an appropriate moment of time, prior to the start of training.

The construction of all buildings and their respective finishes and other room details are given in the following table and they comply with BS 8110, BS 8500, BS 5628, BS 1722, BS 10223, BS 5385, BS 8000, BS 8204, BS 5400, BS 4449, BS 8666, BS 8007, BS 8002, BS 6465, BS EN 12056 and BS 8004 in conjunction with other specifications/standards mentioned General Technical Specification:

7.3.4 Workmanship

The work shall be carried out in accordance with **BS 8102** and **BS 8000**: Particular attention is drawn to the following before installation:

- a) The Works shall be supervised by a competent representative from the Contractor/manufacturer. Inspection shall be carried out and written approval given by this representative for each and every stage of work, from the substructure preparation to the completion of the superstructure. Formal approval for each and every stage of work as mentioned shall be carried out by the Engineer.
- b) The Works shall only be carried out after the concrete has been properly cured for at least 7 days.

- c) Shop drawings shall be produced by the Contractor/manufacturer showing construction details including those at angles, corners, construction joints, pipe intrusions.....etc. for approval and all Works shall be carried out as per the approved shop drawings.

Table 11: Code of building workmanship

BS 8000-1:1989	Workmanship on building sites – Code of practice for excavation and filling
BS 8000-2.1:1990	Workmanship on building sites – Code of practice for concrete work – Mixing and transporting concrete
BS 8000-2.2:1990	Workmanship on building sites – Code of practice for concrete work – Sitework with in situ and precast concrete
BS 8000-3:2001	Workmanship on building sites – Code of practice for masonry
BS 8000-4:1989	Workmanship on building sites – Code of practice for waterproofing
BS 8000-5:1990	Workmanship on building sites – Code of practice for carpentry, joinery and general fixings
BS 8000-6:2013	Workmanship on building sites – Code of practice for slating and tiling of roofs and claddings
BS 8000-7:1990	Workmanship on building sites – Code of practice for glazing
BS 8000-8:1994	Workmanship on building sites – Code of practice for plasterboard partitions and dry linings
BS 8000-9:2003	Workmanship on building sites – Cementitious levelling screeds and wearing screeds – Code of practice
BS 8000-10:1995	Workmanship on building sites – Code of practice for plastering and rendering
BS 8000-11:2011	Workmanship on building sites – Internal and external wall and floor tiling – Ceramic and agglomerated stone tiles, natural stone and terrazzo tiles and slabs, and mosaics – Code of practice
BS 8000-12:1989	Workmanship on building sites - Code of practice for decorative wallcoverings and painting
BS 8000-13:1989	Workmanship on building sites - Code of practice for above ground drainage and sanitary appliances
BS 8000-14:1989	Workmanship on building sites - Code of practice for below ground drainage
BS 8000-15:1990	Workmanship on building sites - Code of practice for hot and cold-water services (domestic scale)
BS 8000-16:1997 +A1:2010	Workmanship on building sites - Code of practice for sealing joints in buildings using sealants

Table 12: Building Requirements and Finishes

Building Type	Plant Rooms
Roofs	Aluminum sheeting
Floors – chlorination building	Screed plus acid resistant tile finish, skirting boards
Floors – pump stations, etc.	Screed plus epoxy floor finish, paint skirting
Floors – electrical distributions	Computer flooring system (double floor)
Floors – kitchens, change rooms, showers, toilets	Screed plus ceramic tile finish, skirting boards
Floors – offices, corridors, other rooms	Screed plus stoneware floor tiles, skirting boards
Floors – Laboratory	Screed plus acid resistant tile finish, skirting boards
Floors – workshops, machine rooms, pump stations, etc.	Screed plus epoxy floor finish, paint skirting
Walls	Block work
Internal wall finish	Plastered and painted
External wall finish	Plastered and painted
Ceilings – general	Plastered and painted
Ceilings – administration, offices, control and welfare rooms	Suspended ceilings
Doors inside admin. building and accommodation building	Hardwood Mninga polished
Doors in process buildings, Pump stations, etc.	Aluminum or steel
Windows – general	Sun exposed windows shall be shadowed by shutters or similar. Good natural light levels during daylight and adequate cross ventilation.
Windows - process buildings, pump stations, etc.	Aluminum, double glazing.
Windows – in buildings with HVAC	Aluminum, double glazing.
All operational rooms and control rooms	HVAC to be provided
Building services	Small power and lighting, Hot and cold-water supplies, sewerage, Rainwater collection and disposal,

Building Type	Plant Rooms
	Internet connection, telephone network, local area computer and telephone network and air conditioning, and furniture.
Buildings for chemical handling	Emergency showers and eyebaths to be installed
All administration, social and control rooms	fully fitted and furnished

7.4 Internal Water Supply

7.4.1 WTP

The drinking/service water pumps/booster station shall be located in the Drinking Water Pump station. The booster station shall provide drinking water to all process and building service points, which shall be designed under phase 1. Allowable water pressure shall be between 3 – 7 bars.

7.4.2 WTP services

The drinking / service water pumps / booster station shall be located in the chlorination building. The booster station shall provide drinking water to all process and building service points, which shall be designed under phase 1. Allowable water pressure shall be 3 bars.

In case the optional Chlorination Building will not be constructed, the drinking / service water network shall be connected at the incoming pressure main, including valve, and PRV (if necessary).

7.5 Hydrant system, firefighting system at WTP

Hydrants will be connected to the main water supply system and shall be sized to cover the firefighting needs in accordance with the local firefighting regulations. It shall be possible to provide firefighting water to all buildings constructed under this contract.

Hydrants shall be provided at appropriate locations with suitable hoses of sufficient length according to Tanzanian legislation. The hoses shall be kept permanently accessible in conspicuous storage units.

Hydrants and hose enclosures shall be painted distinctively red to differentiate them from wash water points. Portable appliances (CO₂ extinguishers, dry powder extinguishers and fire blankets) shall be provided in the various plant areas in accordance with the regulations.

7.6 Protection against Flooding – Storm water drainage

All civil structures shall be built at such a level, or otherwise protected, so that the structures will not be flooded during heavy storms, considering a 100-year return period.

The site shall be graded in such a way as to avoid the occurrence of any flooding from surface water. If required, drainage ditches shall be constructed at areas vulnerable to flooding.

Storm water shall be drained away from roads and buildings, constructed under phase 1, and if possible infiltrated underground. Sealing of surfaces should be avoided as far as possible. Surplus storm water shall not be drained directly into adjacent surface waters, but pass a sedimentation and/or infiltration ditch before entering.

7.7 Wastewater Collection and Treatment Facilities and Waste disposal

Wastewater from the non-process buildings and the process-related buildings containing toilets and wash basins shall be collected by an internal sewage collection system and conveyed by gravity to a two-chamber septic tank system, located at appropriate locations on the sites.

The septic tanks shall be designed for

- 100 population equivalents ("PE") at WTP
- 5 PE at each of the storage tank sites

The overflow of treated wastewater shall be connected to the storm water collection system.

Degraded sludge shall be collected by vacuum tanker and be disposed in a safe manner in accordance with the current valid Tanzanian laws and regulations.

7.8 Perimeter Fencing and another site works

All sites shall be fenced as per BS 1722 and BS 10223 in conjunction with other specifications mentioned in General Technical Specification. The Contractor shall provide suitable, 2.4 m high, permanent security fencing (chain link with coated wire mesh) to all working areas and compounds. Two entrance gates at the WTP site and one gate each at all reservoir sites shall be provided. The residential staff house area shall be fenced separately and receive its own entrance.

Warning signs shall be fixed at suitable intervals along the fence lines to warn the public of the dangers of entering the water treatment plant site.

7.9 Access road during construction

The existing roads and paths shall be improved, to be used during construction as site access.

The Contractor shall construct temporary access roads from the public road to:

- WTP site entrance,
- STs site,

The roads shall partly follow existing roads and their alignments. Principally the roads need to be widened and upgraded to stand the traffic loads, which are expected, due to the operation of the new WTP.

For the STs, the Contractor shall execute the earth works for the permanent access roads (see below) first and pave the access roads temporarily so that they can be used as access roads during construction.

7.10 Construction of the Permanent Access Road

The Contractor shall construct permanent access roads from the public road to:

- WTP site entrance (2 access roads),

The roads shall follow the temporary access roads and alignments.

For the WTP and ST sites upon finalization of Works, the Contractor shall repair any damage to the temporary access roads that may have occurred during construction, remove the temporary pavement and construct the final pavement, ready for taking over.

The roads shall be paved with a combination of hydraulically bound base and top courses, and the width shall be 6.00 m (for the WTP), respectively min. 4.00 m (for ST sites), road gradients shall not exceed 18%.

The road shall be designed for 20-ton trucks (10 t axle) for heavy goods transportation and medium design load (for the WTP), respectively 10-ton trucks (for the ST sites), all in accordance with valid technical standards and regulations. The road shall be designed with cross falls and/or cambers to ensure no standing water on the road. All necessary drainage shall be provided.

Technical requirements follow the relevant sections of this document and the Specifications. Local regulations and other related by-law requirements shall be taken into consideration for the detailed design of the access roads. Contractor design shall comply with Pavement and Materials Design Manual 1999, Laboratory Testing manual 2000, Road Geometry Design Manual 2011 edition and Low volume roads manual 2016 both issued by TANROAD in conjunction with AASHTO specifications.

In the Schedule of Prices quantities for access road earthworks and pavement have been listed. The information is not warranted by the Employer.

7.11 Internal site roads, walkways and car parking

The Contractor shall provide all-weather, paved, internal site roads, to give easy access to all facilities and operational areas of the WTP and the reservoir sites. The scope shall be:

- At WTP site: road construction to access all facilities, constructed under this contract.
- At all other ST sites: road construction to provide access to the storage tanks sites.

The minimum width of the site roads at the WTP shall be 6.00 m designed for 20-ton trucks (10 t axle) for heavy goods transportation and for medium design load. Bends shall be wide enough to allow a double container trailer/truck (80 ft length) to easily maneuver around while delivering consumables to the Plant.

The minimum width of the site roads at the primary reservoir sites shall be 4.00 m, designed for 10-ton trucks.

Hard-standings shall be provided at all the key operational areas, to facilitate personnel and vehicle access for operational and maintenance purposes. All access roads and hard standings shall be designed to prevent standing water and all necessary drainage shall be provided.

The roadway and clearances between and into buildings shall allow a clear and safe approach and maneuvering for all types and size of operational vehicle required for the operation and maintenance of the Works. Vehicle access provisions

shall be provided at all locations where required as part of the Works to facilitate personnel and vehicle access for operational and maintenance purposes.

Where appropriate, ramps shall be provided at equipment access doors to enable a vehicle to be driven into the building. With the exception of access ramps, road gradients shall not exceed 18%.

Turning areas shall permit a vehicle to maneuver into the final discharge or pick-up position. Surface markings shall be provided to identify areas allocated as turning areas and parking/unloading bays.

A minimum of 10 car parking spaces shall be provided at the WTP site and 5 car parking spaces shall be provided at ST sites.

Hard paved, lighted pedestrian walkways 1 meter wide shall be provided to all access points to tanks, chambers and building entrances unless these are beside road ways. The paved routes shall follow logical lines likely to be used by the maintenance workers.

7.12 Landscaping, Green Belt Setting and Area Modelling

Open areas within the site are to be covered with greenery (grass, bushes, trees). Suitable trees and shrubs shall be planted to provide an excellent visual impact of the plant to the general public.

Alongside the chain-link fence a life fence shall be planted, which improves both, the security of the fence and its visual appearance. Plants shall be selected that can cope with the regional climate and need low maintenance.

The Contractor shall design and construct landscaping works to integrate the Works into the surrounding environment. Consideration shall be given to the use of shallow contoured grassy mounds and the provision of local trees and shrubs to mitigate the visual impact of structures.

Where it would be impractical to maintain grass due to shape, slope etc., gravel beds shall be provided. Gravel shall comprise uniform, clean, rounded pebbles (minimum diameter 12 mm).

7.13 Contractor's Personnel

The Contractor shall employ under the Contract only experienced designers, who have profound knowledge in the different fields of engineering required by the task, who are familiar with all aspects of WTP works and be familiar with FIDIC, P&DB (Plant and Design Built).

In particular the Authorized Representative, the Project Manager, the Civil Engineer - site manager, Mechanical Engineer - site manager and Electrical Engineer - site manager shall be familiar with the construction of water treatment facilities, pump stations, pumping mains and pipelines. Good knowledge of modern SCADA and control systems is an asset.

If the Contractor intends to split a specific position, all involved staff shall have comparable experience records as shown in the Table.

Table 13: Staff Qualification

Position	Similar Projects	General Experience in this area of works (years)
1. Authorized Representative	5	20
2. Project Manager	5	20
3. Environmental and Social Safeguards Expert	3	10
4. Health and Safety Manager	3	10
Key Positions Design		
5. Process Engineer Design	4	15
6. Hydraulic Engineer Design	4	15
7. Pipeline Engineer Design	4	15
8. Mechanical Engineer Design	3	15
9. Electrical Engineer, Automation/SCADA Engineer Design	4	15
10. Structural Engineer Design	3	15
Key Positions Construction		
11. Civil Engineer - site manager	3	15
12. Pipeline Engineer – site manager	3	15
13. Mechanical Engineer - site manager	3	15
14. Electrical Engineer Automation/SCADA Engineer - site manager	3	15
Key Positions Operation / Supervision of commissioning initial test Operation Period / Training		
15. Expert in charge for training	3	15
16. Operation Manager during commissioning period	3	15

Change of key personnel shall only be possible upon written approval by the Engineer.

7.14 ESIA Requirements

7.14.1 Specifications for Environmental, Social, Health and Safety

The general Environmental, Social, Health and Safety Requirements have been set-out in detail in the ESHS Specifications, which shall be followed by the Contractor during all implementation phases.

7.14.2 Implementation of Mitigation Measures

An Environmental and Social Impact Assessment was carried out. This ESIA study, together with the mitigation measures therein shall be reviewed, in order to ensure that they will be implemented during the construction works.

The Results of these investigations are shown in the Environmental and Social Impact Assessment Report (ESIA), which forms part of these Tender Documents. The Contractor is requested to implement the Mitigation measures, identified therein. An item in the general Schedule of Prices has been provided and shall be priced.

7.14.3 ESHS-AP and ESMP

Using the ESIA provided in these Tender Documents, the Contractor is required to prepare and submit an ESHS-AP and an ESMP, in accordance with the ESHS Specifications, which shall be monitored during the execution of the works.

These Plans are required assets, respectively by the Authorities of Tanzania and the Donor Agencies and have to fulfil the respective requirements. The plans have many similarities in between themselves and shall be developed together and harmonized before submission to the Engineer.

The Environmental and Social Impact Assessment Study, with its mitigation measures, and the ESHS Specifications have to be read in conjunction and applied.

7.14.4 Assistance during resettlement and compensations for land owners

In addition, it will be necessary that the Employer provides fund to compensate land owners and pay for right of ways.

7.15 O&M – vehicles, spares, workshop, laboratory

7.15.1 Operation vehicles

The Contractor shall supply the following new vehicles for operation of the Water Treatment Plant and the other facilities under this Contract. The vehicles shall not be used by the Contractor during the execution of the construction works.

All trucks, vehicles, and special vehicles necessary for the sound operation of the plant, at least but not limited to:

- Front loader - (e.g. Caterpillar 930G, 110 kW, 13 t, or similar) with bucket and loader rake;
- Forklift – (e.g. Komatsu, AX 50 or equivalent), 1.5 t lifting capacity, fuel, 50 kW, lifting height 2.5 m;
- 1 Light truck (5 ton, 220 HP), with on-board crane,
- 2 Minibus, 9 +1 passengers, engine size 140 HP.

- 2 Pick-up trucks – (e.g. Toyota Hilux or similar), 4+1 passengers, Minimum engine size 2400 cc, diesel, 150 HP, 4X4, AC, radio/CD.
- 1 Sedan Car middle class - (e.g. Toyota Corolla or similar), 4+1 passengers, engine size 130 HP.
- 4 Motorcycles, Honda XR 150 or equivalent, Cubic capacity = 150 cc.

Consumable spares (oil filters, brake pads etc.) shall be supplied for the operation of these vehicles as specified in the schedule of spare parts. All vehicles shall be covered by a three-year guarantee.

7.15.2 Spare parts

The supply of spares shall cover the first two years of normal operation, after the commissioning period and Taking Over. The following spare part areas shall be covered:

- Mechanical Spare parts
- Electrical Spare parts
- Instrumentation, Control and Automation (ICA) Spare parts
- Building Services Spare parts

As a minimum the following spare parts shall be supplied:

- For each offered pump and blower (operating and spare): 2 sets of sealing, 2 sets of bearing spare parts according to the recommendations of the manufacturer (mechanical seals, gland packings, O-rings, V-rings, gaskets, lantern rings, stuffing box rings, shaft sleeve, shaft protection sleeve, etc.)
- 20% of the respective number of built-in LV fuses of each size that is required for the plants (at least 4 pieces per size),
- 10% of the respective number of built-in indicator lights of each color that is required for the plants (at least 10 pieces per color),
- 10% of the respective number of built-in coupling relays, auxiliary relays and isolating amplifiers of each type that is required for the plants (at least 5 pieces per type),
- 10% of the respective number of built-in overvoltage protection units of each type and size that is required for the plants (at least 4 pieces per type),
- 10% of the respective number of built-in miniature circuit breakers of each type that is required for the plants (at least 4 pieces per type),
- 1 complete set of spare parts for each diesel generating set, including all belts, sealings, hoses, flexible connections, bearings, fuel and lubricant filters, rotary diodes for alternator, etc.,
- 20% of the respective number of built-in bulbs of each size and type that is required for the plants (at least 5 pieces per size),
- 50% of the respective number of built-in non-rechargeable batteries of each type and size that is required for the plants,

- For each offered different type of PLC: 1 piece of each installed PLC module, such as CPU, I/O-modules of each installed type, communication-module, interface module, power supply module, etc.

7.15.3 Workshop equipment and furniture

Equipment and furniture for the new Workshop shall be provided according to the list in the Appendixes to these Employer's Requirements.

7.15.4 Laboratory Equipment

Equipment and furniture for the new laboratory shall be provided according to the list in the Appendixes to these Employer's Requirements.

7.15.5 Operation tools and test equipment

The Tenderer shall supply operation tools and test equipment, etc. necessary for the operation and maintenance of the specific WTP and other facilities, proposed by him, under the project. Prices for these items shall be listed in the Schedules of Prices.

7.16 Electrical System and Control System

7.16.1 Power supply to Sites by TANESCO

Tanzania Electric Supply Company, TANESCO will construct the 11 kV dedicated overhead lines from the nearest existing power source (substation) to all Sites. All sites, as listed below, shall be supplied by its dedicated 11 KV OHTL:

- Intake and WTP plus Pumping Station

The tapping point to be determined from the proposed WTP power house/substation.

- Storage tanks.

Storage tanks site will be supplied from the local electrical distribution network. Generally, a 11kV HT, a tapping point will be determined during work construction from the site.

Capacity of each of the TANESCO's feeders will be for maximum capacity of the WTP, calculated by the Works Contractor during his final design. The scope of supply shall include power cables and other terminations accessories.

These lines are for final process purposes only (not for the construction purposes) and will be ready shortly before Taking Over.

It has to be noted that according with TANESCO standards maximum voltage variation along changing momentary load at the end of line is (plus/minus) + 10%.

7.17 Scope of works on electrical systems

7.17.1 General

The Contract shall include the design, manufacture, testing at manufacturers' works, delivery to Site, off-loading, storage, erection, testing and setting to work a complete Electrical installation system at all proposed Sites in the Water Supply Project. Since the construction of the facilities and equipment installation is

foreseen to be in two phases (2025 / 2035), the design of the power supply infrastructure shall consider space demand for 20,000m³/d for Sub-stations, as well as for any electrical building / room that will house 97,000m³/d electrical equipment, however, will be constructed under current contract package.

The scope of Electrical works to be covered in this contract includes, but is not limited to, the following main items:

- A complete design of all electrical power connection / power transformation and distribution system (within the Sites)
- Delivery and installation of
 - o Provisional and temporary power supply to all construction sites from start of contract till issue of Taking Over Certificate.
 - o Medium voltage switchgear, step-down transformers, metering system, power supply cables for both medium and low voltage for each site.
 - o Complete electrical LV installations including UPSs in transformer buildings, distribution panels, VFDs, Electrical Panels-MCC's and control panels, cable trays and building installations.
 - o Complete indoor and outdoor lighting systems, indoor sockets in sufficient number (if not stated anywhere in the Contract), and outdoor socket outlets stations.
 - o Lightning protection / overvoltage systems and installation inclusive protective earthing and equipotential bonding.
 - o Automatic reactive power compensation systems.
 - o Emergency lighting systems in specific rooms.
 - o Diesel generator sets.
 - o Supply of spare parts and special tools.
 - o Elaboration of as-built documentation.
 - o Rendering of training services for the Employer's staff.
- All approvals (design, connections, earthing etc.) and negotiations with local authorities (electricity supply company).

Further details of scope of works are listed in the Electrical Specifications.

7.17.2 Power Connection Point

The Contractor shall indicate the location of the last Tanesco 33 kV pole in his detailed design drawings for each plot. The Contractor shall then continue design and installation from the last Tanesco pole as mentioned above to the MV/transformer station inside the plot.

7.18 Electrical System and Control System.

7.18.1 Power supply to Sites by TANESCO.

Tanzania Electric Supply Company, TANESCO will construct the 33 kV dedicated overhead lines from the nearest existing power source (substation) to all Sites. All sites, as listed below, shall be supplied by its dedicated 33 KV OHTL:

- **Momba Intake**

The tapping point is around 1.2Km from Proposed Momba INTAKE.

- **Momba WTP**

The tapping point is around 1.5Km from Proposed Momba WTP.

- **Ikana 3000m³GBR**

The tapping point is around 900m from Proposed Ikana GBR.

- **Nkangamo BPS**

The tapping point is around 200m from Proposed Nkangamo BPS.

- **Uhuru Park Tunduma 5000m³ GBR**

The tapping point is around 100m from Proposed Uhuru Park GBR.

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Capacity of each of the TANESCO's feeders will be for maximum capacity of the WTP, calculated by the Works Contractor during his final design. The scope of supply shall include power cables and other terminations accessories.

These lines are for final process purposes only (not for the construction purposes) and will be ready shortly before Taking Over.

It has to be noted that according with TANESCO standards maximum voltage variation along changing momentary load at the end of line is (plus/minus) + 10%.

7.19 MOMBA WTP

7.19.1 Middle and Low Voltage Electrical Equipment (33/3.3/0.4 kV)

33/3.3 kV Substation and MV switchgear

A 33/3.3 kV substation shall be constructed at Water Treatment Plant, to feed all pumps.

3.3/0.4 kV Substation and MV switchgear

A 3.3/0.4 kV substation shall be constructed at Water Treatment Plant. The Substations at WTP shall consist of:

- MV Switchgear
- 2 Nos Power Transformers 33/3.3 kV.
- 2 Nos Power Transformers 3.3/0.4 kV.
- Diesel generators
- bursars with bus-coupling switches

- MV MSB (Main Distribution Board)
- LV MDB (Main Distribution Board)

33KV MV Switchgear at WTP

The MV Switchgear shall comply with the required Electrical Specifications and shall consist of:

- Supply and installation of new 33 kV Medium Voltage Switch Gear with one incomer from 33KV supply and two outgoings 33/3.3KV transformers.
- Medium switchgear equipped with circuit breakers with digital protection, remote control and monitoring relay. The relay shall provide protection for but not limited to over-load, single earth fault relay, phase to ground relay and thermal protection.
- The switchgears and their components thereof shall be capable of withstanding the mechanical forces and thermal stresses of the short circuit current of the system without any damage or deterioration of materials.

7.19.2 Power Transformers at WTP

Supply and installation of two 2000kVA 33/3.3 kV transformers; both of them (1W+1S) running parallel during operation.

Supply and installation of two 400kVA 3.3/0.4kV transformers to be used to supply power to Low Voltage equipment.

All power transformers to be designed shall be fully protected for both internal, overload and external cause of faults.

The transformers shall be suitable rated for continuous max. load of the connected consumers under worst climatic site conditions. The transformers shall have capacity to cover 100% of the power. The total consumption shall be spread to 50% on each transformer. The transformers shall work in parallel one duty and other is standby. All transformers shall be from the same manufacturer.

Transformer shall comply with the required Electrical Specifications.

7.19.3 Diesel Generators

The Contractor shall provide generators 3.3KV to be used in case of power supply failure. The generators will be sized to make the WTP, including the treated water pumping station (ONLY two pumps to Ikana 3,000m³ GBR reservoirs, work simultaneously at their maximum capacity)

The generators will start automatically (use of auto transfer switch, ATS) in case of power failure.

Table 14: ATS items

Critical consumer
ONLY Two Pumps.
Valve actuators in filters
Drainage pumps in basements and pump stations
All process instruments which are not backed-up by UPS or battery systems
Indoor and outdoor lighting, indoor power sockets, all other house distribution consumers including ventilators and air conditioners, etc.
Firefighting alarming system
Emergency lights
UPS for PLCs, SCADA rack mounted Professional Servers, and SCADA client work stations
UPS for data transfer facilities.

During stand-by power supply, output voltage of generator must satisfy specified allowable limits (RMS, frequency and distortion) for proper work of connected consumers. The control panel of the generator-set must be able to allow manual and remote start as well.

The diesel generators shall be connected to the substation with Automatic Transfer Switch (ATS). The generators will start automatically (use of auto transfer switch, ATS) in case of power failure.

Diesel Generators shall comply with the required Electrical Specifications.

7.19.4 3.3KV MV Switchgear at WTP

Supply and installation of new 4KV cabinet for 3.3KV with 2 incomers, bus section and 6 outgoing.

The MV MSB mainly consist of the following but not to limit:

- Two incomers from 33/3.3 KV transformers
- One incomer from 3.3KV Generator
- Voltage reference (indication of volt and ampere)
- Surge arrester
- Bus Coupler
- Bus Riser
- Two outgoing to 400KVA 3.3/0.4KV Auxiliaries Transformers
- Energy meter (kWh meter with impulse output.
- Power factor correction connection point.
- 4 outgoing to motor pump VFDs

Supply and installation of Set of 4 VFDs (3.3KV designed with power factor of 0.98), not integrated.

7.19.5 Low Voltage Electrical Equipment

Supply and installation of LV panel which will be used for low voltage installations.

The MV MSB mainly consist of the following but not to limit:

- Two incomers from 3.3/0.4 KV transformers
- Voltage reference (indication of volt and ampere)
- Surge arrester
- Circuit breaker with thermic, magnetic release with motor driver in withdrawable type. All circuit breakers for low voltage switchgear rated above 800A shall be Air circuit breakers (ACB) otherwise shall be Moulded Case Circuit breakers (MCCB)
- Energy meter (kWh meter with impulse output.
- Current transformer for indication and compensation including transmitter 4....20 mA
- Power factor correction connection point.
- 4 outgoing feeder to MCCs
 - i. MCC1- Water treatment process (coagulation, flocculation, clarifiers and filters)
 - ii. MCC2- Sludge treatment processes
 - iii. MCC3- Chemical building
 - iv. MCC4- Treated water pumping station (compressors, service water pumps, crane etc)
- Outgoings feeds for tertiary (administration building, guard house, workshop and staff houses).
- Feeders for PLCs, emergency light, fire system, CCTV camera, telecom cabinets and for UPS etc.
- 20% spare for future extension. each equipped with (the Design report has foreseen 12 feeders):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
 - Spare feeders 20%

In case of a malfunction / breakdown in the substation critical consumers shall be supplied by a diesel generator, which is connected to the substation with Automatic Transfer Switch (ATS):

- Six feeders for PLC, emergency light, fire system, CTV, telecom cabinets each equipped with:
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the water treatment plant including spare feeders.
 - Spare feeders 20%

7.19.6 In-Buildings and external electrical installations

Following in-building and external installation in the all locations shall be foreseen:

- Internal main and emergency lighting
- Floodlighting
- Socket outlets
- Cabling
- Lightning protection
- Earthing

LV equipment shall comply with the required Electrical Specifications.

7.19.7 MCC1 – Water treatment process (coagulation, flocculation, clarifiers and filters)

MCC Panel shall have 20% spare for the future use after installation is completed and when handing over.

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC1 Room shall be located attached to the Rapid Sand Filter Unit. MCC1 shall mainly consist of:

- Income feeder from the LV MSB with:
 - Surge arrester.

- Circuit breaker with thermic-magnetic release in withdrawable type.
- Three current transformers with transmitter 4...20 mA.
- Three Amp. meter, one voltmeter with selector switch.
- One Voltage monitoring device $<U>$, $<f>$.
- One control transformer.
- One network analyzer.
- One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.
- Motor feeders based to 0.4 kV Variable Frequency Drives for blower motors and pumps each equipped with:
 - Current transformers for indication and protection including transmitter 4...20 mA (3 phase).
 - Multifunction protection relay for, phase-sequence, under voltage over voltage, ground detection, instantaneous over current, temperature, supervision, under/over power protection, time over current protection, etc.
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - 0.4 kV Variable Frequency Drives
 - VFD units of <200 kW shall be installed inside of MCC Enclosure, VFD of ≥ 200 kW shall be installed free standing, with its own cubicle. Cos phi compensation (to 0.95) shall primarily be accomplished by the VFD units
 - Indication of Amp and running, fault, individual faults by the display of the protection relay.
 - With all necessary accessories.
 - All open/close/failure/start/stop indications and measuring values must be transferred to the next PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Complete cabling installation material and connections.
- Motor feeders for 0.4 kV for valve actuators each equipped with:
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.

- Phase failure, blocking protection, current value protection
- Failure indication, running hour meter, start counter, Amp meter.
- The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
- The motor feeders for Valve Actuators shall have two DOL starters one for OP (Opening) and one for CL (Closing) with all additional control relays. If the Valve actuator come with build in motor starters than in this MCC panel need to be install only the power feeder describe below as feeder for auxiliary equipment.
- o feeders for all the auxiliary equipment (including LDB for inside lighting and FCP (filter Control panels) and PLC Panel and etc.) each equipped with (the Design report has foreseen 6 motor feeders)):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
- o Spare feeders 20%

7.19.8 MCC2 – Chemical dosing and store

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC2 shall be inside Chemical building. MCC2 shall mainly consist of:

- o Income feeder from the MDB with:
 - Surge arrester.
 - Circuit breaker with thermic-magnetic release in withdrawable type.
 - Three current transformers with transmitter 4...20 mA.
 - Three Amp. meter, one voltmeter with selector switch.
 - One Voltage monitoring device <U>, <f>.
 - One control transformer.
 - One network analyzer.
 - One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.

- Motor feeders for 0.4 kV pumps each equipped with:
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
 - Phase failure, blocking protection, current value protection
 - Failure indication, running hour meter, start counter, Amp meter.
 - The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
 - The motor feeders for motors >than 7.5kW shall be equipped with SS (Soft Starters) and the motor feeders for motors ≤7.5kW shall be equipped with DOL (Direct Online starters)
- Feeders for the auxiliary equipment each equipped with (the design report has foreseen 2 feeders):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
- Spare feeders 20%

7.19.9 MCC3 - Sludge Treatment processes

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC3 Room shall be built in a small house near the sludge sedimentation tank. MCC3 shall mainly consist of:

- Income feeder from the LV MSB with:
 - Surge arrester.
 - Circuit breaker with thermic-magnetic release in withdrawable type.

- Three current transformers with transmitter 4...20 mA.
- Three Amp. meter, one voltmeter with selector switch.
- One Voltage monitoring device $\langle U \rangle$, $\langle f \rangle$.
- One control transformer.
- One network analyzer.
- One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.
- Motor feeders for 0.4 kV motors each equipped with (the Design Report has foreseen 6 feeders):
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
 - Phase failure, blocking protection, current value protection
 - Failure indication, running hour meter, start counter, Amp meter.
 - The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
 - The motor feeders for motors >than 7.5kW shall be equipped with SS (Soft Starters) and the motor feeders for motors ≤ 7.5 kW shall be equipped with DOL (Direct Online starters)
 - The motor feeders for Valve Actuators shall have two DOL starters one for OP (Opening) and one for CL (Closing) with all additional control relays. If the Valve actuator come with build in motor starters than in this MCC panel need to be install only the motor feeder describe below as motor feeder for auxiliary equipment.
- Feeders for the auxiliary equipment each equipped with (the Design report has foreseen 7 feeders):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.

- Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
- o Spare feeders 20%

7.19.10 MCC4 – Treated water pumping station (Surge vessel air compressors, service water pumps, crane etc.)

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC4 shall supply power all auxiliaries of the treated water PS. but not to limit

- Surge vessel air compressors units
- Overhead electric crane
- Service water pumps
- Submersible drainage pumps
- Indoor and outdoor lightings
- Power sockets outlets
- PLC, CCTV Camera, emergency light and firefighting systems

MCC4 shall be a inside the Drinking Water Pump Station. MCC4 shall mainly consist of:

- o Income feeder from the LV MSB with:
 - Surge arrester.
 - Circuit breaker with thermic-magnetic release in withdrawable type.
 - Three current transformers with transmitter 4...20 mA.
 - Three Amp. meter, one voltmeter with selector switch.
 - One Voltage monitoring device $<U>$, $<f>$.
 - One control transformer.
 - One network analyzer.
 - One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.
- o Motor feeders based to 0.4 kV Variable Frequency Drives for different pumps each equipped with (the design report has foreseen 5 motor feeders):
 - Current transformers for indication and protection including transmitter 4 ...20 mA (3 phase).
 - Multifunction protection relay for, phase-sequence, under voltage over voltage, ground detection, instantaneous over current, temperature, supervision, under/over power protection, time over current protection, etc.
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - 0.4 kV Variable Frequency Drives
 - VFD units of <200 kW shall be installed inside of MCC Enclosure, VFD of ≥ 200 kW shall be installed free standing, with its own cubicle. Cos phi compensation (to 0.95) shall primarily be accomplished by the VFD units

- Indication of Amp and running, fault, individual faults by the display of the protection relay.
- Running hours meter, start counter.
- A cos phi compensated to 0.95 primarily by VFDs.
- With all necessary accessories.
- All open/close/failure/start/stop indications and measuring values must be transferred to the next PLC.
- Controlling and interlocking from the low voltage switchgear with respectively PLC.
- Complete cabling installation material and connections.
- Motor feeders for 0.4 kV motors each equipped with (the Design Report has foreseen 6 feeders):
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
 - Phase failure, blocking protection, current value protection
 - Failure indication, running hour meter, start counter, Amp meter.
 - The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
 - The motor feeders for motors >than 7.5kW shall be equipped with SS (Soft Starters) and the motor feeders for motors ≤7.5kW shall be equipped with DOL (Direct Online starters)
 - The motor feeders for Valve Actuators shall have two DOL starters one for OP (Opening) and one for CL (Closing) with all additional control relays. If the Valve actuator come with build in motor starters than in this MCC panel need to be install only the motor feeder describe below as motor feeder for auxiliary equipment.
- Feeders for the auxiliary equipment each equipped with (the Design report has foreseen 7 feeders):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.

- Controlling and interlocking from the low voltage switchgear with respectively PLC.
- Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.

7.19.11 Signage and Labelling

- Safety Signage

All mandatory, caution, prohibition, danger, security and safety signage (e.g., confined space entry)

- Labelling
 - o All equipment shall have the identification number affixed to the equipment to facilitate easy identification for operation and maintenance purposes
 - o All instruments shall be identified with tags mounted adjacent to but not directly on the instrument.
 - o All tags and labels shall be of stainless-steel construction with engraved numbering and lettering
 - o Tags and labels shall be fixed in a prominent location by use of stainless-steel pins.
 - o Piping Identification

Above ground and in-pit pipework shall be provided with identification markers indicating contents and direction of flow

7.20 MOMBA INTAKE

7.20.1 Middle and Low Voltage Electrical Equipment (33/3.3/0.4 kV)

33/0.4 kV Substation and MV switchgear

A 33/0.4 kV substation shall be constructed at Momba Intake, to feed all pumps.

3.3/0.4 kV Substation and MV switchgear

A 33/0.4 kV substation shall be constructed at Water Treatment Plant. The Substations at WTP shall consist of:

- MV Switchgear
- 2 Nos Power Transformers 33/0.4 kV.
- Diesel generators
- bursars with bus-coupling switches
- MV MSB (Main Distribution Board)
- LV MDB (Main Distribution Board)

7.20.2 33KV MV Switchgear at Momba Intake

The MV Switchgear shall comply with the required Electrical Specifications and shall consist of:

- Supply and installation of new 33 kV Medium Voltage Switch Gear with one incomer from 33KV supply and two outgoings 33/0.4KV transformers.
- Medium switchgear equipped with circuit breakers with digital protection, remote control and monitoring relay. The relay shall provide protection for but not limited to over-load, single earth fault relay, phase to ground relay and thermal protection.
- The switchgears and their components thereof shall be capable of withstanding the mechanical forces and thermal stresses of the short circuit current of the system without any damage or deterioration of materials.

7.20.3 Power Transformers at Momba Intake

Supply and installation of two 200kVA 33/0.4 kV transformers; both of them (1W+1S) running parallel during operation.

All power transformers to be designed shall be fully protected for both internal, overload and external cause of faults.

The transformers shall be suitable rated for continuous max. load of the connected consumers under worst climatic site conditions. The transformers shall have capacity to cover 100% of the power. The total consumption shall be spread to 50% on each transformer. The transformers shall work in parallel one duty and other is standby. All transformers shall be from the same manufacturer.

Transformer shall comply with the required Electrical Specifications.

7.20.4 Diesel Generators

The Contractor shall provide generators 0.4KV to be used in case of power supply failure. The generators will be sized to make the Momba Intake, including the Intake pumping station (All Low and High Lift Duty Pumps to Momba WTP, work simultaneously at their maximum capacity)

The generators will start automatically (use of auto transfer switch, ATS) in case of power failure.

Table 15: ATS items

Critical consumer
All Duty Pumps.
Valve actuators in filters
Drainage pumps in basements and pump stations
All process instruments which are not backed-up by UPS or battery systems

Critical consumer
Indoor and outdoor lighting, indoor power sockets, all other house distribution consumers including ventilators and air conditioners, etc.
Firefighting alarming system
Emergency lights
UPS for PLCs, SCADA rack mounted Professional Servers, and SCADA client work stations
UPS for data transfer facilities.

During stand-by power supply, output voltage of generator must satisfy specified allowable limits (RMS, frequency and distortion) for proper work of connected consumers. The control panel of the generator-set must be able to allow manual and remote start as well.

The diesel generators shall be connected to the substation with Automatic Transfer Switch (ATS). The generators will start automatically (use of auto transfer switch, ATS) in case of power failure.

Diesel Generators shall comply with the required Electrical Specifications.

7.20.5 Low Voltage Electrical Equipment

Supply and installation of LV panel which will be used for low voltage installations.

The MV MSB mainly consist of the following but not to limit:

- Two incomers from 33/0.4 KV transformers
- Voltage reference (indication of volt and ampere)
- Surge arrester
- Circuit breaker with thermic, magnetic release with motor driver in withdrawable type. All circuit breakers for low voltage switchgear rated above 800A shall be Air circuit breakers (ACB) otherwise shall be Moulded Case Circuit breakers (MCCB)
- Energy meter (kWh meter with impulse output.
- Current transformer for indication and compensation including transmitter 4....20 mA
- Power factor correction connection point.
- 2 outgoing feeder to MCCs
 - v. MCC1- Low Lift Pumping Station
 - vi. MCC2- High Lift Pumping Station
- Outgoings feeds for tertiary (Operators building, guard house, workshop and Power House/Substation).
- Feeders for PLCs, emergency light, fire system, CCTV camera, telecom cabinets and for UPS etc.

- 20% spare for future extension. each equipped with (the Design report has foreseen 12 feeders):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
 - Spare feeders 20%

In case of a malfunction / breakdown in the substation critical consumers shall be supplied by a diesel generator, which is connected to the substation with Automatic Transfer Switch (ATS):

- Six feeders for PLC, emergency light, fire system, CTV, telecom cabinets each equipped with:
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the water treatment plant including spare feeders.
 - Spare feeders 20%

7.20.6 In-Buildings and external electrical installations

Following in-building and external installation in the all locations shall be foreseen:

- Internal main and emergency lighting
- Floodlighting
- Socket outlets
- Cabling
- Lightning protection

- Earthing

LV equipment shall comply with the required Electrical Specifications.

7.20.7 MCC1 – Low Lift Pumping Station

MCC Panel shall have 20% spare for the future use after installation is completed and when handing over.

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC1 Room shall be located attached to the Rapid Sand Filter Unit. MCC1 shall mainly consist of:

- o Income feeder from the LV MSB with:
 - Surge arrester.
 - Circuit breaker with thermic-magnetic release in withdrawable type.
 - Three current transformers with transmitter 4...20 mA.
 - Three Amp. meter, one voltmeter with selector switch.
 - One Voltage monitoring device $<U>$, $<f>$.
 - One control transformer.
 - One network analyzer.
 - One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.
- o Motor feeders based to 0.4 kV Variable Frequency Drives for blower motors and pumps each equipped with:
 - Current transformers for indication and protection including transmitter 4...20 mA (3 phase).
 - Multifunction protection relay for, phase-sequence, under voltage over voltage, ground detection, instantaneous over current, temperature, supervision, under/over power protection, time over current protection, etc.
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - 0.4 kV Variable Frequency Drives
 - VFD units of <200 kW shall be installed inside of MCC Enclosure, VFD of ≥ 200 kW shall be installed free standing, with its own cubicle. Cos phi compensation (to 0.95) shall primarily be accomplished by the VFD units
 - Indication of Amp and running, fault, individual faults by the display of the protection relay.
 - With all necessary accessories.
 - All open/close/failure/start/stop indications and measuring values must be transferred to the next PLC.

- Controlling and interlocking from the low voltage switchgear with respectively PLC.
- Complete cabling installation material and connections.
- Motor feeders for 0.4 kV for valve actuators each equipped with:
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
 - Phase failure, blocking protection, current value protection
 - Failure indication, running hour meter, start counter, Amp meter.
 - The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
 - The motor feeders for Valve Actuators shall have two DOL starters one for OP (Opening) and one for CL (Closing) with all additional control relays. If the Valve actuator come with build in motor starters than in this MCC panel need to be install only the power feeder describe below as feeder for auxiliary equipment.
- feeders for all the auxiliary equipment (including LDB for inside lighting and FCP (filter Control panels) and PLC Panel and etc.) each equipped with (the Design report has foreseen 6 motor feeders)):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
- Spare feeders 20%

7.20.8 MCC2 – High Lift Pumping Station

MCC Panel shell have 20% spare for the future use after installation is completed and when handing over.

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC1 Room shall be located attached to the Rapid Sand Filter Unit. MCC1 shall mainly consist of:

- Income feeder from the LV MSB with:
 - Surge arrester.
 - Circuit breaker with thermic-magnetic release in withdrawable type.
 - Three current transformers with transmitter 4...20 mA.
 - Three Amp. meter, one voltmeter with selector switch.
 - One Voltage monitoring device $<U>$, $<f>$.
 - One control transformer.
 - One network analyzer.
 - One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.
- Motor feeders based to 0.4 kV Variable Frequency Drives for blower motors and pumps each equipped with:
 - Current transformers for indication and protection including transmitter 4...20 mA (3 phase).
 - Multifunction protection relay for, phase-sequence, under voltage over voltage, ground detection, instantaneous over current, temperature, supervision, under/over power protection, time over current protection, etc.
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - 0.4 kV Variable Frequency Drives
 - VFD units of <200 kW shall be installed inside of MCC Enclosure, VFD of ≥ 200 kW shall be installed free standing, with its own cubicle. Cos phi compensation (to 0.95) shall primarily be accomplished by the VFD units
 - Indication of Amp and running, fault, individual faults by the display of the protection relay.
 - With all necessary accessories.
 - All open/close/failure/start/stop indications and measuring values must be transferred to the next PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Complete cabling installation material and connections.
- Motor feeders for 0.4 kV for valve actuators each equipped with:

-
- Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
 - Phase failure, blocking protection, current value protection
 - Failure indication, running hour meter, start counter, Amp meter.
 - The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
 - The motor feeders for Valve Actuators shall have two DOL starters one for OP (Opening) and one for CL (Closing) with all additional control relays. If the Valve actuator come with build in motor starters than in this MCC panel need to be install only the power feeder describe below as feeder for auxiliary equipment.
 - feeders for all the auxiliary equipment (including LDB for inside lighting and FCP (filter Control panels) and PLC Panel and etc.) each equipped with (the Design report has foreseen 6 motor feeders)):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
 - Spare feeders 20%

7.21 NKANGAMO BPS

7.21.1 Middle and Low Voltage Electrical Equipment (33/3.3/0.4 kV)

33/3.3 kV Substation and MV switchgear

A 33/3.3 kV substation shall be constructed at Nkangamo BPS, to feed all pumps.

3.3/0.4 kV Substation and MV switchgear

A 3.3/0.4 kV substation shall be constructed at Nkangamo BPS. The Substations at Nkangamo BPS shall consist of:

- MV Switchgear
- 2 Nos Power Transformers 33/3.3 kV.
- 2 Nos Power Transformers 33/0.4 kV.
- Diesel generators
- bursars with bus-coupling switches
- MV MSB (Main Distribution Board)
- LV MDB (Main Distribution Board)

7.21.2 33KV MV Switchgear at WTP

The MV Switchgear shall comply with the required Electrical Specifications and shall consist of:

- Supply and installation of new 33 kV Medium Voltage Switch Gear with one incomer from 33KV supply and two outgoings 33/3.3KV transformers.
- Medium switchgear equipped with circuit breakers with digital protection, remote control and monitoring relay. The relay shall provide protection for but not limited to over-load, single earth fault relay, phase to ground relay and thermal protection.
- The switchgears and their components thereof shall be capable of withstanding the mechanical forces and thermal stresses of the short circuit current of the system without any damage or deterioration of materials.

7.21.3 Power Transformers at WTP

Supply and installation of two 1000kVA 33/3.3 kV transformers; both of them (1W+1S) running parallel during operation.

Supply and installation of two 50kVA 3.3/0.4kV transformers to be used to supply power to Low Voltage equipment.

All power transformers to be designed shall be fully protected for both internal, overload and external cause of faults.

The transformers shall be suitable rated for continuous max. load of the connected consumers under worst climatic site conditions. The transformers shall have capacity to cover 100% of the power. The total consumption shall be spread to 50% on each transformer. The transformers shall work in parallel one duty and other is standby. All transformers shall be from the same manufacturer.

Transformer shall comply with the required Electrical Specifications.

7.21.4 Diesel Generators

The Contractor shall provide generators 3.3KV to be used in case of power supply failure. The generators will be sized to make the WTP, including the treated water

pumping station (ONLY One pumps to Uhuru Park 5,000m³ GBR, work simultaneously at their maximum capacity)

The generators will start automatically (use of auto transfer switch, ATS) in case of power failure.

Table 16: ATS items

Critical consumer
ONLY One Pumps.
Valve actuators in filters
Drainage pumps in basements and pump stations
All process instruments which are not backed-up by UPS or battery systems
Indoor and outdoor lighting, indoor power sockets, all other house distribution consumers including ventilators and air conditioners, etc.
Firefighting alarming system
Emergency lights
UPS for PLCs, SCADA rack mounted Professional Servers, and SCADA client work stations
UPS for data transfer facilities.

During stand-by power supply, output voltage of generator must satisfy specified allowable limits (RMS, frequency and distortion) for proper work of connected consumers. The control panel of the generator-set must be able to allow manual and remote start as well.

The diesel generators shall be connected to the substation with Automatic Transfer Switch (ATS). The generators will start automatically (use of auto transfer switch, ATS) in case of power failure.

Diesel Generators shall comply with the required Electrical Specifications.

7.21.5 3.3KV MV Switchgear at WTP

Supply and installation of new 4KV cabinet for 3.3KV with 2 incomers, bus section and 6 outgoing.

The MV MSB mainly consist of the following but not to limit:

- Two incomers from 33/3.3 KV transformers
- One incomer from 3.3KV Generator
- Voltage reference (indication of volt and ampere)
- Surge arrester
- Bus Coupler

- Bus Riser
- Energy meter (kWh meter with impulse output.
- Power factor correction connection point.
- 3 outgoing to motor pump VFDs

Supply and installation of Set of 3 VFDs (3.3KV designed with power factor of 0.98), not integrated.

7.21.6 Low Voltage Electrical Equipment

Supply and installation of LV panel which will be used for low voltage installations.

The MV MSB mainly consist of the following but not to limit:

- Two incomers from 33/0.4 KV transformers
- Voltage reference (indication of volt and ampere)
- Surge arrester
- Circuit breaker with thermic, magnetic release with motor driver in withdrawable type. All circuit breakers for low voltage switchgear rated above 800A shall be Air circuit breakers (ACB) otherwise shall be Moulded Case Circuit breakers (MCCB)
- Energy meter (kWh meter with impulse output.
- Current transformer for indication and compensation including transmitter 4....20 mA
- Power factor correction connection point.
- 1 outgoing feeder to Pumping House MCC
- Outgoings feeds for tertiary (administration building, guard house, workshop and staff houses).
- Feeders for PLCs, emergency light, fire system, CCTV camera, telecom cabinets and for UPS etc.
- 20% spare for future extension. each equipped with (the Design report has foreseen 12 feeders):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.
 - Spare feeders 20%

In case of a malfunction / breakdown in the substation critical consumers shall be supplied by a diesel generator, which is connected to the substation with Automatic Transfer Switch (ATS):

- Six feeders for PLC, emergency light, fire system, CTV, telecom cabinets each equipped with:
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the water treatment plant including spare feeders.
 - Spare feeders 20%

7.21.7 In-Buildings and external electrical installations

Following in-building and external installation in the all locations shall be foreseen:

- Internal main and emergency lighting
- Floodlighting
- Socket outlets
- Cabling
- Lightning protection
- Earthing

LV equipment shall comply with the required Electrical Specifications.

7.21.8 MCC – Pumping House MCC

MCC Panel shall have 20% spare for the future use after installation is completed and when handing over.

Each drive must be equipped with an emergency push-button and common push-button for several units.

MCC1 Room shall be located attached to the Rapid Sand Filter Unit. MCC1 shall mainly consist of:

- Income feeder from the LV MSB with:
 - Surge arrester.
 - Circuit breaker with thermic-magnetic release in withdrawable type.
 - Three current transformers with transmitter 4...20 mA.
 - Three Amp. meter, one voltmeter with selector switch.

- One Voltage monitoring device $<U>$, $<f>$.
- One control transformer.
- One network analyzer.
- One bus coupling with circuit breaker with thermic magnetic release in withdrawable type.
- Motor feeders based to 0.4 kV Variable Frequency Drives for blower motors and pumps each equipped with:
 - Current transformers for indication and protection including transmitter 4...20 mA (3 phase).
 - Multifunction protection relay for, phase-sequence, under voltage over voltage, ground detection, instantaneous over current, temperature, supervision, under/over power protection, time over current protection, etc.
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - 0.4 kV Variable Frequency Drives
 - VFD units of <200 kW shall be installed inside of MCC Enclosure, VFD of ≥ 200 kW shall be installed free standing, with its own cubicle. Cos phi compensation (to 0.95) shall primarily be accomplished by the VFD units
 - Indication of Amp and running, fault, individual faults by the display of the protection relay.
 - With all necessary accessories.
 - All open/close/failure/start/stop indications and measuring values must be transferred to the next PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Complete cabling installation material and connections.
- Motor feeders for 0.4 kV for valve actuators each equipped with:
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - Current transformer with transmitter 4...20 mA.
 - Protection according to the mechanical equipment,
 - Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
 - Phase failure, blocking protection, current value protection
 - Failure indication, running hour meter, start counter, Amp meter.

- The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.
- The motor feeders for Valve Actuators shall have two DOL starters one for OP (Opening) and one for CL (Closing) with all additional control relays. If the Valve actuator come with build in motor starters than in this MCC panel need to be install only the power feeder describe below as feeder for auxiliary equipment.
- o feeders for all the auxiliary equipment (including LDB for inside lighting and FCP (filter Control panels) and PLC Panel and etc.) each equipped with (the Design report has foreseen 6 motor feeders)):
 - Control and protection device according to the mechanical equipment
 - Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
 - With all necessary accessories.
 - All running and fault indication, open/close data / failures and all measuring values must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
 - Feeders for UPS and for any additional mechanical equipment which is installed in the intake station including spare feeders.

Spare feeders 20%

7.22 IKANA 3,000m³ GBR

7.22.1 MV Switchgear / Transformers / LV Distribution – IKANA GBR

Scope of works

(a) Incoming power supply

Reservoirs site will be supplied from the local electrical distribution network. Generally, a 33KV HT, a tapping point is located around 900m from the site.

Transformer shall be pole mounted 33/0.4 kV 50 kVA transformers. Capacity of the cable link and the transformer shall cover the total service areas coverage. The Overhead MV Line switch shall be installed on the pole above the transformer and shall be for Load Break Operation for 33kV overhead line and with current capacity related to the particular Transformer Size of 50kVA.

(b) Switchboard.

The switchboard is to be located inside the security enclosure, Containing metering, main distribution section, PLC and telemetry equipment. switchboard shall be located at but not to limit:

- o Guard house panel, supply to street lights, security lights and tertiary of the building

7.23 Uhuru Park 5,000m³ GBR

7.23.1 MV Switchgear / Transformers / LV Distribution –Uhuru Park GBR

Scope of works

(c) Incoming power supply

Reservoirs site will be supplied from the local electrical distribution network. Generally, a 33KV HT, a tapping point is located around 100m from the site.

Transformer shall be pole mounted 33/0.4 kV 50 kVA transformers. Capacity of the cable link and the transformer shall cover the total service areas coverage. The Overhead MV Line switch shall be installed on the pole above the transformer and shall be for Load Break Operation for 33kV overhead line and with current capacity related to the particular Transformer Size of 50kVA.

(d) Switchboard.

The switchboard is to be located inside the security enclosure, Containing metering, main distribution section, PLC and telemetry equipment. switchboard shall be located at but not to limit:

- Guard house panel, supply to street lights, security lights and tertiary of the building
- Chlorine preparation and Dosing pumps.

7.23.2 MCC for Uhuru Park Balancing Reservoir

Each feeder shall be equipped with Circuit breaker with thermic magnetic release and earth-fault protection in withdrawable type.

The MCC shall have 20% spare for the future installation.

Each drive must be equipped with an emergency push-button and common push-button for several units.

The MCC panels shall mainly consist of:

- Income feeders from the /0.4 kV transformer
 - Surge arrester.
 - Current transformers for indication and compensation including transmitter 4...20 mA.
 - Circuit breaker with thermic, magnetic release with motor drive.
 - kWh meter with impulse output.
 - Indication of Volt and Ampere.
 - Monitoring of <U> and <f>.
 - Circuit breaker with thermic, magnetic release with motor drive in withdrawable type.
 - A cos phi of 0.95 must be guaranteed.
- Motor starters for Dosing pups and Chlorine mixer agitators and two fans for the chlorination room based to 0.4 kV DOL each equipped:

- Circuit breaker with thermic-magnetic release in withdrawable type motor operated.
- Current transformer with transmitter 4...20 mA.
- Protection according to the mechanical equipment,
- Multifunctional motor protection relay over/under load current imbalance, temperature, supervision, phase-sequence, under voltage over voltage, under/over power protection, time over current protection, etc.
- Phase failure, blocking protection, current value protection
- Failure indication, running hour meter, start counter, Amp meter.
- The following hardwired protection must be foreseen: dry running protection by electrode, flow protection by flow switch for each pump.

The chlorination power supply and control system with motive water pump motor starters and dosing control may be purchased together with the Electrical panel, complete as a package unit. Also, the Drinking water booster pump set may be purchased as a package unit with its own Electrical panel. In both cases the motor starters described above shall be replaced with Power Feeders.

- Power feeders for all the auxiliary equipment (LDB for outside lighting, LDB for inside lighting, PLC, emergency light, fire system, telecom cabinets and spares) each equipped with:
 - Circuit breaker with thermic-magnetic release, motor operated.
 - With all necessary accessories.
 - All fault indication, must be transfer to the PLC.
 - Controlling and interlocking from the low voltage switchgear with respectively PLC.
- Spare feeders 20%

7.24 Scope of the works on ICA / SCADA System

The Contract shall include the design, manufacture, programming, testing at manufacturers' works, delivery to Site, off-loading, storage, erection, testing and setting to work a complete ICA system at all proposed Sites in the Water Supply Project and of a complete SCADA system located at the WTP operational center.

Supervisory control and data acquisition (SCADA) system shall be designed for control, monitoring the entirely system from the Momba Intake (raw water intake) to the Momba WTP, Ikana 3,000m³ GBR, Nkangamo booster pumping station, Uhuru Park GBR, and their entire facilities. Where the central control room should be located at Momba WTP I administration building.

The system of control/command and the PLC will have full redundancy to ensure the continuity of operation in case of dysfunction of anyone of its components.

The data received from WTP will be at minimum but not to limit:

- water flowrate and pressure,

- position and control of all valves,
- water level,
- alarms on levels of water.

The plant main monitored data parameters will be at least:

- Process flow rates;
- Process monitoring both water treatment and sludge treatment processes;
- Tanks liquid levels;
- Pump delivery pressure;
- Water quality control (turbidity, pH, chlorine, etc.).
- Energy consumption parameters (kWh, kVA, Kvar and power factor etc.)
- General piping and instrumentation

The SCADA system to be supplied under this contract shall consist of four primary subsystems:

- Field sensors (instrumentation) which shall include pressure switches/transmitters, level transmitters and measure process variables.
- Programmable logic controllers (PLCs) /controllers with multiple input/output (I/O) signals capable of executing control strategies.
- Communications systems, which shall include Ethernet switches, cables, fibre-optic networks, and radio links.
- Monitoring system (Human-machine interface (HMI)).

The system should include the following facilities as a minimum:

- Two PC's and atleast two 42" display screens in SCADA room.
- Two Printers.
- Standard SCADA system software.
- Software for monitoring of the plant.
- Power backup systems
- Alarm handling.
- Graph function.
- Collection of historical data, backups and archiving.
- Report of measured values.
- Simple maintenance function.

Contractor shall clearly indicate in the designed system but not limited to the following:

- Design layout showing location of main components and wiring etc.
- Applicable standard uses.
- Form of separation of the compartments according to informative Annex D of Standard BS EN 60439, Form 4b being the minimum required.
- Degree of protection of the panels.
- Full technical specification of all main electrical equipment such as switchboards, switch gear, cables, fixtures, alarm and control system, instrumentation, transformers, etc.
- Schematic electrical layout of the total installation in the form of single line diagrams showing power supply lines, switchboards, control boards and main switching equipment.
- Wiring control systems in switchboards in the form of single line diagrams and key diagrams.

- Drawings of face for switchboards.
- List of components indicating manufacture, type and component number, ordering numbers, etc.
- Tag numbering system.
- List of cables.
- List of signals.
- All electrical network, wiring, cables, cables channels, transmission lines, all telecommunication network.

The scope of ICA / SCADA works to be covered in this contract includes, but is not limited to, the following main items:

- A complete design of all instrumentation, control and automation system.
- Delivery and installation of
 - All process and electrical instrumentation including cables and cabling for each site.
 - All PLCs, control units, RTUs, panels, UPS, etc. including programming ready for operation:
 - The ICA / SCADA system shall only integrate the systems to be procured and installed in whole project area - above, but as well the systems procured and installed in the Works Contract for dam raising, as follows. Details of the ICA for these systems are contained in the P&ID diagram:
 - Momba Intake
 - Ikana GBR, Nkangamo BPS and Uhuru Park GBR
 - Momba WTP
 - SCADA computers/servers, printers, work stations, screens, etc. including programming ready for operation.
 - Elaboration of process description, FDS, SCADA, As-built documentation.
 - Supply of spare parts.
 - Rendering of training services for the Employer's staff.
- All approvals and negotiations with local authorities (radio telemetry and GSM communication).

Each above listed Sites shall be equipped with a RTU (Remote Telemetry Unit) to communicate with the central SCADA at the WTP via radio / GSM. At the all sites except the WTP, the RTU shall be combined with a suitable PLC that will execute all controls and automation at the specific site (refer to SCADA block diagram).

At the WTP the following sub-PLCs shall be connected to the SCADA system by a cable bus:

- PLC1- Water Treatment Process (Coagulation, flocculation, clarifiers and filters)
- PLC2- Chemical Dosing
- PLC3- Sludge Treatment
- PLC4- Treated water pumping station

- PLC5- MV Switchgear

The SCADA Center for Water Treatment Plant shall be the same time the Central SCADA center for the entire water supply scheme.

The PLCs / RTUs shall monitor and control the instruments and field devices as it is indicated (but not limited) in the P&ID Drawings of the Design Report for CP2 equipment / pumps plus 20% spare capacity on the max. no. of I/O signals of full capacity of the WTP. The final number of I/O shall be defined and approved by Engineer during detail design to be executed by the Contractor.

Further requirements are contained in the Specifications for Electrical Works.

7.25 Uninterruptible Power Supply (UPS)

An UPS shall be provided for switchboards that have critical equipment. The UPS will generally have the form of an ELV power supply with batteries attached. The UPS shall generally provide power for:

- Control system equipment;
- Radio/communications equipment;
- Instrumentation;

7.26 Instrumentation

- ***Electromagnetic Flow Meters***

Bi-directional magnetic flow meters shall with a minimum conductivity of $\geq 5 \mu\text{S/cm}$ and meet at least the following requirements but not to limit:

- Flow meter shall be installed entirely as per the manufacturer's instructions with particular attention to minimum straight distance before and after the flow meter relative to tees, bend and other obstruction;
- The flowmeter shall not be direct buried and shall not be installed in pits without adequate drainage.
- Flow meters shall be entirely suitable for accurate and reliable measuring of the process liquid under pressure over the entire expected flow range;
- The transmitter/sensor is to be remote mounted;
- The lining of the flowmeter is to be elastomer;
- Potential equalising fluid contact rings shall be used for metal pipe connections;
- The flow meter shall comprise an electromagnetic detector, power supply and converter providing an overall system accuracy of $\pm 1\%$. The flow meter shall utilise bi-polar pulse technology and shall have a configurable pulse output pre-set to 100 L/pulse;
- ***Level Floats***
 - Two Flygt ball level floats shall be supplied, connected and commissioned for the purposes of alarming during high- and low-level conditions.
 - The high-level float shall be positioned 500 mm below the overflow level of the reservoir and be 'normally closed'.
 - The low-level float shall be positioned 500 mm above the floor of the reservoir and be 'normally open'. The alarms shall be sent to telemetry and SCADA.
 - The preferred equipment is the Flygt Xylem ENM-10 Level regulator.

- Level floats shall be terminated in a stainless steel/marine grade aluminium IP66 junction box adjacent to the maintenance hole entry for the reservoir as per the radar.
- Labels shall be affixed to each ball float cable leaving the junction box before entering the reservoir with "High Float" and "Low Float" to allow for rapid identification during replacement/maintenance.
- In addition to overflow switch, it is recommended to mark the overflow level on the wall inside the reservoir near the internal access ladder.
- **Level Sensor**
 - The reservoir shall be provided with a radar type level transmitter.
 - The level reading shall be relayed through to the SCADA and RTU for remote continuous level measurement and raise SCADA/RTU alarms when the level reaches a level alarm high (overflow level) and a level alarm low (empty level).
 - The equipment shall include all the necessary documentation, fittings, antenna, housing, cables, and electrical connections required to complete the installation.
 - All cables and cores shall be uniquely and clearly labelled to match the electrical drawings
 - All level sensors shall be installed in a location to allow ease of access for maintenance and shall be connected to the telemetry system.

7.27 Area lighting and power

- Sufficient lighting facilities shall be provided within the reservoir security structure (Floodlights area recommended) and for the reservoir roof access stairways and platform (Pierlite Contempo RVS 250 or similar accepted).
- Internal lighting shall comply with the requirements of AS/NZS1680. Road-way lighting shall comply with the requirements of AS/NZS1158.
- External lighting will be operated by a switch on the Distribution section of the site switchboard.

7.28 Security system

- Perimeter Fencing
 - Fencing shall be erected around the perimeter of the reservoir to restrict unauthorized access to the site.
 - The reservoir shall be protected using heavy gauge chain wire mesh fencing with fixed bottom rail and topped with three strands of barbed wire set on a 45-degree outward crank. Coastal fringe and corrosion prone environments, Black PVC coated heavy gauge wire is to be used with all posts and rails powder coated black to match.

- Perimeter Fencing Entry Points.

The entry point shall be secured via gate of construction to match fence type. Gate shall be secured via chain and padlock arrangement. Chain welded to gate frame and 10mm chain to be used.

- External Security Lighting

Lighting shall be provided to site entry gate and reservoir access points.

- Video Surveillance system

Network of cameras, monitors/display units and recorders. It should operate for 24/7.

7.29 Signage and Labelling

- Safety Signage

All mandatory, caution, prohibition, danger, security and safety signage (e.g., confined space entry)

- Labelling

- All equipment shall have the identification number affixed to the equipment to facilitate easy identification for operation and maintenance purposes
- All instruments shall be identified with tags mounted adjacent to but not directly on the instrument.
- All tags and labels shall be of stainless-steel construction with engraved numbering and lettering
- Tags and labels shall be fixed in a prominent location by use of stainless-steel pins.

- Piping Identification

Above ground and in-pit pipework shall be provided with identification markers indicating contents and direction of flow.

8. Site Working Arrangements

8.1 Mobilization to Site

Mobilization to site shall include the erection of all site facilities and the supply of Contractor's equipment necessary for carrying out the Works under this Contract. The Contractor shall make his own arrangements for his site compound and shall set them up at the locations agreed with the Engineer. Mobilization shall be concluded one month after commencement date.

8.2 Demobilization from Site

The demobilization includes the removal of all site facilities and temporary installations, the demobilization of all equipment from site, the removal of all surplus materials, the reinstatement of all damaged or worn facilities used by the Contractor (including temporary access roads) and the cleaning up of the construction site(s). It also includes the re-shipment of construction equipment, which has been imported for construction purposes.

8.3 Quality Assurance Procedure

A Quality Assurance Procedure shall be implemented in accordance with Clause 4.9 of the Conditions of Contract. The Contractor shall develop the Quality Assurance Plan relying on all his internal competencies in the matter of quality control in the design phase, the work phase, setting to work, commissioning, training and handover stages. The Quality Assurance Plan shall be subject to approval by the Engineer.

The procedures as stated in the Quality Assurance Plan shall be implemented by the Contractor during the Contract. The Quality Assurance Plan shall be available at Commencement Date.

The Engineer has the right to audit the Quality Assurance Procedure of the Contractor, at any time.

8.4 Restrictions to Operations imposed by the Site

8.4.1 The Site

The Employer will hand over the site to the Contractor for work purposes. Subject to this, the Contractor shall actively assist the Employer in:

- obtaining any necessary approvals and permits from governmental authorities, by providing information and preparing documentation including letters, submissions, designs, details and special procedures reports, drawings or any other information, free of charge;
- responding to notices and letters of authorities in connection with the occupation of sites, construction of the temporary and permanent works, and approvals and permits relating to design or execution of the Project, free of charge.

The Contractor shall keep the Employer fully informed of progress on all aspects of this work and shall copy to him all submission and written communications with owners, occupiers, authorities, utilities, and the like, and all permits and agreements received.

Working sites shall be used solely for the construction of the works.

The Contractor shall comply with the requirements of the police and other traffic authorities, regarding the parking of and movements of his vehicles, including constructional plant near working sites.

The Contractor shall make his own arrangements for all additional storage sites near the works or elsewhere.

Before entering to any additional working sites, the Contractor shall obtain and forward to the Engineer a copy of written consent of the owner and occupier or authority having charge of the land, stating the purposes for which such land is to be used by the Contractor, and defining the extent and periods of his occupation for which such consent is granted. Notwithstanding the requirements of such consent, additional working sites shall be protected by fences or barriers or other works as required by the Engineer. Wherever practicable, a passage shall be maintained for vehicles and pedestrians along public roads and to all adjacent properties.

8.4.2 Working Hours for Construction

Site working hours shall be as listed in the Appendix to Tender.

The Contractor's program and methods of working must be made on the assumption that the working hours will not be varied.

Any proposal by the Contractor to work outside of these hours shall be submitted to the Engineer for approval giving at least 7 days' notice. A clear definition of the work to be carried out and the reasons for the request shall also be provided.

Unplanned deviation from the normal working hours will normally be limited to emergencies only and the Engineer shall be informed, at the earliest opportunity.

8.4.3 Access by Officials

Authorized government and municipal officials shall, at all times, have access to the work whatever it is in preparation or progress, and the Contractor shall provide proper facilities for such access and for inspection.

8.4.4 Interference with Existing Apparatus

The Contractor shall give strict and specific instructions to all his employees that no valves or other fittings, not forming part of the Contract, are to be operated, adjusted or interfered with in any way without the specific agreement of the Engineer.

8.4.5 Existing Services

According to Employer's knowledge and investigations, there are existing utilities on this virgin land sites of the WTP and the reservoirs. During pipeline and road construction existing facilities might be encountered.

The Contractor shall survey the sites and determine/confirm the non-existence of utilities. Any utilities found shall be adequately marked and protected throughout the construction period.

The Contractor shall execute the works in such a manner that he does not damage or interfere with existing services on or near the site. If damage or interference happened, the Contractor shall make good the damage at his own cost.

The Contractor shall make his own arrangements for any diversion or removal of services, which he may require for his own works. He shall obtain the prior approval of the Engineer to such arrangements.

8.4.6 Street cleaning and sprinkling during Construction

The Contractor shall clean all spilled dirt, gravel, or other material caused by the construction operations from all streets and roads on a daily basis, or as instructed by the Engineer.

8.4.7 Protection against Damage

The Contractor shall take all necessary precautions to avoid causing any unwarranted damage to roads, lands, properties, trees, etc. and shall deal promptly with any complaints by owners or citizens.

8.4.8 Temporary Works

The Contractor shall make provision at his expense for temporary works like staging, timbering, strutting, sheet piling, coffer dams, dealing with water, bridge-ways, scaffolding and other works as may be necessary and required for the safe and efficient performance and construction of the works.

8.4.9 Abatement of Nuisance

The Contractor shall adopt such measures as the Engineer may consider reasonable and necessary to minimize nuisance from dust, noise, or other cause. During periods of dry weather, the Engineer may require that roads in frequent use with heavy traffic shall be sprinkled regularly and as agreed with the Engineer.

8.4.10 Temporary Access Roads

The Contractor shall construct at his own cost temporary access roads to and on the sites. The existing roads can principally be used for traffic from and to the sites. Such roads shall be temporarily arranged to have sufficient width, quality and stability to allow the movement of all vehicles and equipment. In steep slope sections, the upper surface layer shall be made of gravel material, which prevents slippery conditions after rainstorms.

The Contractor shall temporarily improve and maintain, during the whole contract period, existing roads and paths to the WTP and construct new unpaved roads to the remote reservoir sites, to be used during construction as site access. The temporary access roads shall be fit for Contractor's construction purposes.

The Contractor shall ensure that damage to any public or private roads, footpaths and tracks caused by his vehicles or equipment shall be made good to the satisfaction of the Engineer and the owners and/or controlling authorities.

All temporary roads shall be maintained by the Contractor in good condition throughout the construction period. If the Contractor fails to maintain the roads as required, the Engineer may instruct a third party to carry out the work at the cost of the Contractor.

The Employer and the Engineer and their employees as well as other contractors (if any) working on the site for the Employer shall at all times have free use of the temporary roads prepared by the Contractor.

8.4.11 Use of Explosives

Except if specifically approved by the Engineer in writing, the Contractor shall not use explosives.

8.4.12 Fire Prevention and Protection

The Contractor shall perform all work in a fire-safe manner. He shall supply and maintain on the site adequate fire-fighting equipment. The Contractor shall comply with all current applicable fire regulations.

8.4.13 Accidents, Extraordinary Events

The Contractor shall give immediate written notice to the Engineer of any accident or extraordinary event occurred on site giving details of the same, regardless if such an accident or event affects the progress of work. The Contractor is also obliged to report on any measure taken.

8.4.14 Advertisements

The Contractor shall not undertake or allow bill posting or advertising of any kind concerning the works without the written consent of the Engineer.

8.4.15 Clearance of Site on Completion

A Performance Certificate will not be issued before the Contractor has removed all his machinery, equipment, plant, waste material from the site and the site reinstated to the satisfaction of the Engineer.

8.5 Record of Site Conditions before Construction

The Contractor shall record together with the Engineer, prior to his first entry on site, the surface topography and ground conditions of the working site and the immediate surrounding area, to the extent considered necessary by either the Engineer or the Contractor. The agreed record shall include photographs and spot levels as necessary and written descriptions of the site conditions.

8.6 Setting Out Points, Lines and Levels of Reference

The Contractor shall supply to the Engineer details of the temporary bench marks and reference points, which he proposes to use.

The Contractor shall submit to the Engineer for approval details of the Contractor's system for checking locations and levels of the temporary bench marks and reference points. This will not relieve the Contractor of this responsibility for the accurate measurements of temporary bench marks and reference points.

The Contractor shall provide all survey and measuring instruments necessary for his own use in the execution of the works.

8.7 Health and safety

The Contractor shall comply with all relevant H&S legislation of Tanzania and the ESHS Specification at all times, during the period of the Contract.

The Contractor shall be responsible for the safe conduct of the Works. He shall ensure that all operations are carried out safely and that any person made responsible for the safe conduct of any part of the operations carries out his duties in a proper manner.

The Contractor shall submit to the Engineer the ESMP, including a H&S section, within 28 days after Commencement date. This HS Plan section shall comply with international good practice, local safety codes, laws and regulations including a comprehensive AID/HIV prevention plan for the construction site.

Where any part of the Works is not covered by the Contractor's Safety Policy, or is a high-risk activity, the Contractor shall, before commencing work on that part of the Works or activity, submit to the Engineer a work-safety Method Statement for approval. No related work shall be commenced unless the Engineer has approved the proposed methods.

The Contractor shall nominate a responsible person for the organization and supervision of first-aid measures and facilities, throughout the contract period. The Contractor shall conduct H&S-training for all his personnel on a regular basis, familiarizing them with H&S measures.

The Contractor shall introduce an "Operator Permit to Work" system after adequate operator training to ensure that the operator is adequately trained to carry out the required work on any equipment.

On site a first-aid facility, in accordance with the ESHS Specifications shall be provided. Medical supplies for first-aid provision shall be clearly marked and labelled, easy to identify and easily accessible.

The Contractor shall provide for emergency arrangements, in order to mobilize quickly labor-force outside the normal working hours to carry out any work needed for an emergency associated with the works. The Engineer shall be provided at all times with a list of addresses and telephone numbers of the Contractor's staff who are currently responsible for organizing emergency works. The Contractor shall familiarize himself and his employees with any relevant, existing, local arrangements, dealing with emergencies.

8.8 Electricity, water, gas and other services

No utilities are available at the site and the Contractor has to make his own arrangements, for water, sewage, power, communications, etc., at his own costs for all services, necessary for the undisturbed construction and all supplies for his work force, offices and Engineer's and Employer's offices.

All costs and fees with respect to such supply shall be borne by the Contractor. All such facilities shall be removed by the Contractor prior to final acceptance of the Works.

8.9 Sanitary Arrangements and Waste Disposal

The Contractor shall provide all necessary medical, sanitary and other facilities for the welfare of his employees on Site. The facilities shall be at least in accordance

with the standards required by law, local regulations or agreements between Trade Unions and Employers. A high standard of hygiene and cleanliness shall be maintained always. The Contractor shall be responsible for the H&S training of both his personnel and the Engineer's staff.

The Contractor shall provide sufficient chemical type toilets, of suitable type, at remote working locations and maintain the facilities in good condition always. Chemical type toilets shall be of adequate construction so that no contamination of the area will be caused. Upon completion of the work, the temporary sanitary facilities shall be removed and the areas restored to their original condition.

8.10 Site Tidiness

The Contractor shall confine his operations within the allocated Site, or such other areas of land, as agreed between the Engineer and Contractor.

The Contractor shall maintain the site in a clean, tidy and safe condition during the period of construction and commissioning. The Contractor shall remove from site any disused materials and other debris. The Plant shall not be taken over until such material has been removed.

The Contractor shall prevent vehicles entering or leaving the site disposing-off mud or other debris on the surface of the adjacent roads or footpaths.

8.11 Tipping Facilities

No tipping facilities will be provided by the Employer for the Contractor and he shall make his own arrangements for the disposal of all surplus excavated material at his own cost and in agreement with the Engineer. Unauthorized tipping shall not be permitted.

8.12 Site Sign Board

The Contractor shall provide, erect and maintain weather resistant signboards at the entrance to the Intake and WTP site and to the junction of each storage tank access road and every 10 km along the main route of the transmission pipelines.

The sign-board writing shall be in English language. The sign-board shall be a minimum of 4 m width by 2.5 m height and provide the project name, the name of the beneficiary, the funding agencies, the Contractor and the Engineer.

The signboard, precise information therein and exact location shall be subject to approval by the Engineer, prior to erection.

8.13 Temporary fencing of the sites

The Contractor shall be responsible for ensuring that the sites are adequately fenced. The Contractor shall regularly inspect and maintain all such fencing, any defects shall be made good without delay.

Temporary fencing shall be provided at all construction sites and shall remain in position until it is replaced by permanent fencing. Access shall be provided in temporary site fencing (if necessary) for the use of the occupiers of adjacent lands. All fenced areas shall be provided with lockable gates.

All open trench excavations for pipelines, outside the WTP and reservoir sites, shall be protected ensuring the safety at work.

The type and height of temporary fencing shall be aluminum or steel corrugated sheets 2400 mm high fixed on timber, concrete or steel posts, all well braced to withstand wind damage. Alternatively, if agreed by the Engineer, the Contractor may use chain link fencing or galvanized wire netting.

8.14 Facilities for the Engineer

8.14.1 Site offices

The Contractor shall erect offices and storages for his own use at a location agreed by the Engineer. Within the office area, the Contractor shall provide separate office space for the use of the Engineer's team with separate hygiene facilities. The office shall have a net floor area and consist of rooms and offices as detailed in the Appendices.

The Contractor shall provide, furnish, equip and maintain, for the duration of the Contract, site offices for the sole use of the Engineer's resident site staff at the water treatment works site. The Contractor shall make available to the Engineer and his representatives all services provided for his own offices including road access, fencing, hard standing, water, power, telephone, internet and sewerage, etc. Each door shall be provided with a lock and two keys. The clear height of all offices between floor and ceiling shall be 2.4 m minimum. Windows shall be able to open the full window area.

The Contractor shall provide and install air-conditioning units as directed by the Engineer. The air-conditioning unit shall be an electrically operated compressor type with closed circuit, and not an evaporation type. The capacity of the air-conditioning units shall be at least 2.2 kW each.

The office shall be substantially built, weatherproof and well lit. It shall be properly secure. The offices shall not be removed from the Site until the completion of the commissioning period and the commencement of operation or until the Engineer, by notice in writing, calls upon the Contractor to do so.

The siting and orientation of all offices shall be to the Engineer's satisfaction and shall be decided upon in consultation with him and confirmed in writing before erection.

The Contractor shall not order any housing, materials, equipment or fittings based on what is specified or scheduled without written confirmation from the Engineer. No buildings shall be erected without the Engineer's written instructions as to the exact position and orientation of the buildings.

The Contractor shall take all reasonable precautions to prevent unauthorized entry to the offices and to ensure the general security of offices and laboratories.

The access around the offices shall be surfaced with bitumen macadam. They shall be well-drained and kept trafficable and free from mud always.

The Contractor shall provide site personnel with all the welfare requirements, to meet the applicable statutory provisions and supply all necessary safety equipment and clothing.

8.14.2 Sanitary Arrangements

The Contractor shall be responsible for providing all sanitary services necessary to keep all offices and quarters in a clean, neat and hygienic condition.

The Contractor shall provide septic tanks as necessary for the disposal of sewage. The tanks shall be emptied on a regular basis and the removed fecal sludge shall be disposed-off safely.

The Contractor shall also provide for the removal and disposal of all rubbish and household solid waste from offices and quarters with regular collections being made at least twice weekly.

Where the construction of septic tanks or water-borne sewerage is not feasible, the Contractor shall send an alternative proposal to the Engineer for approval.

8.14.3 Water, Electricity and Gas

The Contractor shall provide a constant supply of clean potable water suitable for human consumption, as well as the necessary electric power at 220 V AC to the offices.

Electric power shall be supplied to the offices during working hours and if necessary extended, for working purposes.

8.14.4 Maintenance

The Contractor shall provide all labor, equipment and material which may be necessary to keep all accommodation in a neat and clean condition and any repairs shall be done immediately upon the request of the Engineer.

8.14.5 Furniture and Equipment

The offices shall be furnished and equipped as detailed in the Annexes.

8.14.6 Instruments for the Engineer

The Contractor shall provide on loan for the use of the Engineer and his representatives all survey and other instruments necessary for checking the Contractor's work, such instruments to be maintained in good order by the Contractor.

Requirements for these instruments are listed in an Annex1 to these Employer's Requirements.

8.14.7 Transport for the Engineer

Means of transport for the Engineer's staff has to be provided in accordance with the list, attached to the Appendixes. All means of transport for the Engineer shall be handed over to the Employer at the end of the Contract.

8.14.8 Communication for the Engineer

The Contractor shall arrange for the installation of telephones and extensions thereto, a powerful internet connection, as directed by the Engineer and for his sole use. 5 cell phones from reputable manufacturers and local SIM cards shall also be provided.

The Contractor shall pay for such equipment and for local calls made by the Engineer and/or Employer and his representatives.

8.14.9 Assistance for the Engineer

The Contractor shall provide every assistance to the Engineer and his representatives in carrying out their duties, and shall provide a sufficient supply of pegs, poles, paint, lines, spirit levels and other materials and small tools for checking and setting out and for the measurement of work. When required by the Engineer, the Contractor shall submit samples for approval.

All such articles shall remain the property of the Contractor; they shall be repaired or replaced, if necessary, due to normal wear and tear.

8.15 Contractor's Equipment

Details of all Contractor's equipment to be used by the Contractor for the execution of the works shall be submitted to the Engineer prior to its use.

The Engineer's consent to use the Contractor's equipment will not be reasonable withheld, but if in the Engineer's opinion circumstances arise which make it desirable that the use of the Contractor's equipment should be suspended either temporarily or permanently, the Contractor shall change the method of performing the work affected and he shall be deemed to have no cause for claims against the Employer on account of having to carry out the work by another method, nor shall he be deemed to have cause for claim if any order issued by the Engineer results in the Contractor's equipment having to stand idle for a period of any duration whatsoever or having to be removed.

Where the use of excavators is impossible, due to the proximity of existing structures or services, manual excavation shall be applied. This will not provide grounds for claims of the Contractor.

8.16 Delivery and Storage of Plant

8.16.1 Packaging and Marking

Upon the successful completion of the factory-testing and inspection and prior to dispatch from the place of manufacture, all items of plant shall be thoroughly protected against corrosion and incidental damage, including effects of insects, vermin, strong sunlight, rain, frost, high temperatures, humid and salty atmosphere, etc.

All items intended to be painted shall receive the specified treatment and all unpainted parts exposed to corrosion (except for the electrical equipment) shall be thoroughly coated with a high melting point grease or tallow (free from acid) or other protective substance, which is easily removable during or after erection.

In addition, all electrical equipment shall be suitably protected against corrosion and incidental damage to the satisfaction of the Engineer. Electrical equipment with anti-condensation heaters shall be powered, when the plant is in store, if the climatic conditions do require.

All items shall be packed to withstand land and sea transportation, and rough handling in transit. Packages shall be suitable for outside-storage. The Contractor is responsible for the packing of equipment, which shall reach its destination intact and undamaged. Packages containing dangerous or breakable goods shall be packed and marked, in accordance with the legal requirements and international good practice.

All crates and packages shall be correctly and adequately marked as follows:

- Employer's name;
- Title of contract;
- Contractor's name;
- Work location;
- Designation and item number complying with agreed schedules;
- Usual commercial markings.

8.16.2 Delivery

The Contractor shall be responsible for the delivery of items to the site, including their handling and safekeeping.

The Contractor shall keep the Engineer fully informed of the state of deliveries of items, and shall mail copies of all shipping documents to the Employer and to the Engineer, in accordance with instructions to be issued to him from time to time by the Engineer. All shipping documents shall be accompanied by copies of test certificates as required.

Delivery of items shall consist (but not be limited) of the following:

- Loading and transport of items by sea, land or air freight, including insurances.
- Taking charge of items at the port of entry or rail or air terminal in eastern Africa, including payment of port charges, wharfage, demurrage, intermediate storage, off-loading at terminal points, reloading and transporting to the site.
- Delivery of items to the site, off-loading and appropriate temporary storage in store buildings or under cover, until they are required for installation.

The Contractor shall examine all equipment upon delivery at the site. Should the Contractor discover any damages he shall draw the attention of the Engineer thereto, who will inspect and give instructions for further treatment of the case. Items with short shelf-life such as chemical reagents shall be delivered in the shortest possible time before use with the plant.

Any item rejected by the Engineer shall be immediately removed and replaced by the Contractor.

8.16.3 Storage

The Contractor shall provide all facilities for the safe and proper storage and handling of all equipment as recommended by the manufacturers, with particular consideration given to temperature, rain, sunlight, wind and ground conditions.

Items shall be stored in a programmed order, to facilitate their retrieval for use.

Stacked items shall be protected from damage by spacers on load distributing supports and shall be safely arranged. No metalwork shall be stacked directly on the ground.

Small items shall be held in suitable bins, boxes or racks and shall be clearly labelled.

Items shall be handled and stored so that they are not subject to excessive stress and that their protective coatings will not be damaged.

8.17 Environmental Impact Clauses

The ESMP sets out the requirements during construction and has been attached to these Tender Documents.

9. Testing, Commissioning and Taking over

9.1 General

The Contractor shall perform all necessary tests to demonstrate compliance of the Works with the Specifications, the plant performance requirements and the process performance guarantees. All tests specified herein and/or found necessary for the Works shall be carried out under responsibility and at the cost of the Contractor.

The procedures required to be undertaken and documentation necessary shall be in accordance with the requirements described in the Conditions of Contract. During the testing the Contractor shall demonstrate to the satisfaction of the Engineer that:

- The Works are suitable to produce the required water quality standard and that the sludge dewatering system meets the specified performance requirements;
- The power and chemical consumption do not exceed the specified limits.
- The transmission and storage system are able to convey and store the specified water quantities.
- The Works complies with the Specifications.

The Contractor shall demonstrate that the Plant operates satisfactorily when operated in both manual and automatic control.

The testing shall include, but not be limited to:

- Quality Control and testing during construction;
- Inspection and testing at the manufacturers' premises;
- Tests on Completion.

In addition to any specific obligations for sampling and testing, the Contractor shall be responsible for routine inspection sampling and testing of all materials, workmanship, plant and measuring devices, in order to control the quality of work and to ensure compliance with the Specification and approved samples.

- Tests after Completion for water treatment plant.

9.2 Water sampling in general

Discrete representative water samples shall be taken by the Contractor from identified sampling points in the Plant. The sample location and sampling procedure shall be set out in the Contractor's method statement and approved by the Engineer. Sampling shall be undertaken throughout the day, 7 days per week on an hourly basis the first week of each tests period and as long as performance is not demonstrated, and on a 3 hours basis for the remaining of the testing period.

Inline monitoring equipment results shall be reported together with the laboratory results.

The contractor is responsible for conducting the water sampling based on **BS ISO 5667: Part 1,3,5 and 13**. As a consequence, the contractor will cover the costs

of the overall tests, even if more samples and analyses are necessary in case of contradictions between inline monitoring equipment and laboratory results.

EWURA regulations will be followed (refer to "water and wastewater quality monitoring guidelines for water supply and sanitation authorities, 2nd Edition, March 2020") and completed with some additional parameters:

- **Raw water pipe/ inlet WTP**
 - Turbidity
 - Apparent and True Colour
 - pH
 - Conductivity
 - Oxygen at inlet and outlet of the intake
 - Once a week only:
 - Faecal and Total Coliforms
 - Iron
 - Manganese
 - Sulphate
 - Chloride
 - Calcium hardness
 - Alkalinity
 - Temperature
- **Coagulation**
 - pH
- **Clarified Water on each sedimentation tank**
 - Turbidity
 - Apparent and True Colour
 - Total Aluminium
 - Soluble Aluminium
- **Filtered Water on each filter**
 - Turbidity
 - Apparent and True Colour
 - pH
 - Total Aluminium
- **Final Water**
 - Turbidity
 - Apparent and True Colour
 - pH
 - Total Aluminium
 - Total and Free Chlorine
 - Once a week only.
 - Faecal and Total Coliforms
 - Iron
 - Manganese
 - Sulphate
 - Chloride
 - Calcium hardness

- Alkalinity
- Temperature

9.3 Testing during Construction

During construction, the Engineer will inspect the works on a regular basis, in the presence of the Contractor, to establish conformity with the requirements of the Specification. Any deviations found shall be corrected to the satisfaction of the Engineer.

The Contractor shall be responsible for establishing and maintaining procedures for quality control during construction/installation works, which will ensure that all aspects of the Works comply with the requirements of the Contract.

The Contractor shall provide a fully equipped site laboratory to carry out the specified tests of materials to be used in the permanent works as stipulated in CoC clause 7.4 (testing by the contractor). Qualified and experienced staff shall be provided to carry out the tests and record the results.

If the submission of samples is required by the Engineer, they shall be submitted to the Engineer, at the expense of the Contractor, not less 21 days prior to the time that the materials shall be used for the works. Samples shall be subject to approval by the Engineer, and respective material shall not be manufactured and delivered to the site, nor used for any work without prior approval. Samples shall be carefully preserved for this purpose on site by the Contractor until permission is given by the Engineer for their disposal.

Individual tests shall be carried out in the presence of the Engineer and shall include, but not be limited, the following:

- Soil tests
- Leakage of structures and tanks;
- Weather proofing of buildings;
- Adequacy of access, ventilation and illumination;
- Connection tightness;
- Hydraulic pressure;
- Alignment;
- Correct lubrication;
- Electrical insulation;
- Electrical conductivity;
- Direction of rotation;
- Functional operation;
- Interlock tests;
- Trip circuit operation;
- Operational reliability;
- Thermal loading;
- Interchangeability;

- Calibration;
- Other tests requested by the Engineer.

Tests of equipment, plant and materials shall be in accordance with the requirements of the Specifications and in accordance with the requirements of the applicable Reference Standard.

9.4 Testing at Manufacturer's premises

Whenever the Contractor is ready to carry out any such factory tests, he shall submit a test program to the Engineer, which is subject of approval. If the Engineer does not attend any such test, then the Contractor shall carry out the test in the absence of the Engineer and the certified copies of the test results shall be deemed to be a correct record thereof.

Where equipment is of identical size and duty the Engineer may chose not to witness all tests and the Contractor shall assume that the performance tests on all items have been carried out.

No inspection shall release the Contractor from any of his obligations under the Contract.

Engineer and Employer will require to witness factory tests. These tests shall be performed at the Manufacturer's premises in the presence of the Engineer and Employer. The Contractor shall give 28 days' notice of his intention to test specific equipment.

The Contractor's shall bear all associated cost, i.e. air travel, (business class), appropriate hotel accommodation, allowances and subsidiary expenses for the Engineer and Employer (up to a maximum of 2 persons) for max. 3 trips for the following equipment that shall be witness tested:

- Pumps and other equipment with motors over 50 kW
- Main pipes (raw water, drinking water)

All instruments used for testing purposes shall have been calibrated by an independent accredited testing authority and shall have a valid calibration certificate. The calibration validity period shall not be longer than 12 months. The test instruments shall be recalibrated when requested by the Engineer.

For any and every factory test which the Contractor is required to carry out, the Contractor shall produce written records as agreed by the Engineer, certifying that the test has been carried out. The Contractor's representative who carries out such tests shall sign the test record. The tests which are witnessed by the Engineer shall be certified by the Engineer upon the same record provided by the Contractor.

The Contractor shall provide the Engineer with three certified copies of the test results within two weeks after the tests have been carried out. On each test certificate sufficient information including the Contract number and title shall be given to enable the Engineer to check for compliance with the Specification. It shall include all test results, calculations, performance graphs and curves and shall be signed by representatives of the manufacturer and the Contractor. Copies of all test certificates shall be included in the operating and maintenance manual.

Following any inspection or testing of material or plant, the Engineer shall, within a reasonable time, notify the Contractor in writing either that the item has passed

the tests, or that the item on any part thereof is defective, or not in compliance with the Specification or has not met guaranteed performance or efficiency requirements and is rejected, stating the grounds on which the decision is based.

Should the test(s) fail then the Contractor shall be required to rectify the failure and arrange for the test to be repeated, again giving two weeks' notice and providing travel facilities as previously provided.

9.5 Commissioning

Following the construction period, a three months commissioning period has been foreseen by the Contract.

The Contractor shall prepare a comprehensive commissioning program and details of the procedures and commissioning test he proposes to employ in performing commissioning.

The commissioning and start-up shall be carried out:

- upon approval of the draft FDS,
- Upon submission of the As built drawings/records
- upon submission of the draft O&M Manual,
- when the Engineer is satisfied that all tests before commissioning as described above have been successfully completed and that the plant is ready for start-up.

Commissioning may be started for parts of the works which are ready for commissioning and can be commissioned independently of the reminder of works. However, tests on completion shall be carried out for the whole of the Plant.

During commissioning the Contractor shall show that the equipment complies with the mechanical, electrical, instrumentation, control and automation requirements of the Specifications. During commissioning amongst others, the following shall be executed:

- Demonstration that all hard-wired and soft-wired interlocks and program loops are working, in particular pump dry running protections;
- Optimization of plant automation, in particular optimization of set values for hard-wired and soft-wired control loops and timers, minimum but not limited to pump operation, filter backwash operation, chemical dosing, pre-chlorination and disinfection;
- Checking of data transfer functions;
- Test simulations of emergency shutdown procedures including electricity power failure, and of automatic start-up procedures;
- Test simulations of alarm conditions;
- Test of water hammer and surge protection devices.

9.6 Tests on Completion

General

Prior to the commencement of tests on completion the Contractor shall submit for approval the following:

- Site Acceptance Test Documents;
- Factory Acceptance Test Documents;
- As-Built Drawings/Records;
- Operation and Maintenance Manuals.

Tests on Completion shall not be commenced until the aforementioned documents are approved.

9.7 Dry Test Requirements

9.7.1 General

As a minimum requirement the following dry tests shall be carried out as a general requirement:

- A general inspection to check for correct assembly and quality of workmanship;
- A check on the presence of lubricant, cooling medium, electrolyte, etc.:
- A check on adequacy and security of plant fixing arrangements;
- A general check to ensure that all covers, access ladders, water proofing, guard railings etc are in place;
- A check on damp-proofing, rust-proofing and vermin-proofing and particularly the sealing of apertures between building structures, chambers etc. and the outside;

9.7.2 Pipe Line Works

As a minimum requirement presence of foreign bodies in pipe work and structures shall be checked.

9.7.3 Mechanical works

As a minimum requirement preliminary running checks shall be carried out on all the mechanical works as far is permitted by circumstances in order to ensure smooth operation of Plant.

9.7.4 Electrical Works

The following commissioning tests are to be carried out on all the equipment/systems, as applicable.

- Insulation resistance measurement of equipment, accessories, cabling/wiring etc.
- Dielectric tests on equipment, accessories, cabling/ wires etc.
- Phase sequence and polarity
- Voltage and current ratios
- Vector group
- Resistance measurement of winding, contacts etc.
- Continuity tests
- Calibration of indicators, meters, relays, etc.
- Control and interlock checks
- Settings of equipment and accessories

- Checking of accuracy/error
- Checking of operating characteristics, pick-up voltages and currents, etc.
- Operational and functional tests on equipment, accessories, control schemes, alarm/trip/indication circuits, etc.
- Measurement of guaranteed/approved design values including lighting levels, earth resistance measurement, etc.
- Complete commissioning checks of the system

9.7.5 Instrumentation & Control System

The following dry tests shall be carried out on the instrumentation & control system:

- Continuity check on all signal/ control/ power supply cables.
- Check instrument loop integrity, functionality and calibration; A written report on each instrument in the format required by the Project Manager/ Purchaser shall be provided certifying that the instruments have been calibrated to published specified accuracy.
- Check functionality of the instrument control panels/ consoles.

9.7.6 Hydraulic Wet test requirements

Hydraulic wet tests shall be carried out on completion of dry tests. Raw water or treated water (if available) shall be used for hydraulic wet tests. The purpose of the tests is to prove as far as practical the hydraulic performance of the Works. In order to demonstrate this, the Contractor shall ensure that each part of the Works is hydraulically loaded to its rated throughput for a period of at least six hours. In order to remove doubt, the following tests inter alias shall be carried out:

- Pressure testing of all piped systems laid direct in ground in accordance with the relevant standards;
- Fill all structures and check for leaks;
- Filling of all storage vessels to check for leaks and distortion;
- Running of all pumped systems in order to check for:
 - ✓ Correct functionality;
 - ✓ Absence of leaks;
 - ✓ Correct running temperatures
 - ✓ Smoothness of running and the absence of undue vibration or stress;
- Carry out calibration of instruments where appropriate;
- Hydraulic testing of primary/ impulse tubing. Carry out valving, diversions etc. To full hydraulic load (or where there is a requirement to withstand an over load to overload) each process element;
- Demonstrate correct functionality of electrical, control and instrumentation systems.

The Contractor shall simulate the conditions that will prevail when operating as a process in order to demonstrate the correct functionality of process control loops etc. During these tests a check on the performance of Plant shall be made, as far as site facilities will allow, to compare its site performance with the factory test data and to identify any constraints on performance due to site conditions.

9.7.7 Filter wet tests

The Contractor shall carry out the following tests on each of the completed filters:

9.7.8 Pressure Test

All the nozzle outlets shall be plugged and the filter under drainage system shall then be filled with water leaving an air gap of between 25 mm and 50 mm between the water surface and the underside of the filter floor. The top of the filter floor shall be covered with water to a depth of 50 mm above the top of the nozzle plugs.

The underside of the filter floor shall then be pressurized to 0.5 bar, by either utilizing the Plant air scour blowers or by providing and connecting up a temporary air blower. The test shall be maintained for one hour and all leaks identified. The pressure shall then be relieved, the water removed from the filter and the detected leaks rectified. The filter shall then be retested and the procedure repeated until all leaks are rectified to the satisfaction of the Employer's Representative.

On rectification of all leaks the filter's strength shall be tested by applying the above-mentioned test pressure for a period of 48 hours. Any leaks or minor structural damage shall be rectified and the filter retested to the satisfaction of the Employer's Representative. If, in the opinion of the Employer's Representative, structural damage has occurred to the filter floor, which cannot be satisfactorily repaired without affecting the structural integrity or performance of the filter, the entire floor shall be replaced.

During the above tests the Contractor shall set the pressure relief valve on the air blower system to relieve pressure above the test pressure. He shall also install a standpipe connected to the under drain to ensure that over pressurization cannot take place. The bottom end of the standpipe shall be below the air/water interface under the filter floor and the top end shall be at the level equivalent to the test pressure.

9.7.9 Air Distribution Test

The empty filter shall be filled with water to a level of 25 mm above the top of the nozzles. Air shall be introduced into the under-drain system at such a rate that the system can be seen to pass air through the nozzles over the whole filter area. Any discrepancies in distribution over the filter shall be rectified and the test repeated until the Employer's Representative is satisfied that the air distribution is uniform over the entire filter with all nozzles blowing.

9.7.10 Combined Air Scour and Backwash Distribution Test

The filter shall be tested for combined air and water scour at the design rates in the same manner as described above for the air distribution test. Air shall be brought on first and allowed to fully develop before the water is introduced.

Any discrepancy in distribution over the filter shall be rectified and the test repeated until the Employer's Representative is satisfied that the distribution is uniform over the entire filter with all nozzles blowing. If for any reason it is necessary to remove a nozzle dome after it has been fully tightened, it shall not be re-used but replaced by a new one. As soon as possible after satisfactory completion of the above tests the filter shall be charged with filter media.

9.7.11 Pumps, Piping and Valves

The tests on the different mechanical and electrical equipment shall include but not be limited to:

- Leakages tests shall be carried out on all erected pipe work, pumps and valves immediately after erection and where possible before being built in.
- Operating tests shall be conducted on valves.
- The pumps shall be tested for mechanical performance. The vibration and noise levels shall be checked to be within the specified limits.
- Pump performance tests shall be conducted for all clear water pump sets. Each pump shall be tested at a time.
- To check and verify efficiency of each pump and vibration of pump.

9.8 PUMP MOTORS

- Condition of winding insulation be tested and insulation values shall be restored to required level by suitable heating arrangements locally.

9.8.1 Surge Protection System

- Leakages tests shall be carried out for the erected vessels and associated piping. The leak test for airline shall also to be carries out.
- The compressor shall be checked for erection and performance.
- Installation and operational test for the instrumentation provided for the surge system.

The pump trip test shall be conducted to ensure the surge pressures within the limits.

9.8.2 Detailed Test on Completion

Test on completion shall be foreseen during the 3 months commissioning period and the testing stages shall follow the stipulations of the CoC – Clause 9.

The Contractor shall prepare a comprehensive testing program and details of the inspection and testing procedures he proposes to employ in performing the Tests on Completion.

The program shall indicate the quantities of raw water and power required. Details shall include the manner and order in which each item of plant will be tested and the associated flow rates and duration of flows. In addition, the program shall take into consideration the progress of other contracts with its interface to SCADA system.

The Works may be subdivided into pipeline and Site sections which may be tested separately.

The Contractor shall provide all necessary labor, materials, chemicals, electricity, fuel, stores, apparatus, instruments and indicators necessary to carry out the tests as per CoC clause 7.4.

Before starting the performance tests, it shall be ensured that the period for continuous operation will not be interrupted for adjustment or breakdown. Any such interruption, which are in the responsibility of the Contractor, will render the test void. In case of unreliable power supply, Employer, Engineer and Contractor

together will have to assess feasible testing procedures, which take account of the situation.

The performance tests shall be conducted continuously during a period of 21 days, uninterrupted by other tests. The objective of the tests shall be to obtain a comprehensive set of data to show that the system performance meets the requirements set-out in the performance guarantees. After a period of performance stabilization, all parameters shall be measured and recorded for comparison against performance guarantees made.

- Water samples shall be taken and tested in accordance with the national regulations for drinking water plants.
- Drinking water spot samples: At minimum, 95% of the drinking water spot samples taken during the tests shall comply with the specified drinking water quality.
- Combined drinking water samples: Combined 24h drinking water samples shall comply to 100% with the specified drinking water quality.
- Sludge samples: Per 5m³ dewatered sludge one sample shall be taken and analyzed. 100% of the samples shall comply with the value, specified in these Employer's Requirements. Test procedures, in accordance with the relevant standard of the selected dewatering equipment, shall be issued to the Engineer for approval together with the Testing Program.
- Test of daily production capacity: The daily drinking water production capacity shall be tested and checked, using the daily average flow meter records of the Drinking Water Pump station, generated by the SCADA system.
- Test of annual production capacity: The annual drinking water production capacity shall be tested and estimated, using the flow meter records of the Drinking Water Pump station, generated by the SCADA system. In case of lower production requirements at the beginning of the operation period, Employer, Engineer and Contractor will have to assess methods, which allow for testing, simulating the annual drinking water production.
- Tests of the raw water and drinking water pumps: Properties have to comply with the Specifications and the Guarantee Schedules.

Any deficiencies or deviations from the guaranteed or specified performance of the facilities disclosed by the tests shall be corrected by the Contractor and tests shall be repeated as necessary until the achieved test results comply with the Specifications.

The Contractor shall submit to the Engineer for consent the test records in such format as the Engineer may require, and after approval, shall provide the Employer with copies thereof. The Contractor shall be responsible for collecting and listing all data for such records.

Satisfactory completion of the tests on completion is the prerequisite for the issuance of the Taking-Over Certificate.

9.9 Verification of treatment performance during trial operation

9.9.1 Effluent standards

The effluent standards to be fulfilled are stated in table 17 of this document. The verification of the fulfilment of these effluent standards shall be done within the trial operation period before the Engineer's operational acceptance.

9.9.2 Compliance criterion

The compliance criteria are described in table 17 of this document.

9.9.3 Trial operation period analyses and measurements

Compatibility of the works with the performance guarantee during the trial operation period shall be confirmed to the full satisfaction of the Engineer.

i) Test failure criteria - water quality

Guarantee for the treated water quality is the most important one. The details about the standards of the treated water quality are given in table 16 of this document. This test shall be conducted as shown in Section 7 of this document during the trial operating period, where analyses are made on both raw water, water after each treatment unit taken at places agreed with the engineer and on treated water. Raw water analyses shall be compared to analyses in table 17 and if any unacceptable excess is found, the Contractor shall propose treatment adjustments to be approved by the Engineer. The pre-requisite of the operational acceptance, is compliance to guaranteed water quality of treated water. Final acceptance will be after defects notification period during which period, analyses of water will be made according to Tanzanian regulations.

The results for each parameter for each sample location shall be judged separately as follows:

- **Clarified Water**

The test on the clarified water shall fail if any of the following occurs:

- greater than 0.5% discrete samples exceed the maximum value in any week;
- greater than 1.5% discrete samples exceed the maximum value during the 28 days test period;
- any single discrete sample exceeds 10 NTU turbidity;
- greater than 10% discrete samples exceed the target value in the 28 days test period.

- **Filtered Water**

The test on the filtered water shall fail if any of the following occurs:

- any single discrete sample exceeds the maximum value;
- greater than 10% discrete samples exceed the target value in the 28 days test period.

- **Final Water**

The test on the final water shall fail if any of the following occurs:

- $\geq 5\%$ of the samples taken during the 28 days test period exceeds the effluent standard defined under Guarantee Schedule.

- any single discrete sample taken during the 28 days test period exceeds the effluent standard defined under Guarantee Schedule by 10%.

Table 17: Definition of target and maximum values, clarified and filtered water

Parameter	Units	Target	Maximum
Clarified water			
Turbidity	NTU	2	5
Total Aluminium	mg/l Al	1.0	2.0
Soluble Aluminium	mg/l Al	0.05	0.2
True Colour	TCU	-	15
Filtered water			
pH		6.2 – 7.2	
True Colour	TCU	5	10
Turbidity	NTU	0.5	1
Total Aluminium	mg/l Al	0.05	0.2

Table 18: Definition of minimum water quality parameters and maximum limits to be analysed for final treated water as per Guarantee Schedule

Indicators	Maximum Limits
Turbidity	< 0.1 NTU
Colour	< 10 TCU
Smell	No smell
Fe	< 0.3 mg/l
pH	6.5 – 8
Residual free Cl	≥ 0.5 mg/l

ii) Filter Back Flush and other service Consumption

The quantity of water abstracted and the quantity of treated water produced shall be monitored in order to assess whether the volume of water losses through washing filters and de-sludging clarifiers is greater than that guaranteed by the Contractor.

The test shall fail if the volume of water losses exceeds the values guaranteed by the Contractor in Schedule of performance guarantees

Plant configuration – recycling and direct filtration, shall not change over the entire duration of the tests. Should plant configuration be changed the Contractor shall continue the test in one specific configuration for at least 28 days and 7 days for Test after Completion.

iii) Chemical Consumption

From the start of the trial run period, records shall be kept about the raw water and treated water quality, chemical dispensing and all chemical consumptions. This information shall be used during this period for each and every chemical in determining the correlation between the raw water quality and the average amounts and in determining the quarterly and annual averages.

The test shall fail if the chemical consumption exceeds the guaranteed value stated by the Contractor in the Schedule of performance guarantees.

iv) Sludge Production

After the record keeping was started for the chemical consumption, which is explained in the item (iv) above, measurements for the sludge production shall start and records shall be kept. Following these measurements, the amount of the solids produced, the chemical consumption of these solids and their relation with the raw water quality shall be determined. The concentration of the solids in the sludge, which is taken from the sludge thickeners, shall be measured and recorded regularly. The amount of the solid at the mouth of the dewatering equipment shall be measured and recorded regularly.

v) Pump efficiency

The pump efficiency shall be recorded using pump flow meters, pressure gauges and electrical meters as shown in Schedule of performance guarantees

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vi) Power consumption

Suitable equipment, including approved power meters, energy meters and data loggers, shall be provided to allow monitoring of the following plant at each site:

- Raw Water delivery system
- Treated water delivery system
- Sludge recirculation pumps for clarification units
- Backwash facilities, pumps and blowers
- Recycling facilities, pumps and horizontal mixers
- Sludge disposal facilities, pumps
- Service and motive water pumps
- Works main incomers (total plant consumption)

I, V, $\cos \Phi$ and P shall be monitored for at least 24 hours on each equipment. Parameters shall be monitored when starting-up – running and shutting down the Plants.

Power consumption will be assessed over an average seven-day period during each of the Tests after Completion when the raw water quality is within the range stated in Schedule of Technical Information.

The test shall fail if the power consumption exceeds the guaranteed value stated by the Contractor in the Schedule of performance guarantees.

vii) Failure of performance test

In the event of failure of a Performance Test the Contractor shall prepare and submit to the Engineer for approval proposals for modifications to the Plant to enable the performance requirements to be attained.

On completion of the modifications to the Plant, the normal operation of the works shall be resumed and the Performance Tests shall be repeated as follows:

- Test after Completion (water quality and backwash water quantity): To be repeated in their entirety during the next 28 days period when the raw water is expected to be of the same or similar quality. If necessary, further modifications shall be carried out until the tests are successfully concluded;
- Test after Completion (power consumption and chemical consumption): To be repeated in their entirety during the next 7 days period when the raw water is expected to be of the same or similar quality (in the case of power consumption the Engineer may at his discretion permit the test to be repeated as soon as possible).

In the event of any of the Tests after Completion for power consumption or chemical consumption failing a second time the Liquidated Damages for failing Tests after Completion shall be applied.

Quality performance tests shall be made, defects corrected and the test procedures repeated until compliance before the final acceptance request is issued. List of Acceptance Tests given below.

Table 19: List of Acceptance Tests

Period of Pre-commissioning and Commissioning by the Contractor	Period for Preparing for Operation Before Granting Certificate of Operational Acceptance (two months Trial Operation)	Defects Notification Period Before Granting the Final Acceptance
Off-site and on-site tests made according to test plan made by the Contractor and approved by the Engineer	Blower, rapid mixers, water flow distribution to the clarifiers, clarifiers performance, performance equality of the clarifiers, adequacy of the filter backwash, tests specified in the specifications to be conducted on the clear water reservoir, tests of mechanical-, electrical equipment and SCADA all as specified in contract documents	Final test during last 28 days of Defects Liability period
	Monitoring the quality of raw, clarified, filtered and purified water	
	Quantity performance tests	
	Samples and tests according to Section 8	
	Monitoring of consumption of chemicals, power, service water etc. as specified. Monitoring of production of sludge	
	Consumption of filter backwash water	

9.9.4 Remedial action

In case of compliance failure, the Contractor shall execute the improvement as quickly as technically possible with reasonable consideration to all occurrences. The work must be completed within maximum three (3) months. The Contractor pays all costs related to improvement works.

After a new trial operation of maximum one (1) month from completion of improvement works, a new analysis of the control requirements shall be carried out. If results show that the control requirements are not met, the above procedure shall be repeated.

9.10 Tests after Completion

Tests after completion are foreseen after Taking Over, during the DNP, on a regular basis. The Employer will carry out the tests within the normal plant operation in the plant's laboratory. The Contractor may attend the tests. If the Contractor doesn't attend the tests, the readings shall be considered accepted by him as accurate.

- Water samples shall be taken and tested in accordance with the national regulations for drinking water plants.
- Drinking water spot samples: At minimum, 95% of the drinking water spot samples taken during the tests shall comply with the specified drinking water quality.
- Combined drinking water samples: combined 24h drinking water samples shall comply to 100% with the specified drinking water quality.
- Sludge samples: Per 5m³ dewatered sludge one sample shall be taken and analyzed. 100% of the samples shall comply with the value, specified in these Employer's Requirements. Test procedures shall be equal to the Tests on Completion.
- Test of daily production capacity: The daily drinking water production capacity shall be tested and checked, using the daily average flow meter records of the Drinking Water Pump station, generated by the SCADA system.
- Test of annual production capacity: The annual drinking water production capacity shall be tested and checked, using the flow meter records of the Drinking Water Pump station, generated by the SCADA system. In case of lower production requirements at the beginning of the operation period, Employer, Engineer and Contractor will have to assess methods, which allow for testing, simulating the annual drinking water production.
- Tests of the raw water and drinking water pumps: Properties have to comply with the Specifications and the Guarantee Schedules

9.11 Test Certificates

Where certificates are required by the Specifications or relevant Reference Standard, the original and one copy of each such certificate shall be provided by the Contractor, unless otherwise specified.

Certificates shall be clearly identified by serial or reference number, referring to the certified material, and shall include information required by the relevant Reference Standard or Specification clause.

Unless otherwise specified, the timing for submittal of certificates shall be as follows:

- Manufacturers and supplier's test certificates shall be submitted as soon as the tests have been completed and, in any case, not less than seven calendar days prior to the time that the certified materials will be used for the permanent works.
- Certificates of test carried out during the construction or on completion of parts of the permanent works shall be submitted within seven (7) days after completion of the test.

9.12 Training of Employer's Personnel

The Contractor shall at his cost prepare, submit for approval and execute a comprehensive training program for future operation and maintenance personnel. The Contractor shall submit the training program to the Engineer for approval 6 months prior to the start of training.

9.12.1 Training Requirements

The Contractor shall provide comprehensive training for the different categories of the Employers operation and maintenance staff. Training shall fall into two main types which are 'off the job' and 'on the job'. Off the job training shall take place in the class room on the job training shall be carried out on the running plant.

9.12.2 Off the Job Training

The Contractor shall prepare formal training documentation for distribution to the trainees. Visual aids shall be used where possible to illustrate the points being made and to make the training programme as interesting and enjoyable as possible for the participants. Off the job training shall comprise the following:

- i. on the simple chemistry and process principles involved in the operation of the Works;
 - ii. health and safety;
 - iii. plant safety procedures;
 - iv. on the use of the SCADA.
- Off the Job Training Programme for Operators
 - i. On the operation of individual items of plant and sections of the Works including automatic operation and manual operation in the event of say automatic control failure;
 - ii. On the day-to-day operation of the Works and procedures;
 - iii. On a comprehensive list of 'what if' scenarios dealing with the actions to be taken in the event of potential process problems, alarms, plant failures overflows, power failures etc.;
 - iv. On first line mechanical maintenance;
 - v. Safe methods of work general;
 - vi. On safety procedures to be followed in operating, maintaining and cleaning the plant;
 - vii. Special precautions to be followed in the event of chlorine leak.
- Off the Job Training Programme for Electrical Maintenance staff
 - i. on the configuration, construction and operation of the electrical Plant;
 - ii. on the electrical maintenance requirements of the Works;
 - iii. on the switching and safety procedures to be followed;
 - iv. safe methods of working;
 - v. on fault finding and repair procedures.
- Off the Job Training Programme for Control and Instrumentation Maintenance Staff
 - i. On the configuration, construction and operation of the Plant;
 - ii. On the control and instrumentation maintenance requirements of the Works;

- iii. On fault finding and repair procedures;
 - iv. Safe methods of working;
 - v. Special training on the use of the PC and associated programming software for fault finding on PLC based control systems;
 - vi. Special training on the use and performance of the central MMI and software and other specialist hardware and software systems used on the plant;
- Off the job training programme for Mechanical Maintenance
 - i. On the routine mechanical maintenance requirements of the Works;
 - ii. On lubrication requirements of the Works;
 - iii. On fault finding, repair and overhaul procedures;
 - iv. Safe methods of working.
 - Off the job training programme for water treatment Management staff
 - i. Water treatment process management techniques;
 - ii. Water treatment plant cost management;
 - iii. Water treatment plant laboratory management;
 - iv. Safe methods of work general;
 - v. On safety procedures to be followed in operating, maintaining and cleaning the plant.

9.12.3 On the Job Training

The Contractor shall utilize the Operations and Maintenance Manuals as the primary training aid in carrying out the on-the-job training. Short comings, omissions and errors identified in the O & M Manuals during the training shall be rectified prior to final acceptance of the O & M Manuals. On the Job training programme for all trainees

- i. Plant familiarization tour;
 - ii. On the use of the SCADA system;
 - iii. Health and safety;
 - iv. Identify areas where special safety precautions are necessary.
- On the Job Process Training Programme for Operators
 - i. under operational conditions on the operation of individual items of operation in the event of say automatic control failure;
 - ii. illustrate by example the day-to-day operation of the Works and procedures;
 - iii. illustrate by example the actions to be taken in the event of potential process problems, alarms, plant failures overflows, power failures etc. (as identified in the 'what if' scenario off the job training);
 - iv. illustrate by example the first line mechanical maintenance;
 - v. illustrate by example safety procedures to be followed in operation, maintenance and cleaning of the Works.

- On the Job Training Programme for Electrical Maintenance Staff
 - i. Carry out detail tour of the electrical plant;
 - ii. Illustrate by example the operation of the electrical Plant;
 - iii. Illustrate by example the electrical isolation and maintenance procedures;
 - iv. Illustrate by example fault finding and repair procedures;
 - v. Illustrate by example switching and safety procedures to be followed;
 - vi. Illustrate by example safe systems of work.

9.12.4 Training for Control and Instrumentation Maintenance Staff

- i. Illustrate by example the operation of the Works;
- ii. Illustrate by example the control and instrumentation maintenance requirements of the Works;
- iii. Illustrate by example fault finding and repair procedures
- iv. Illustrate by example fault finding on PLC based control systems, the central SCADA hardware and software and other specialist hardware and software systems used on the plant;
- v. Illustrate by example safe systems of work.

9.12.5 Training for Mechanical Maintenance Staff

- i. Illustrate by example the routine mechanical maintenance requirements of the Works;
- ii. Illustrate by example lubrication procedures;
- iii. Illustrate by example fault finding, repair and overhaul procedures.
- iv. Illustrate by example safe systems of work.

It is Contractor's obligation to inform/remind the Employer 6 months prior to start of training to start the recruitment process for the necessary O&M personnel.

9.13 Taking Over

The Taking-Over Certificate will be issued when the following obligations have been fulfilled by the Contractor:

- The satisfactory completion of the tests on completion.
- The handing over of all as-built documentation, O&M manuals, other documents, equipment and material as specified by the Contract Documents, including tools, spare parts, etc.

9.14 Defects Notification Period

The Defects Notification Period shall start after substantial completion of the physical works and issuance of the provisional TOC.

Following the testing and commissioning period of 3 months and the issuance of the TOC the Employer shall provide operating and maintenance personnel for the works.

The Contractor shall comply with the Contract during the DNP and shall be responsible for the following:

- Attending routine visits, as agreed by the Engineer; e.g. for plant tests, assessing the consumption of power and performance.
- Making good any defects in accordance with the Contract.

10. Provisional Sums

Provisional Sums are allowed for and are shown in the Schedules of Prices. Provisional Sums shall be used strictly in accordance with the Contract.

Provisional sums are foreseen for the following items only:

- Construction of all Overhead Transmission Lines to the Plant sites-TANESCO
- Optical Fibres-TTCL

ANNEXES

ANNEX 1: PROVISIONS FOR THE ENGINEER

Site offices for the Engineer:

The following offices with a total plan area of 100 m², complete with furnishings and equipment, all as specified, is required at the water treatment works site:

- 4 Nos. offices: 4 m x 5 m each
- 2 No. offices: 3 m x 4 m
- 1 No. conference room: 7 m x 6 m

Office Furniture and Facilities:

- 1 No desk (minimum 2000 mm x 900 mm) with six lockable drawers and swivel armchair to suit.
- 6 Nos. desks (minimum 1800 mm x 900 mm) with six lockable drawers and swivel armchair to suit.
- 2 Nos. desks (minimum 1200 mm x 750 mm) with four lockable drawers and swivel armchair to suit.
- 1 No conference table(s) (minimum 6000 mm x 2000 mm).
- 2 Nos. tables to match desk (minimum 2000 mm x 1000 mm).
- 12 Nos. upholstered upright chairs to suit conference table.
- 4 Nos. kitchen chairs and 1 kitchen table to match.
- 3 Nos. plan chests with six drawers to suit AO size drawings.
- 5 Nos. dustproof lockable cupboards (1800 mm x 900 mm x 540 mm) with four shelves.
- 5 Nos. lockable filing cabinets each with four drawers.
- 4 Nos. bookcases (1200 mm x 450 mm x 900 mm height).
- 4 Nos. Pinboards (3 sqm).
- 6 Nos. wastebaskets.
- 3 Nos. clothes lockers 0.5 m x 0.5 m x 1.80 m.
- 1 worktop, minimum 4 m long for viewing drawings.

Adequate Toilet Facilities, in two separate Bathrooms including:

- 2 Nos. European type W.C.s.
 - 2 No urinal.
 - 2 Nos. hand basins.
 - 1 No shower unit complete with shower tray, mixing taps and curtain.
 - 2 Nos. mirror's wall mounted (1500 mm x 1000 mm).
 - 1 No sanitary disposal unit.
 - 4 Nos. towelling facilities.
-

Adequate Kitchen Facilities, including:

- 1 No four-burner gas stove with oven, complete with two 25-kg gas bottles.
- 1 No single bowl sink with draining board.
- 1 No 340-liter capacity refrigerator.
- 1 No drinking water fountain.
- Crockery and cutlery (8 sets each, minimum), ironmongery, and utensils.
- 2 Nos. first-aid lots.
- 4 Nos. flashlights.

Office Sundries:

- 2 No Photocopy/scanner machine (wireless, automatic feeding, automatic 10 bin sorting).
- 1 No Server and network cables for networking of computer system.
- 1 No Binding machine.
- 3 Nos. Telephone (operator and devices).
- 5 Nos. Mobile Phones (reputable make, digital camera)
- Powerful Internet connection

Stationeries:

For all Offices the Contractor shall supply all necessary Office supplies, stationery and consumables, including requirements of pens, pencils, paper, erasers, notebooks, surveying books, sharpeners, staplers, adhesive tape, drawing accessories, chemicals for copying machine, bottled gas, cleaning materials, etc. for the duration of the contract to the requirements of the Engineer.

Computing Facilities as follows:

(a) 4 Laptop Computers

- 3 x USB 3.0
- DVD-RW
- 8T Hard Drive
- Processor: min. Intel Core i9
- 64 GB RAM
- 17" Full HD monitor
- WLAN, Bluetooth
- Windows 11, pre-installed
- MS Office Professional 2021 perpetual, AutoCAD Civil 3D pre-installed, and maintained during the entire Contract period

(b) 1 Nos Server Computer

- 4x USB 3.0
 - DVD-RW
-

- 10T Hard Drive
- 1x external hard disk (6 TB) for Back-ups
- Processor: min. Intel Core i9
- 32 GB RAM
- 32" full-HD monitor
- WLAN, Bluetooth
- Network equipment
- Enhanced keyboard and MS-compatible mouse
- Windows Server, pre-installed
- MS Office professional, pre-installed, maintained during entire Contract period

the Server computer shall be equipped with UPS.

- (c) 3 Nos. Desktop Full-color Printer / Copier / Scanner (A3/A4),
- sorter,
 - separate paper tray A3/A4,
 - full color support,
 - wireless,
 - HDD 16 GB,
 - Memory 4GB,
- (d) 2 No A4 printer (LaserJet, scanner -Kyocera).
- (e) 2 No Inkjet Printer (A3 and A4 Color Printer-Kyocera)
- (f) The electronic equipment shall be connected to electronic stabilizers to ensure data and hardware safety.
- (g) Windows Office for all laptops and computers and Special Software. Standard texts and calculations shall be executed with MS Office programs.

ANNEX 2: TRANSPORT FOR ENGINEER

The following means of transport shall be provided:

7 Nos.:

Land Cruise 76

Toyota; land cruise Hardtop – equivalent

5 doors

Cubic capacity = 2,400 cc, Diesel, Maximum power (bhp/rpm): 129/3800, 4X4

A/C unit

Radio/CD

- Operation and maintenance costs (including fuel) are to be borne by the Contractor.

ANNEX 3: TRANSPORT FOR COUNTERPARTS

The following means of transport shall be provided:

- 4 No.:
 - Pickup truck
 - Toyota Hilux – double cabin or equivalent
 - 4 doors
 - Cubic capacity = 2,400 cc, Diesel, 150 HP, 4X4
 - A/C unit
 - Radio/CD
- Operation and maintenance costs (including fuel) are to be borne by the Contractor.

ANNEX 4: EQUIPMENT FOR THE ENGINEER

The Contractor shall provide and maintain the following surveying equipment:

- 2 No GNSS/ Total Station (LEICA Smart Station or equivalent) with tripods and ancillary equipment (Leica GNNS RTK Rover or equivalent)
 - 1 No automatic digital level (LEICA or equivalent) completes with tripods.
 - 1 No 5 m extending levelling staves, with detachable staff bubbles and footplates.
 - 1 No 4 m extending levelling staves, with detachable staff bubbles and footplates.
 - 3 Nos. 30 m steel tapes.
 - 3 Nos. 30 m linen tapes.
 - 6 Nos. 5 m pocket tapes.
 - 6 Nos. Safety equipment (safety-leather shoes, rubber boots, signal jackets, helmets, ear-protection, eye-protection, waterproof protective clothing, torches, etc.).
-

ANNEX 5: WORKSHOP AND STORAGE EQUIPMENT

The Contractor shall provide and maintain the following equipment.

All equipment shall be of capacity, lifting capacity, and dimensions, typically needed for repairing pumps, machines, equipment, pipes, and metal parts to be procured and installed under the Contract. All machines shall be of high quality, heavy-duty, and easy to be used. The below specification gives only an indication.

- **Lathe machine:**

- with precise ground cylindrical steel guideways

- 3-jaw lathe chuck Ø 100 mm with turning and drilling jaws

- The main spindle bore Ø 20 mm

- Center distance 500 mm

- Center height 110 mm

- Workpiece diameter max. 220 mm

- Variable drive allows for 30 - 2000 rpm

- **Pillar drilling machine:**

- Digitally controlled threading

- Precise adjustment of the thread cutting depth by digitally controlled depth indicator

- Emergency-OFF punch button

- Electrical drill guard

- Table height adjustment with rack

- Drill height adjustment with rack

- Tapping capacity max. M8

- Drill depth 60 mm

- **Machine hacksaw:**

- With Coolant system

- Fully hydraulic feed control

- Pendulum vice

- Cutting range 90°: 240x190 mm

- Cutting range + 45°: 150 x 100 mm

- Average cutting speed: 16 m/min

- Stroke speed: 62 m/min

- **Pedestal grinding machine:**

- Closed guards at both sides

- Rigid and adjustable tool support

Equipped with two adjustable eye shields

Wheel diameter 175 x 25 x 32 mm

Speed: 2,800 rpm

- **Milling Machine:**

Vertical milling machine with dovetail guides and with universal tool holder

Longitudinal travel X-axis 260 mm

Transverse travel Y-axis 150 mm

Vertical travel Z-axis 280 mm

- **Electric arc welding set:**

Automatic control of stable arc regardless of the input voltage and changes in distance between the electrode and the metal

Capable of welding 1.6 mm to 5 mm electrodes

open circuit voltage: 100V

- **Oxy-Acetylene gas welding/cutting equipment:**

Wheeled trolley to carry industrial gas cylinders

Flashback arrestor for oxygen and for acetylene

Gas regulator for oxygen and acetylene

Fitted hose set with safety valves, 5-meter-long

Combined welding- and cutting shank, welding mixer, and set of welding nozzles

- **Small electric tools:**

2 electric hand drillers (2-speed), 600 W, including steel, concrete, wood drills, and spare parts;

2 electric battery-operated screw drivers with different chucks;

1 electric angle grinder, 210 W, including set of stone cutting, metal cutting, grinding, discs and cup wire brush, and spare parts;

1 electric circular hand saw, 1,600 W, cutting depth minimum 65 mm, including set of saw blades with point teething, wolf teething;

2 machine vices with prisms base and plain base with soft vice jaws

- **Mobile air compressor:**

Capacity: > 200 l

Pressure: 8 - 12 bars

Vessel volume: > 200 l

including accessory kit comprises:

Spray guns

Inflating gun

Air duster

Degreasing gun

Hoses

- **Mobile diesel generator, mounted on trailer:**

- 20 kVA prime power rating

- Tank capacity sufficient for 24 hours run time

- Fully automatic control

- Fully encapsulated with noise hood

- One axle trailer, with springs, dampers, brakes, lights

- **Lifting equipment:**

- 1 manual hydraulic pallet truck, capacity: 2-ton, max. lift height: 200 mm,

- 1 fast hydraulic lifting jack, capacity: 5 ton

- 1 mobile gantry, can be dismantled for easier transport, capacity: 2-ton, span: 3 m, clear height: 2.5 m, incl. manual hoist fitting to beam, and 2 chain slings

- **Complete tool set:**

- incl. all sizes of spanners, wrenches, screwdrivers, cutters, pliers, saws, vices, files, squares, rulers, hammers, brushes, pipe bender and cutter, set of chrome finish steel vernier calipers, gas burner, necessary to equip a professional mechanical workshop;

- **Mobile pumps;**

- 2 submersible electric drain pumps 3.5 kW, 400 V, with floater and 15 m cable including 15 m hose with coupling;

- Set of water and air hoses, gasket material for each grade of thickness and flange type;

- Oil handling equipment

- 5 oil measuring/filling cans,

- 5 oil funnels,

- 2 grease guns,

- 2 oil hand pumps,

- 2 grease hand pumps;

- **High-pressure cleaner:**

- Cold water high-pressure cleaner, with an Easy Press trigger gun and servo press controller for pressure and water flow regulation

- Robust brass cylinder head and ceramic piston

- Incl. high-pressure hose, 10 m and spray lance, 1,000 mm

- Flow rate: up to 1,000 l/h

- Working pressure: 150 bar

- **Multi-purpose vacuum cleaner/ wet and dry vacuum cleaner**

- Air volume: 55 l/sec.

- Vacuum: 220 mbar

- Capacity: 60 l

- **Safety tools:**

- 5 Inspection Lamps;
- 1 aluminum ladder 6 m long,
- 1 foldable aluminum ladder 3 m height,
- Tripod, lightweight aluminum with telescopic legs,

- **Complete electrical tool set:**

- 1 Industrial multi-meter for AC and DC, ampere-meter supplied with test leads;
- 1 multi-range insulation tester;
- 1 Digital industrial multi-meter;
- 1 Analogue industrial multi-meter;
- 1 Voltage tester 2 poles, testing range 4.5 – 380 V, AC/DC, with lighting indicator;
- 1 Universal voltmeter 4.5 - 380 V for direct and alternate current;
- 1 Amperage tester;
- 1 Digital AVO clip-meter;
- 2 sets of soldering / de-soldering, 100 W, (with inter changeable tip), pistol type;
- 1 Auto earth resistance tester
- 1 Complete set of pliers and wire strippers, screw drivers, necessary to equip a professional electrical workshop;

- **Consumables:**

- for 1 year for the workshop such as standard lubricants, gaskets, nuts and bolts, standard paint, etc.

ANNEX 6: LABORATORY EQUIPMENT

This laboratory will be provided with equipment, glassware and chemicals enabling the determination of the main bacteriological, physical and chemical parameters of the water.

The contractor shall supply all items necessary for operating the laboratory along the guidelines set out below so that the Authority can operate the laboratory without providing any additional items other than those supplied by the Contractor under this Contract.

Glassware shall be supplied with a maximum of 50% of spares compared to the equipment necessary to perform all analyses.

The supply shall include also a provision for one year's consumption of all chemicals and consumable materials.

The Contractor shall supply the necessary equipment for performing the tests detailed in the following sections, as well as operation manuals in English describing clearly the procedures for each test.

Physical Examination

- Color (Hazen Unit),
- Turbidity (NTU),
- Conductivity (micro mho/cm),
- pH value,
- Jar Test,
- Temperature.

Chemical Analyses (mg/l)

- Ammoniacal Nitrogen (as N),
 - Nitrate Nitrogen (as N),
 - Nitrite Nitrogen (as N),
 - Total Alkalinity (as CaCO₃),
 - Total Acidity (as CaCO₃),
 - Calcium Hardness (as CaCO₃),
 - Magnesium Hardness (as CaCO₃),
 - Total Hardness (as CaCO₃),
 - Total solids,
 - Total Dissolved Solids,
 - Suspended Solids,
 - Organic matter (KMnO₄ test),
 - Chloride (as Cl),
 - Phosphate (as PO₄),
 - Sulphate (as SO₄),
 - Iron (as Fe),
 - Manganese (as Mn),
 - Magnesium (as Mg),
 - Calcium (as Ca),
 - Residual chlorine (free and combined),
 - Dissolved oxygen,
 - Silicates,
 - Carbonates,
 - Bicarbonates.
-

Microbiological analysis:

- Total coliform
- Thermotolerant (fecal) coliforms
- faecal streptococci
- Heterotrophic plate count
- Salmonella
- Enteroviruses

The Contractor shall provide the following equipment:

Table 20: Laboratory Equipment

Item	Description	Quantity
I	Microbiological analysis and Microscope test	
1	Digital Incubators - 80 liter with Thermostat and thermometer (+/- 0.2°C)	1
2	Water bath or heat sink incubator, 44.5° (+/-0.2°) C	1
3	Vertical Autoclave 65 liter	1
4	GW-Bunsen Bunsen Burner Complete	2
5	Digital Hot Air Sterilizing Oven 160 to 170°C.	1
6	Refrigerator (2 -10°C) - 300 lt.	1
7	Inoculation Equipment	1
8	Vacuum pump complete with suction pipes	1
9	Sampling Thermometer 0-50 °C	2
10	Water ladle thermometer with substitute thermometer	1
11	Biosafety Cabinet-Level 2	1
12	pH/Cond/O2: HQ40d Multimeter, pH	1
13	Triple beam balance	1
14	AC Filter Membrane 0.45um,47mm 100P	each /for 2 years
15	Advantec Vacuum Filtration Manifolds with 3 valves	1
16	Dispersing machine with dispersing tool	1
17	Chlorine Test Kits	1
18	Indicator paper pH 0-14	
19	Emergency shower and eyewash	1
20	Metallic forceps	2 pairs

Item	Description	Quantity
21	Bi-Distillation unit with accessories	1
I.1	Microscope Test -Scale room	
1	Microscope and Light Source (10X to 15X with a diffused light developed by cool white fluorescent lamps) with camera	1
2	ACN Filter Membranes,0.45um,47mm 100P	5
3	Colony counter	1
4	Fume Hood with air extraction	1
5	Ultraviolet UV Sterilizer Cabinet	1
6	Moisture analyzer (Accuracy of the weighing 1 mg; Reproducibility: Initial sample weight > 1 g; (%): $\pm 0,20$; Readability: 0,01 %	1
7	Analytical balance: Capacity: 230 g, Readability: 0.1 mg	1
8	Emergency shower and eyewash	1
I.2	Reagent for microbiological analysis like:	
1	Dehydrated lauryl sulfonate tryptose (LST) broth -Reagent grade water	for 2 years
2	HiCrome E. Coli Agar	for 2 years
3	E. Coli Agar 90mm Plates 10 Pack	for 2 years
4	MFC Agar 65mm Plates 10 Pack	for 2 years
5	Ethanol 99.9% 2.5L	for 2 years
6	Adsorbent pads	for 2 years
7	Acetone	2 liters
8	Quantity tray sealer for Colilert method	for 2 years
9	Rubber sealing tray holders (quanti disc) - big trays	1
10	Rubber sealing tray holders (quanti disc) - small trays	1
11	UV light viewer-Colilert method	1
12	Quanty tray	pack of 100
13	Quanty tray 2000	pack of 100

Item	Description	Quantity
14	Colilert 18 media	pack of 200
I.3	Glassware for Microbiological Analysis	
1	Draining rack of wire	4
2	Corrosion resistant test tube racks	4
3	Petri dish	100
4	Erlenmeyer flask, 50 ml,	2
5	Erlenmeyers flask, 100 ml	2
6	Erlenmeyers flask, 250 ml	2
7	Erlenmeyers flask, 500 ml,	2
8	Erlenmeyers flask, 1000 ml	2
9	Measuring Cylinder 1 Liter	2
10	Measuring Cylinder 50 ml	1
11	Measuring Cylinder 100 ml	2
12	Measuring Cylinder 250 ml	2
13	Measuring Cylinder 500 ml.	2
14	Measuring Cylinder 1000 ml.	2
15	Glass Measuring Pipette - 1ml	2
16	Glass Measuring Pipette - 5ml	2
17	Glass Measuring Pipette - 10ml	2
18	Safety pipette filler universal for bulb/graduated pipettes all sizes	2
19	Durham Tubes	20
20	Glass Beakers, 50ml	2
21	Glass Beakers, 100 ml	4
22	Glass Beakers, 250 ml	4
23	Glass Beakers, 500 ml	4
24	Glass Beakers, 1000 ml	2
25	Watch glass	3
26	Stirring rods	2
27	Borosilicate glass Burettes 10 ml	1
28	Borosilicate glass Burettes 25 ml	1
29	Borosilicate glass Burettes 50ml	1

Ite m	Description	Quantit y
30	Borosilicate glass Burettes 100 ml	1
31	Borosilicate Glass Culture Tubes	20
32	Fermentation Tubes	3
33	Dilution Bottles closed with glass stoppers or plastic screw caps (Pyrex), 100ml	2
34	Dilution Bottles closed with glass stoppers or plastic screw caps (Pyrex), 250 ml	2
35	Dilution Bottles closed with glass stoppers or plastic screw caps (Pyrex), 500 ml	2
36	Dilution Bottles closed with glass stoppers or plastic screw caps (Pyrex), 1000ml	2
37	Sampling Bottles 500 ml	30
38	Sampling Bottles 1000 ml	30
39	Transfer loops	2
40	Bottle Wash 500ml	2
41	Sampling stick 2,3 m with beaker 1000 ml	1
42	Vacuum pump complete with suction pipes	1
II	Physical -Chemical analysis	
1	Photometer complete with accessories (last generation)	1
2	Spectrophotometer complete with accessories (last generation)	1
3	BOD 5 Day system complete with spare bottles and chemicals. Spares for 2 years	1
4	COD Digestor 25 place with cover	2
5	Fume Hood with air extraction for Physical-chemical	1
6	Fume Hood with air extraction for Reagent preparation room	
7	Moisture analyzer	1
8	Refrigerator (2 -10°C) - 300 lt.	1
9	Portable Multimeter or similar with spare probes and calibration kit	1
10	Portable Meter for pH, ORP, Ion, Conductivity, TDS, Salinity, Resistivity, Dissolved Oxygen, Temperature. With Spare probes, maintenance and calibration Kit.	1 each
11	Sediment cone set with rack	1
12	Dispersing machine with dispersing tool	1

Item	Description	Quantity
13	Desiccator	
14	Drying oven	1
15	BOD Refrigerated Incubator	1
16	Bi-Distillation apparatus with accessories -	1
17	Hot plate	1
18	Chlorine Test Kits	1
19	Jar Testing Equipment with accessories	1
20	Indicator paper pH 0-14	1
21	Analytical balance Capacity: 520g, Readability: 0,1 mg	1
22	Thermometers 0-100 Celsius	2
23	Sampling thermometer 0-50°C	2
24	Turbidity Meter: HF – Micro 100 WL Turbidity Meter	1
25	Open balance 2 kg.	1
26	Lovibond comparator with Chlorine discs. 0- 5.0mg/l	1
27	Emergency shower and eyewash	1
28	Electric stirrer	1
29	Digital titrator	1
30	Flow meter	1
31	Van dorn water sampler	1
II.1	Reagent for Physical Chemical analysis like:	
1	Ammonia range 0.2 +B96:B109-2.5 mg/l N 100 vials	3
2	Ammonia vial test reagents set COD VIALS Vario > 0 - 150mg/l 25 vials	for 2 years
3	COD vials Vario > 0 - 1 500mg/l	for 2 years
4	Vario NITRA X Reagent Unit set	for 2 years
5	Nitrite range: 0.01 to 0.5 mg/l N Nitrite tablets	for 2 years
6	Nitrite LR Tablets	for 2 years
7	Phosphate, ortho: Range: 0.05 to 4 mg/l PO4 tablets L/R 1	for 2 years
8	Phosphate, No. 2 LR tablets	for 2 years

Item	Description	Quantity
9	Phosphate, ortho, Range: 1 - 80 mg/l PO ₄ Phosphate HR P1 tablets	for 2 years
10	Aluminum test, range: 0.01 - 0.3 mg/l Al CELL & LID - PC CHECKIT	for 2 years
11	Aluminum No. 1 Tablets	for 2 years
12	Ammonia test, Range: 0.02 - 1.0 mg/l N CELL & LID - PC CHECKIT 12 Pack	for 2 years
13	Ammonia No. 1 Tablets	for 2 years
14	Chloride T1 Tablets 100 Tests	for 2 years
15	Chloride free range: 0.01 - 6 mg/l Cl, DPD No.1, tablets	for 2 years
16	Hardness (Total), Range: 2 - 50; 20 - 500 mg/l CaCO ₃ Cell & Lid - PC CHECKIT	for 2 years
17	Hardcheck tablets	for 2 years
18	Manganese, Range: 0.2 - 4 mg/l Mn CELL & LID - PC CHECKIT	for 2 years
19	Manganese LR NO 1 Tablets	for 2 years
20	Iron II, III Soluble, Range: 0.1 - 3mg/l Fe - W 100/OG/50mm CELL & LID - PC CHECKIT	for 2 years
21	Iron total Ferro 10ml sample, Powder Pillows	for 2 years
22	Nitrate Range: 0.5 - 14 mg/l N, Nitrate Cuvette test unit	for 2 years
23	Vario Nitrite Powder Pillows 100 Powder Pillows	for 2 years
24	Sulphate, range: 2 to 100 mg/l SO ₄ , CELL & LID - PC CHECKIT	for 2 years
25	Sulphate Vario sulpha 4 F10- Powder Pillows	for 2 years
26	D.P.D Tablets No.3	for 2 years
27	D.P.D Tablets No.1	for 2 years

Item	Description	Quantity
28	EDTA 0.02Mol solution standard 5l	for 2 years
29	Bromocresol Green 25g	for 2 years
30	Methyl red sodium salt 25g	for 2 years
31	Phenophtalyne disoduim salt 25 g	for 2 years
32	Sodium thiosulphate 250g	for 2 years
33	Sodium Hydroxide 250g	for 2 years
34	Sodium sulphite 100g	for 2 years
35	Sodium Carbonate 250g	for 2 years
36	Ammonium Hydroxide 250g	for 2 years
37	Potassium dihydrogen phosphate 250g	for 2 years
38	Methyl Orange powder 50g	for 2 years
39	Ammonium chloride 250g	for 2 years
40	Silver Nitrate 250g	for 2 years
41	TITRISOL Solutions.	for 2 years
42	Potassium permanganate 1 Normal for 500ml	for 2 years
43	Hydrochloric acid 0.1 Molar for 500ml	for 2 years
44	Nitric acid 0.1 Molar for 500ml	for 2 years
II.2	Chemicals	
1	Hydrochloric acid 33% x 2.5 l	for 2 years
2	Ammonia solution	for 2 years

Ite m	Description	Quantit y
3	Chloroform 2.5 l	for 2 years
4	Nitric acid 2.5l	for 2 years
5	Sulphuric acid 2.5l	for 2 years
6	Acetic acid	for 2 years
7	Isopropyl alcohol 2.5l	for 2 years
II.3	Glassware Physico	
1	Wide neck bottle, PP 100ml	2
2	Wide neck bottle, PP 250 ml	2
3	Wide neck bottle, PP, 500 ml	2
4	Wide neck bottle, PP, 1000ml	2
5	Wide neck bottle, glass 100ml	2
6	Wide neck bottle, glass 250 ml	2
7	Wide neck bottle, glass, 500 ml	2
8	Wide neck bottle, glass, 1000ml	2
9	Sampling Bottles 500 ml	30
10	Sampling Bottles, 1000ml	30
11	Distilled water wash bottles	2
12	Beaker, PP low form 100 ml	5
13	Beaker, PP low form 250 ml	5
14	Beaker, PP low form 500 ml	5
15	Beaker, PP low form 1000 ml	5
16	Beaker, glass low form 100 ml	5
17	Beaker, glass low form 250 ml	5
18	Beaker, glass low form 500 ml	5
19	Beaker, glass low form 1000 ml	5
20	Spatula spoon	2
21	Aluminum weighing dishes	4
22	Filter flask	1
23	Vacuum Flask	1

Ite m	Description	Quantit y
24	Vacuum pump	1
25	Filter pads	for 2 years
26	Volumetric wide tip pipettes, 25 ml	2
27	Volumetric wide tip pipettes, 50, ml	2
28	Volumetric wide tip pipettes, 100 ml	2
29	Borosilicate glass Burette 50 ml capacity	4
30	Burette capacity 25 ml	
31	Micro burette capacity 10 ml	2
32	Burette stand	2
33	Burette stand clamps double	2
34	Volumetric flask 50ml	
35	Volumetric flask 100ml	2
36	Volumetric flask 250ml	2
37	Volumetric flask 500ml	2
38	Volumetric flask 1000ml	2
39	Volumetric flask capacity 250 ml	2
40	Volumetric flask capacity 100 ml	2
41	Pipettes capacity 5ml	2
42	Pipette capacity 10 ml	2
43	Pipette capacity 25 ml	2
44	Beakers capacity 100ml	2
45	Beakers capacity 250ml	2
46	Beakers capacity 500ml	2
47	Beakers capacity 1000ml	2
48	Measuring cylinders 50ml	2
49	Measuring cylinders 100ml	2
50	Measuring cylinders 250ml	2
51	Measuring cylinders 500ml	2
52	Measuring cylinders 500ml	2
53	Dropper bottles 75ml	3
54	Dropper bottles 100ml	3
III	Others	

Item	Description	Quantity
1	Certified Safety Glasses (for eye protection)	10
2	Acid Resistant Neoprene Gloves (package of)	10
3	First Aid Kit in a Weather-proof Container (as per the NWT General Safety Regulations)	4
4	Cooler	3
5	Ice packs	10
IV	Furniture	
6	PC latest generation, for office purpose, incl. Basic software (Win, MS Office, antivirus, ACAD preview), internet available	9
11	Telephone tables	9
12	Telephone units	9
13	Telephone exchange (9 lines)	9

ANNEX 7: FURNITURE

The Contractor shall provide and place the following furniture for the Administration Building, Workshop, Guard houses, and Staff houses, at the various sites:

Table 21: Furniture

Building/room	Area (m2)	Features/equipment / furniture
1. WTP		
Administration Building		Clear height: 3.4 m (top ready floor slab till the bottom of the ready ceiling)
Corridor	15	1 fire extinguisher, wall mounted, 1 first aid kit, wall mounted.
Office	12	1 office desk ≥ 2 m ² , with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 1 armchair with 5 rolling casters, 2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, and indoor conditions to be achieved: max. air temperature 25°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 4 grounded sockets.

Building/room	Area (m2)	Features/equipment / furniture
Kitchen/mess	12	1 base cabinet L = 3.00 m, with kitchen counter on top, covered with stone tiles, including two recessed double stainless-steel sinks, battery (hot/cold water), 1 wall cabinet L = 3.00 m, 1 Microwave 20 l, 1,000 W, 1 Double hot plate, in total 1,500 W, 1 refrigerator, 100 l, 1 eating table, 1.2 m x 1.2 m 4 eating chairs 4 grounded sockets.
PLC room	12	1 office desk ≥ 2 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 2 armchairs with 5 rolling casters, 2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, and indoor conditions to be achieved: max. air temperature 25°C, air humidity: 60-80% (cooling capacity not less than 1,000 W) 4 grounded sockets, connected to the circuit for UPS
Shower room	12	3 shower basins 80 cm x 80 cm, with swivel arm and soap dish and cabin with lockable door 3 open racks (4 shelves) 80 cm x 200 cm, 40 cm deep, 1 hot water boiler, capacity: 50-liter, 2.5 kW elec.
Change room	12	8 lockers for clothes with 2 shelves, made of steel, including locks.
Toilet	12	2 standard western toilets including flush box and toilet roll holder, 2 Urinals with separation walls, manual flush valve, 2 wash basins with mirror, plate shelf, and soap dish.
Guard House		Clear height: 2.8 m (top ready floor slab till the bottom of the ready ceiling).
Office	5	1 office desk ≥ 1 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 1 armchair with 5 rolling casters, 1 bookshelf (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 4 grounded sockets.
Toilet	2	1 standard western toilet including flush box and toilet roll holder, 1 wash basins with mirror, plate shelf and soap dish.

Building/room	Area (m2)	Features/equipment / furniture
Accommodation building		Clear height: 2.4 m (top ready floor slab till bottom of ready ceiling). List applies for one semi-detached house only
Kitchen	12	1 base cabinet L = 3.00 m, with kitchen-counter on top, with recessed double kitchen sink covered with stone tiles, including two recessed double stainless-steel sinks, battery (hot / cold water), 1 base cabinet L = 3.60 m, with built in gas cooker / oven with 4 stove tops, 1 wall cabinet L = 3.60 m, 4 grounded sockets.
Shower + toilet	9	1 standard western toilet including flush box and toilet roll holder, 1 wash basin with mirror, plate shelf and soap dish, 1 shower basins 80 cm x 80 cm, with swivel arm and soap dish, 1 hot water boiler, capacity: 25-liter, 1.5 kW elec.
Bedroom	14	3 grounded sockets.
Living room	14	3 grounded sockets.
2. WTP		
Administration Building		Clear height: 3.4 m (top ready floor slab till the bottom of the ready ceiling)
Corridor Workers Section	30	1 fire extinguisher, wall mounted, 1 first aid kit, wall mounted.
Corridor Administrative Section	40	1 fire extinguisher, wall mounted, 1 first aid kit, wall mounted.
Secretary office	12	1 office desk ≥ 2 m ² , with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 1 armchair with 5 rolling casters, 2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 4 grounded sockets.
Archive	8	2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 4 grounded sockets.
Kitchen	8	1 base cabinet L = 3.00 m, with kitchen-counter on top, covered with stone tiles, including two recessed double stainless-steel sinks, battery (hot / cold water), 1 wall cabinet L = 3.00 m, 1 Microwave 20 l, 1,000 W, 1 Double hot plates, in total 1,500 W, 1 refrigerator, 100 l, 4 grounded sockets.

Building/room	Area (m2)	Features/equipment / furniture
Toilet Female	8	2 standard western toilet including flush box and toilet roll holder, 2 wash basins with mirror, plate shelf and soap dish.
Toilet Male	12	2 standard western toilet including flush box and toilet roll holder, 2 Urinals with separation walls, manual flush valve, 2 wash basins with mirror, plate shelf and soap dish.
Laboratory Office	12	1 office desk ≥ 2 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 1 armchair with 5 rolling casters, 2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 25°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 8 grounded sockets.

Building/room	Area (m2)	Features/equipment / furniture
Laboratory 1 (Phys / Chem)	22	Note: laboratory equipment is specified in other Sections in this document. 1 laboratory worktop covered with acid and lye proof ceramic tiles (pointed acid and lye proof), length: 400 cm, height: 85 cm, depth: 70 cm, 1 laboratory worktop covered with acid and lye proof ceramic tiles (pointed acid and lye proof), with double ceramic sinks (produced in one piece), acid and lye resistant batteries (hot / cold water), length: 240 cm, height: 85 cm, depth: 70 cm, 1 laboratory worktop covered with acid and lye proof ceramic tiles (pointed acid and lye proof), acid and lye resistant batteries (hot / cold water), length: 240 cm, height: 85 cm, depth: 70 cm, 6 floor units (length = 80 cm) below worktops, with 1 to 3 shelves and doors, 4 floor units (length = 80 cm) below worktops, with 4 drawers, 1 laboratory refrigerators, 160 l, 1 fume cupboard with glass side walls and glass front window, length: 60 cm, height: 100 cm, depth: 70 cm, to be placed onto the worktop, including electric connection and fan, 2 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (total cooling capacity not less than 2,000 W), 1 hot water boiler, capacity: 100-liter, 5.0 kW elec., 8 grounded wet room sockets, 4 grounded wet room sockets connected to the circuit for UPS, 1 emergency shower head with ripcord, 1 emergency eye washing unit, 1 fire extinguisher, wall mounted, 1 first aid kit, wall mounted.

Building/room	Area (m2)	Features/equipment / furniture
Laboratory (Biol)	2 15	Note: laboratory equipment is specified in other Sections in this document. 1 laboratory worktop covered with acid and lye proof ceramic tiles (pointed acid and lye proof), length: 400 cm, height: 85 cm, depth: 70 cm, 1 laboratory worktop covered with acid and lye proof ceramic tiles (pointed acid and lye proof), with double ceramic sinks (produced in one piece), acid and lye resistant batteries (hot / cold water), length: 240 cm, height: 85 cm, depth: 70 cm, 5 floor units (length = 80 cm) below worktops, with 1 to 3 shelves and doors, 3 floor units (length = 80 cm) below worktops, with 4 drawers, 1 laboratory refrigerators, 160 l, 2 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (total cooling capacity not less than 1,000 W), 6 grounded wet room sockets, 2 grounded wet room sockets connected to the circuit for UPS, 1 emergency shower head with ripcord, 1 emergency eye washing unit, 1 fire extinguisher, wall mounted, 1 first aid kit, wall mounted.
Office head of operations / manager	16	1 office desk ≥ 2 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 1 low table ≈ 1 m2, 1 armchair with 5 rolling casters, 2 chairs, 1 two-seater leather sofa set 3 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 8 grounded sockets.

Building/room	Area (m2)	Features/equipment / furniture
Meeting room	22	1 meeting table ≥ 8 m2, consisting of 2 m2 elements, 4 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 12 chairs, 1 overhead projector and roll-down screen 2 waste baskets, 2 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (total cooling capacity not less than 2,000 W), 8 grounded sockets.
SCADA room	12	Note: SCADA equipment is specified in other Sections in this document. 2 office desks ≥ 2 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 2 armchairs with 5 rolling casters, 2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 8 grounded sockets, 8 grounded sockets, connected to the circuit for UPS.
Server room	12	Note: Server and PLC equipment is specified in other Sections in this document. 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 8 grounded sockets, 8 grounded sockets, connected to the circuit for UPS.

Building/room	Area (m2)	Features/equipment / furniture
Maintenance office	16	2 office desks ≥ 2 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 2 armchairs with 5 rolling casters, 4 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 8 grounded sockets.
Office shift leaders	16	2 office desks ≥ 2 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 2 armchairs with 5 rolling casters, 4 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 1 AC unit of capacity sufficient to the gross volume of the room complete with all supports, electrical and plumbing work, indoor conditions to be achieved: max. air temperature 30°C, air humidity: 60-80% (cooling capacity not less than 1,000 W), 8 grounded sockets.
Kitchen Workers	16	1 base cabinet L = 3.00 m, with kitchen-counter on top, covered with stone tiles, including two recessed double stainless-steel sinks, battery (hot / cold water), 1 wall cabinet L = 3.00 m, 1 Microwave 20 l, 1,000 W, 1 Double hot plate, in total 1,500 W, 1 refrigerator, 100 l, 2 eating table, 1.2 m x 1.2 m 8 eating chairs 8 grounded sockets.
Toilet workers	16	3 standard western toilet including flush box and toilet roll holder, 2 Urinals with separation walls, manual flush valve, 2 wash basins with mirror, plate shelf and soap dish.
Change room	20	30 lockers for clothes with 2 shelves, made of steel, including locks.
Shower room	20	6 shower basins 80 cm x 80 cm, with swivel arm and soap dish and cabin with lockable door 6 open rack (4 shelves) 80 cm x 200 cm, 40 cm deep, 1 hot water boiler, capacity: 200-liter, 5.0 kW elec.

Building/room	Area (m2)	Features/equipment / furniture
Workshop and Store		Clear height: 3.4 m (top ready floor slab till bottom of ready ceiling).
Store	70	24 m heavy duty steel open steel shelves, 4 shelves levels, total height: 300 cm, depth: 60 cm.
Workshop, total (mechanical and electrical)	50	Note: Machines and tools are specified in other Sections in this document
Guard House		Clear height: 2.8 m (top ready floor slab till bottom of ready ceiling).
Office	5	1 office desk ≥ 1 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 1 armchair with 5 rolling casters, 1 bookshelf (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 1 coat rack, 1 waste basket, 4 grounded sockets.
Toilet	2	1 standard western toilet including flush box and toilet roll holder, 1 wash basins with mirror, plate shelf and soap dish.
Accommodation building		Clear height: 2.4 m (top ready floor slab till bottom of ready ceiling). List applies for one semi-detached house only
Kitchen	12	1 base cabinet L = 3.00 m, with kitchen-counter on top, with recessed double kitchen sink covered with stone tiles, including two recessed double stainless-steel sinks, battery (hot / cold water), 1 base cabinet L = 3.60 m, with built in gas cooker / oven with 4 stove tops, 1 wall cabinet L = 3.60 m, 4 grounded sockets.
Shower + toilet	9	1 standard western toilet including flush box and toilet roll holder, 1 wash basin with mirror, plate shelf and soap dish, 1 shower basins 80 cm x 80 cm, with swivel arm and soap dish, 1 hot water boiler, capacity: 25-liter, 1.5 kW elec.
Bedroom	14	3 grounded sockets.
Living room	14	3 grounded sockets.
3. Each Storage Tank		
Guard House		Clear height: 2.8 m (top ready floor slab till bottom of ready ceiling).

Building/room	Area (m2)	Features/equipment / furniture
Office	10	2 office desks ≥ 1 m2, with mobile file drawer (on 4 rolls) and drawers, all lockable incl. keys, 2 armchairs with 5 rolling casters, 2 bookshelves (4 shelves, lower 2 shelves with doors), 80 cm x 200 cm, 40 cm deep, 2 coat rack, 2 waste baskets, 8 grounded sockets.
Toilet	4	2 standard western toilet including flush box and toilet roll holder, 2 wash basins with mirror, plate shelf and soap dish.
